
A RANDOM FOREST APPROACH FOR CROP FERTILIZER RECOMMENDATION USING NPK SENSOR

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In Partial Fulfillment of the Requirements
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DEDICATION

This paper is dedicated in good faith to all these people, who supported, encouraged, and guided this project to completion. We are coming to extend our utmost gratitude to our families, mentors, classmates, and friends who strongly believed in us, supported us, and shared their own experiences that helped us to be even more determined in the course of our journey.

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ABSTRACT

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This study addresses the limitations of conventional fertilizer recommendation systems, which often provide generalized advice that fails to consider soil nutrient variability and crop-specific requirements, resulting in inefficient fertilizer use and reduced crop yields. The objective of this research is to develop a fertilizer recommendation system that supports informed decision-making through real-time soil nutrient analysis and machine learning techniques. The system utilizes an NPK sensor to measure nitrogen, phosphorus, and potassium levels in the soil. The sensor is interfaced with an Arduino microcontroller that collects and transmits nutrient data to a web-based application developed using the Flask framework. A quantitative analytical approach was employed to process soil nutrient data and evaluate the performance of a Random Forest machine learning model, which identifies nutrient deficiencies and generates appropriate fertilizer recommendations. Model performance was evaluated using standard statistical metrics, including accuracy, precision, recall, and F1-score. The results achieved an accuracy of 67.49%, precision of 67.72%, recall of 67.49%, and an F1-score of 64%, demonstrating satisfactory predictive performance. Overall, the prototype confirms that combining sensor technology with machine learning can deliver timely fertilizer recommendations, enhance fertilizer efficiency, improve crop productivity, and promote sustainable agricultural practices.

TABLE OF CONTENTS

Approval Page.....	i
Panel of Examiners.....	ii
Dedication.....	iii
Acknowledgments.....	iv
Abstract.....	v
List of Tables.....	ix
List of Figures.....	x
Chapter 1: Introduction.....	1
Background of the Problem.....	1
Statement of the Problem.....	2
Objectives of the Study.....	3
General Objective.....	3
Specific Objectives.....	3
Significance of the Study.....	4
Scope and Limitation.....	5
Project Dictionary.....	6
Notes.....	11
Chapter 2: Related Literature and Studies.....	14
Review of Related Literature and Studies.....	14
Soil Nutrient Management and Fertilizer Recommendation.....	14
Machine Learning in Agriculture.....	15
Random Forest Algorithm.....	16
Machine Learning Approaches for Fertilizer Recommendation (Focusing on NPK).....	17
Role of NPK Soil Data in Predictive Modeling.....	19
Synthesis of the State-of-the-Art.....	20
Gap Bridged by the Study.....	23
Notes.....	25
Chapter 3: Methodology.....	29
Research Design.....	29
Theorems, Algorithms And Mathematical Models.....	30

Equation.....	31
Pseudocode.....	31
Materials and Statistical Tools.....	33
Prototype Flowchart.....	34
Device Setup.....	36
Prototype Software Application Development.....	40
Research Process.....	42
Model Training.....	42
Data Collection.....	43
Synthetic Data Generation.....	44
Data Processing.....	44
Data Scaling.....	45
Model Evaluation.....	45
Cross-Validation.....	47
IoT Integration.....	48
Deployment(Flask).....	49
Recommender System.....	49
Theoretical Framework.....	50
Statistical Learning Theory.....	51
Ensemble Learning Theory.....	51
Bootstrap Aggregating.....	51
Decision Tree Theory:.....	52
Data-Driven Decision Making (DDDM) in Agriculture.....	52
Notes.....	54
Chapter 4: Results and Discussion.....	55
Training and Development of the Random Forest Model.....	55
Dataset and Exploratory Data Analysis (EDA).....	55
Synthetic Data.....	57
NPK Relationships and Correlation Matrix.....	59
Outliers and Missing values.....	61
Heatmap.....	62
Crop Distribution.....	64
Importance of Feature Analysis.....	66
Model Development.....	68
Model Evaluation.....	68
Accuracy.....	69
Precision.....	69
Recall.....	70
F1-Score.....	70
Overall Assessment.....	70
Confusion Matrix.....	71
Error Analysis.....	73
Soil Testing, System and Prototype Output.....	74

Prototype and User Soil Testing.....	74
Dashboard.....	75
Confirmation Button.....	76
End Page.....	78
Fertilizer Recommendation Output.....	79
Crop Ranking.....	82
Mobile Responsive.....	85
Chapter 5: Summary of Findings, Conclusion and Recommendations.....	87
Summary.....	87
Findings.....	88
Conclusion.....	89
Recommendations.....	90
Bibliography.....	91
Appendices.....	95
Appendix A: Language Editing Certification.....	96
Appendix B: Secretary's Certification.....	97
Appendix C: CRD Certificate.....	98
Appendix D: Joint Affidavit of Undertaking.....	99
Appendix E: Relevant Code Snippets	100
Appendix F: Role Acceptance Form	113
Appendix G: Hearing Form	116
Appendix H: Consultation Log Form	119
Appendix I: ACM	121
Appendix J: User Manual	127
Appendix K: Panels Recommendation, Suggestions and Comments (RSC) . . .	130
Appendix L: Certification of Dataset	138
Appendix M: Non-Disclosure Agreement Form	139
Appendix N: Project Team Assignment Form	145
Appendix O: Pre Proposal Title Form	146
Appendix P: Final Project Title Form	148
Vita	149

LIST OF TABLES

Table 1	Crop-Specific NPK Requirements (kg/ha) [1].....	34
Table 2	Synthetic Dataset Samples.....	58

LIST OF FIGURES

Figure 1	Prototype Flowchart	35
Figure 2	Device Setup	37
Figure 3	Research Process	42
Figure 4	Theoretical Framework	50
Figure 5	NPK nutrient distributions by crop type	56
Figure 6	Relationships and correlations across different crop types	60
Figure 7	Correlation Heatmap	63
Figure 8	Crop Distribution	65
Figure 9	Feature Importance	66
Figure 10	Evaluation Metrics	69
Figure 11	Confusion Matrix	71
Figure 12	Prototype	74
Figure 13	Dashboard	76
Figure 14	Confirmation Buttons	77
Figure 15	End Page Interface	78
Figure 16	Crop Selection Interface Urea	79
Figure 17	Crop Selection Interface Organic Compost	80
Figure 18	Crop Display Interface for Crop Recommendation	82
Figure 19	Crop Display Interface Crop Results	83
Figure 20	Mobile Application Interface	85

CHAPTER 1

INTRODUCTION

This chapter consists of the following: including the project purpose, the problem that solved by the researchers, and the objectives that achieved. It also elaborates on the importance of the study, scope, and limitations encountered.

Background of the Problem

Agriculture was an important component of the Philippine economy as it generates employment and guarantees food security in the Country. However, the increasing population and environmental complications have triggered the need for immediate adoption of environmentally sustainable and efficient farming practices. In the Philippines, there is still a strong reliance on out-dated, non-specific techniques for the application of fertilizers in conventional crop production. These methods tend to neglect particular nutrient requirements of the soil and particular crop needs, thus causing soil degradation, over fertilizing, and decreased crop yields and increased cost of production [3, 4].

Fertilizer use, which is frequently guided by standardized guidelines given out by agricultural advisers, does not take into account site specific characteristics. This one-size-fits-all approach will not suffice against the evolving climate patterns and variety in the make-up of the soils in regions. There is an increasing requirement for intelligent, site specific systems that maximize fertilizer recommendations while constituting for soil fertility, climate variability and type of crop [2, 17, 19].

The study of Alex, Sini Anna and Kanavalli, Anita(May 2023) developments in technology have filled this void. For example, a recommendation system based on Hadoop Distributed File System (HDFS) was designed for analyzing attributes of farm soil, envi-

ronmental variables and farmers' welfare indicators. It used feature extraction, dual-layer-computing, and cluster analysis for the formulation of fertilizer recommendations, which showed favourable results in terms of precision, recall, and overall accuracy for crops such as papaya, bananas, and leafy vegetables [1].

Similarly in India where agriculture is also a significant service area researchers have combined with IoT sensors, Hybrid XGBoost Classifier Algorithms that help determine the dynamic needs of fertilization. Such data-driven strategies have gone a long way in improving decision-making and brought noticeable improvements in crop productivity [12].

Another research tested an IoT-based system that has real-time soil monitoring and uses a Decision Tree algorithm to provide the recommendations on the type and quantity of fertilizer to be used. The model showed good predictive powers with the values of standard deviation ranging between 0.06 and 0.7, acceptable for real world scenarios [7].

This study has created a software platform to merge soil database information with learning algorithms and easy-to-use operating tools. A technology system seeks to provide precise and efficient sustainable recommendations compatible with farming practices and environmental objectives. By creating this system, research supports the country's food production agenda while protecting farmers' long-term health.

Statement of the Problem

In the Philippines, a large number of smallholder farmers continue to use the traditional method of applying the fertilizer without taking into consideration the actual nutrient composition of the soil or nutrient requirements of varieties grown. This generalized approach tends to lead to soil nutrient imbalance, and over-fertilization, higher input cost and low crop productivity. In spite of the technologies available out there, there is still an absence of effective and data-driven systems that can help the farmers in making accurate decisions regarding fertilizers with reference to the local soil conditions [21].

The traditional practice for fertilizer recommendation would often be manual, rule-based, or rely on periodic soil testing that involves great expenses, takes too much time, and cannot be scaled for small-scale agriculture. Besides, most of the existing systems do not take real-time soil nutrients data into account, which reduces their efficiency in addressing location-specific and crop-specific fertilization needs [16].

This study solved the problem by developing a web based/mobile friendly fertilizer recommendation system that uses in real time NPK sensor readings and Random Forest algorithm for site specific fertilizer recommendations. The idea was to optimize fertilizer performance, promote sustainable agriculture and strive for the promotion of high crop yield through the use of machine learning and sensor-based technologies.

Objectives of the Study

General Objective

The primary objective of this research was to develop and evaluate a Random Forest model using soil nutrient data for Nitrogen, Phosphorus, and Potassium to generate precise fertilizer recommendations.

Specific Objectives

The Smart Fertilizer Recommendation System was a prototype agricultural decision-support system that assisted farmers in making the best fertilizer decisions, based on nutrient analysis of soil. Three objectives were involved in the creation of the system:

1. To trained a machine learning model of a Random Forest to predict appropriate fertilizer recommendations using the available soil nutrients of Nitrogen (N), Phosphorus (P), and Potassium (K), in order to predict the appropriate fertilizer requirements of different crops.

2. To performed classification analysis on the performance of the model based on classification metrics including accuracy, precision, recall, and F1-score.
3. Created a functioning prototype that combined live data from NPK sensors with the trained program to produce an automated solution of fertilizers using a user interface.

Significance of the Study

The used of Random Forest Algorithm, the Smart Fertilizer Recommender System provided a modern solutions to traditional Philippine agricultural problems related to fertilizer applications and decision making. The research starts as a sustainable time-based fertilizer recommendations through a system that merges data processing algorithms with crop-specific parameters. The implementation of research reveals vital features that affect different sectors as follows.

Farmers - This system helped the farmers access exact fertilizer recommendations that leverage the real NPK values of their soils and crop requirements. The system replaces traditional guessing methods and general practices, which leads to improved productivity together with reduced costs and stronger crop health. The system helps to promote sustainable agriculture because it reduces both the dangers of excessive use of nutrients and environmental damage.

Department of Agriculture - The system functioned as a digital instrument to help the Department of Agriculture achieve precision farming objectives. The system delivers scientific information about fertilizer usage which enables maximum efficiency in both fertilizer distribution and directive operations. The efficiency of resource management improves through this system due to its support of government food security initiatives, which protect natural resources.

Camarines Sur Polytechnic Colleges - Camarines Sur Polytechnic Colleges (CSPC) can employ the system to deliver academic opportunities across agriculture, environmen-

tal studies, and computer science in educational and research applications. Students gained practical experience by using this system to understand agricultural machine learning applications, and their faculty members can perform additional research and innovation through it. The system functioned as a tool for school farming initiatives to enhance the management of program lands.

Community - Healthier crops, together with increased production at the local level and reduced market prices, benefit the larger community. The environmental benefits include clean water resources and healthy ecosystems that arise from reducing the risk of fertilizer runoff associated with over application. Sustainable agriculture happens through providing necessary support for lasting food safety, together with climate resistance.

Researchers - This project enabled us researchers to bridge academic principles of machine learning and data analysis with system development to solve actual challenges in agricultural practices. The project enables us to address complex problems by implementing multiple discipline methods, which ultimately helps improve smart farming technologies through innovative solutions.

Future Researchers - This system creates the research base that future researchers will use for precision agriculture development. The system's functionality extends beyond current specifications because researchers can add more information assets, including climate data and pest observations, while simultaneously using mobile or automatic procedures for operation. The project provides researchers with a useful basis to build from when creating innovations in agricultural technology and sustainable farming systems in the future.

Scope and Limitation

Smart Fertilizer Recommendation System is an agricultural decision-support prototype that assists farmers to make the best decisions regarding fertilizer application depending on the analysis of the soil nutrients. In the prototype, the primary soil factors such as Nitrogen

(N), Phosphorus (P) and Potassium (K) are manipulated with the Random Forest algorithm to produce data-driven fertilizer recommendations that will maximize the crop yield and minimize fertilizer wastage. The prototype targeted at the farmers, agriculturalists and the researchers who would like to analyse the soil data using NPK sensors or manually keying the data into the system. The project is aimed at creating a working prototype web based prototype that can be utilized by the potential customers of the project, the farmers of Iriga City, Camarines Sur that produce 10 major crops in the Philippines namely: Tomato, Eggplant, Chili Pepper, Cabbage, Okra, String Bean, Carrot, Chayote, Ampalaya and Patola. The information applied in the development of the models is obtained based on the published records of agriculture and research studies and this makes the recommendations of the system credible.

The prototype has a number of limitations as opposed to the benefits. It is based only on the NPK values and fails to consider other important variables like the weather conditions, pest attacks, irrigation rates and soil PH all of which are very critical in fertilizer performance. The prototype is not fully automated as well, so in case the sensor integration is not possible, a user will be required to manually enter NPK readings. In addition, the model could be inaccurate depending on the type of soil, geo-location, and the type of crop used since the recommendations are guided by pre-determined datasets. Lastly, the fact that the results of the study were confined to the Iriga City narrows down the ability to extrapolate the prototype findings to other large-scale agricultural conditions in the Philippines.

Project Dictionary

The Project Dictionary contains the technical terms that define the concept and operation of this study:

Accuracy: Accuracy measures the proportion of correct predictions made by the model out of all predictions [22]. In this project, accuracy is applied to evaluate system.

Confusion Matrix: A table used to evaluate classification models by comparing predicted and actual labels, showing true positives, true negatives, false positives, and false negatives to assess performance metrics like accuracy, precision, and recall [9]. By comparing the model to actual fertilizer label data, this tool was utilized in this study to assess the accuracy of the fertilizer recommendation model for different fertilizer classes and NPK.

Crop Recommendation: A system that offers a novel approach to crop management by leveraging NPK sensor technology and machine learning [5]. Here, this system uses crop suitability predictions based on soil conditions, NPK, and machine learning results to determine which crops are best to plant.

Data-Driven Decision Making (DDDM): Data-Driven Decision Making (DDDM) is the process of making informed choices in agriculture by relying on algorithm-generated insights from ongoing soil and crop data instead of traditional practices or guesswork [10].

In this project, DDDM is applied by using soil NPK sensor data from Iriga City as input to a Random Forest classification model. The system analyzes these data and generates recommendations for the top 10 most suitable crops, providing farmers with evidence-based guidance rather than relying on conventional or experience-based decision-making.

Evaluation Metrics: Evaluation metrics are quantitative measures used to assess the performance and accuracy of machine learning models [18]. In this study, evaluation metrics such as accuracy, precision, recall, and F1-score are applied to validate the effectiveness of the Random Forest model in providing correct fertilizer recommendations.

F1 Score: F1 Score is the harmonic mean of precision and recall, providing a single measure that balances both metrics [14]. In this project, the F1 Score is used to assess how well the fertilizer recommendation system maintains a balance between precision and recall when analyzing NPK values. This measure was quite helpful in terms of assessing the model performance in case of imbalanced distributions of nutrient-classes. It offers only, one good measure that punishes excessive bias in either false positives or false negatives.

Feature Importance: Feature Importance refers to identifying which input variables contribute most significantly to a machine learning model's predictions. In the Random Forest approach for agriculture, soil nutrients such as Nitrogen, Phosphorus, and Potassium play the most influential roles in determining the correct fertilizer choice [13].

Fertilizer Recommendation System: A web-based application built during this study to allow users or sensors to provide NPK data and determine which fertilizer type (Urea, Organic Compost, or NPK 14-14-14) best suits their chosen crop [6]. This project provides optimized fertilizer suggestions aimed at enhancing crop yield and minimizing excessive fertilizer use.

Flask Framework: Flask is a lightweight web framework in Python used to develop and manage the backend of the fertilizer recommendation system. It handles user requests, integrates machine learning models, and generates appropriate web interface responses [20]. In this project, Flask connects the Random Forest model with the user interface, ensuring smooth interaction between soil data input, model processing, and recommendation output.

IoT in Agriculture: IoT in Agriculture refers to the use of sensors and wireless systems to collect real-time data from the field, enabling smarter and more efficient farming practices [7]. In this project, IoT is applied by integrating an NPK sensor with an Arduino device, which transmits soil nutrient data over the Internet to the web application for real-time fertilizer predictions.

Mean Absolute Error (MAE): Mean Absolute Error (MAE) measures the average magnitude of the errors between the predicted and actual values, regardless of their direction [22]. It evaluates how close the system's predictions are to the true nutrient requirements by averaging the absolute differences between the predicted and observed fertilizer values. In this project, MAE is used to assess the overall accuracy of the Random Forest model in predicting fertilizer recommendations based on soil NPK values. With lower MAE values indicating more precise and reliable fertilizer dosage predictions.

Nitrogen (N): Nitrogen is one of the essential macronutrients required for plant growth. Promotes chlorophyll production, leaf development, and overall vegetative growth [11]. In this study, Nitrogen serves as one of the key soil nutrients analyzed to determine the appropriate fertilizer recommendation.

NPK Sensor: An NPK sensor measures the current amount of three essential soil nutrients—Nitrogen (N), Phosphorus (P), and Potassium (K) [11]. In this project, the NPK sensor provides real-time soil nutrient data that enables the system to generate fertilizer recommendations based on up-to-date soil analysis.

Phosphorus (P): Phosphorus is a vital nutrient that supports root growth, energy transfer, and flowering in plants [11]. In this project, Phosphorus is included as one of the primary soil parameters used in generating fertilizer recommendations.

Potassium (K): Potassium enhances plant disease resistance, improves drought tolerance, and helps regulate water uptake [11]. In this study, Potassium plays a significant role in balancing soil nutrients to ensure optimal fertilizer application.

Precision: Precision measures how many of the predicted positive cases are actually positive [14]. In this project, precision evaluates how accurately the system's fertilizer recommendations align with correct nutrient requirements based on NPK values.

Precision Agriculture: Precision Agriculture is a modern farming approach that utilizes advanced technologies such as sensors and machine learning to improve decision-making and optimize input efficiency [10]. In this project, Precision Agriculture is applied by implementing data-driven fertilizer planning based on soil NPK sensor readings and machine learning analysis.

Random Forest Algorithm: The Random Forest Algorithm is a machine learning method that constructs multiple decision trees and combines them for classification and regression tasks [13]. In this study, it analyzes soil nutrient data (N, P, K) to generate fertilizer recommendations through historical soil data analysis. By leveraging past nutrient patterns, the system is able to identify optimal fertilizer requirements.

Recall: Recall measures the proportion of actual positive cases that are correctly identified by the model [14]. In this project, recall determines how well the system detects and recommends the correct fertilizers that match the soil's NPK nutrient profile.

Root Mean Squared Error (RMSE): Root Mean Squared Error (RMSE) represents the square root of the average squared differences between predicted and actual values [22]. Unlike MAE, it penalizes larger errors more heavily, making it more sensitive to significant deviations. In this study, RMSE is applied to evaluate the robustness and reliability of the fertilizer recommendation system when estimating the nutrient requirements derived from soil NPK data.

Supervised Machine Learning: Supervised learning is a type of machine learning where models are trained on data that include both inputs and corresponding outputs [8]. In this project, the Random Forest algorithm is applied as a supervised learning model, classifying the best fertilizer recommendation among N, P, and K based on soil nutrient data.

Training Dataset: A training dataset is a collection of data used to teach a machine learning model, enabling it to recognize patterns and make predictions based on learned relationships [15]. In this project, the training dataset consists of soil and fertilizer records, including NPK nutrient values, associated crops, and corresponding fertilizer labels.

Web-Based Application: A web-based application is a user interface that can be accessed through the Internet using any standard web browser [20]. In this study, the system is developed as a web-based application using Flask, allowing both farmers and agricultural experts to share soil information and view automated fertilizer recommendations on any device.

Notes

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CHAPTER 2

RELATED LITERATURE AND STUDIES

This section of the study contained the review of the related literature that was relevant to the study of A Random Forest Approach for Crop Fertilizer Recommendation using NPK Sensor.

Review of Related Literature and Studies

Soil Nutrient Management and Fertilizer Recommendation

As stated by Reddy et al. (2024) a machine learning-driven system has been designed that combines the data nesting of soil and satellite images, to render more precise suggestions on the use of fertilizers and timing of irrigation. The authors stated that these models can deliver recommendations based on user-specific conditions and are able to process large amounts of data to do it [20]. On the same note, Sethi and Lakhina (2024) suggested a real-time crop suggestion framework that employs the use of soil nutrient parameters (N, P, K, Fe, Zn, and others) in addition to climate data. Several machine learning methods were used to estimate the appropriate crops and the amount of fertilizers needed to cut down on the possibility of random and overuse of fertilizers [23]. Sarkar et al. (2021) used different classification algorithms to analyze the optimal choice of fertilizers in the paddy and wheat planting in West Bengal. Their strategy involved the use of Soil Test Crop Response (STCR) system and gave several fertilizer prescriptions looking at the financial capabilities and availability of agricultural inputs by the farmers to the farmers [21].

The nutrient levels in the soil should be well controlled in order to keep the soil healthy and yield more crop. Farmers had in the past applied the fertilizers inaccurately because

they were not well aware of the nutrient status of the soil. The increasing environmental issues and the necessity of precise agriculture have promoted the introduction of data-driven methods, specifically machine learning, to nutrient management. Following the mentioned processes, the given work seeks to streamline the fertilizer needs with the help of a Random Forest model and NPK sensor information [20, 23].

The results of the research carried out by other scholars prove the idea that machine learning is an efficient method to enhance nutrient management and find the right courses of fertilizer strategies. Despite the variations in methods, the outcomes are the same that, with the help of the data-driven fertilizer practices, which are informed by the soil and consider the environmental impact, the crop productivity can be increased, whereas environmental impact can be minimized. Nonetheless, there is an existing gap in terms of availability of real-time systems, which are able to process sensor data in offline conditions. The proposed NPK sensor-integrated Random Forest model is a solution to this limitation, as demonstrated in the article by Sethi et al. (2024) (Intelligent) and Sarkar et al. (2021) (Intelligent). [23, 21].

Machine Learning in Agriculture

One of the studies by Panchal and Jain (2024) talked about how the intelligent farming systems enabled by machine learning are able to assess the weather, soil characteristics, and status of crops to predict future yield, optimize the use of fertilizers, and identify plant diseases. The authors highlighted that these systems can also help local agricultural industries to cut down on the cost of inputs and also increase their efficiency [18]. Liakos et al. (2018) analyzed a broad scope of machine learning applications in the Agriculture industry, such as machine learning in crop production, machine learning in disease and weed identification, machine learning in soil management, and machine learning in livestock tracking. Their results emphasized that the combination of machine learning and sensors and preci-

sion farming approaches present real-time insights and support decision-making to farmers [14]. In a similar manner, Yadav et al. (2024) explored the application of machine learning to increase the production of food at the minimum waste and reduced environmental pollution. In their work, sustainable agriculture has developed in the context of Internet of Things (IoT) devices, drones, and machine learning being one of its technological building blocks [28].

Machine learning has helped enhance precision of crop management, and it takes advantage of data-driven knowledge. The problem of climate variability and shortage of labor by farmers has been addressed by applying machine learning as real-time analysis and automation. The field of study is also an important one, whereby machine learning models are capable of prescribing the correct amount of fertilizer application depending on the data collected through NPK sensors [14, 28]. Literature has established that machine learning is one of the technologies that have become central to the contemporary agricultural sector, where it is used to monitor crops and make well-informed choices about the respective farms. In spite of the fact that the use of real-time sensing and automation continues growing, such issues as the availability of data and the interpretability of models have not been overcome. This work has a contribution to the field as it aimed at discussing how machine learning algorithms could be used in conjunction with sensor technologies to facilitate targeted and effective fertilizer application [14, 28].

Random Forest Algorithm

Using the analysis conducted on the crop growth and environmental data, Geetha et al. (2020) used the Random Forest (RF) algorithm in predicting crop production in different weather conditions. The findings were high predictive accuracy and strength, which implies that the model can be used in real-time crop monitoring and agricultural decision-making. One model was developed by Tynchenko et al. (2024), and it was an RF-based model focused on measuring water quality and its effects on agro-industrial productivity

[8]. Their methodology examined physical and chemical water parameters and was able to determine the factors that affected the health of soil and crops [27]. and so RF proved to be effective in monitoring the environment. In the same vein, Sayyad et al. (2024) suggested an all-encompassing system comprising of comprising of IoT sensors and the Random Forest algorithm as an indication of environmental conditions (temperature and humidity) to predict crop diseases. The system had an accuracy of 92 and this indicates the ability of RF in real-time sensor based agricultural systems [22]. Elavarasan and Vincent (2021) came up with a more advanced form of a Random Forest model that uses reinforcement learning when creating a tree in prediction of crop yields. The results of their research showed that they were able to achieve high accuracy of 92.2% with lower error rates than traditional methods of machine learning [5]. Random Forest is an ensemble learning algorithm that is widely used and it combines many decision trees to enhance the prediction accuracy and reduce overfitting. It is especially applicable to agricultural application due to its capability of dealing with non-linear relationships, coping with large sets of features, and working with heterogeneous data. The subsection identifies the application of Random Forest in predicting crop yield, water quality, and detection of crop diseases [8, 22]. In general, the Random Forest algorithm has demonstrated to be a viable tool in the process of predicting the results of agricultural activity, measuring environmental trends, and assisting farmers in making timely decisions. Random Forest is an ideal fit in the quality of accuracy in precision agriculture systems like the proposed fertilizer recommendation system which uses NPK sensor readings due to its adaptability, precision and compatibility with sensor-generated data [22, 5].

Machine Learning Approaches for Fertilizer Recommendation (Focusing on NPK)

Gottemukkala et al. (2023) suggested how a sustainable crop recommendation system would be designed to use real-time data collected through NPK sensors. In the research, the correlation between the level of nutrients in the soil and crop yield was studied using

the machine learning method. The findings showed that the system has a potential of providing precise and eco-friendly nutrient advice [9]. On the same note, Thapa et al. (2024) designed a smart agricultural system by incorporating machine learning as well as a deep learning system with NPK sensors. Random Forest was utilized in crop and fertilizer recommendation, whereas LightGBM with Conv2D was utilized in insect identification. The suggested platform helps monitor the health of the soil and optimize the use of fertilisers in real-time, which is especially effective in resource-deficient rural areas [26].

Tanaka et al. (2024) compared the output of different machine learning systems, such as Random Forest, XGBoost, and Support Vector Machines (SVM), of proposing economically optimal input rates (EDIRs) by using NPK fertilizer experiments. Their results showed that machine learning models can be used to accurately predict crop yields, but the results of EOIR recommendations are highly dependent on the choice of algorithms and feature sets, which makes model choice and deployment a very sensitive issue [25]. Munezero et al. (2024) used a number of machine learning models, including the Random Forest, Decision Tree, and Gradient Boosting, to forecast NPK fertilizer demand of cassava plants in Rwanda. Random Forest model was found to be effective in terms of accuracy of prediction 90% when used in managing fertilizers in real time in the context of using IoT sensor data [17]. Moreover, Komala Devi Kumar (2024) presented an ensemble machine learning-based system with an assistant to recommend the crops and fertilizer according to the NPK and environmental factors. The system obtained up to 96.5% of accuracy in different soil and climatic conditions, which supported the strength of ensemble methods in agricultural decision support [12].

Sustainable agricultural practices require that appropriate recommendations of fertilizer must be done. The major primary macronutrients are nitrogen (N), phosphorus (P), and potassium (K) also referred to as NPK, which are needed to ensure optimum growth in plants. Combining machine learning methods with NPK data has been demonstrated to be an effective method of producing site-specific fertilizer prescriptions based on different

soil conditions. In this section, the author discusses machine learning-based systems which aid the fertilizer management process by examining the NPK-related data [9, 26]. There is a possibility to choose relevant types of fertilizers and application rates with the help of machine learning methods, like Random Forest and ensemble models, relying on the NPK measurements. Such techniques are very precise, flexible and can be adopted in real-time. Nevertheless, as the performance of models may differ depending on the location and conditions, they have to be carefully calibrated and localized to provide reliable information. The current work is based on the previous literature as it builds on existing methods by using real-time data provided by NPK sensors and a localized Random Forest model to provide precise and context-specific fertilizer application recommendations [25, 12].

Role of NPK Soil Data in Predictive Modeling

Monica et al. (2023) have designed an Extreme Gradient Algorithm (EGA) model to determine the soil NPK content depending on environmental conditions, including the temperature, pH, humidity and rainfall. The model was much better at predicting nutrient levels than the old ones, indicating that accurate NPK prediction will result in more efficient crop management and crop optimization [16]. Siddique et al. (2024) suggested a system that combines the NPK sensor data with machine learning algorithms, such as a Random Forest and a Naive Bayes, to forecast the crop in real-time. Their research indicated that predictive models which consider real-time NPK values improve the decision-making process on the field and do not involve the use of the laboratory-based testing of the soil [24]. Dattatreya et al. (2024) assessed the traditional and the latest soil sensing technologies and highlighted that the combination of NPK sensors data with machine learning is a ground-breaking method in the agricultural field. The investigations point to the fact that the models constructed on the basis of sensor-based NPK measurements have the potential to give fast and precise fertilizer recommendations [4]. Adhikary et al. (2024) used neural networks

to predict the nutrient content in soils using the measurements of the soil using different outdoor sensors. Their method enhanced the accuracy of sensor measurements and proved that the accuracy of NPK predictions can improve the performance of smart agriculture systems [1]. Aniyen et al. (2024) have created an Android application that uses the data of machine vision and NPK sensors to identify the type of soil and propose the suitable crops. The program used Support Vector Machines and Random Forests allowing to make predictions not only by the measurements of NPK but also by the soil photos [2].

Application of soil NPK data is vital in predictive agriculture especially in the creation of systems that pertain to crop advisory, soil analysis and optimal regulation of fertilizers. Precision agriculture NPK sensor data has received a lot of importance in encouraging sustainable farming practices particularly in real-time. The studies reviewed reveal that more NPK data will enhance model accuracy and reliability in the agricultural forecasting and decision support models [16, 24].

Synthesis of the State-of-the-Art

Modern farming is moving towards more data-driven soil management and that specifically requires a proper measurement of nitrogen (N), phosphorus (P), and potassium (K). Correct evaluation of NPK allows to optimize the use of fertilizers, maintains soil health, reduces the loss of nutrients, and decreases the dangers of the environment, including soil erosion and acidification. These findings were demonstrated in a number of studies such as Reddy et al. (2024), Sethi and Lakhina (2024) and Sarkar et al. (2021) who noted that soil management systems based on machine learning could provide site-specific fertilizer recommendations, which consequently led to reduced resource use and better crop performance. The results confirm that the incorporation of soil nutrient information with intelligent decision-making systems can be very effective in improving the productivity and sustainability [23, 21, 20].

Machine learning (ML) has been at the center of the Agricultural modernization. Liakos et al. (2018) emphasized the effectiveness of ML in forecasting yields, evaluating soil, and pests. Further investigations by Panchal and Jain (2024) and Yadav et al. (2024) extended the scope of the use of ML by adding the use of IoT sensors, drones, and environmental monitoring systems to make predictions in real-time and control fields automatically. The technologies alleviate human mistakes, streamline the use of labour, and enhance the ability to adapt to climate fluctuations [14, 18, 28]. Together, these works indicate that smart agricultural systems will always enhance the precision of the decision, cost-effectiveness, and flexibility.

The Random Forest (RF) algorithm became prominent in the field of handling nonlinear and high-dimensional agricultural data as one of the ML methods. Geetha et al. (2020), Tynchenko et al. (2021), and Sayyad et al. (2024) in the studies found that RF is a strong predictor of the following tasks: crop yield estimation, disease detection, and water quality assessment. Elavarasan and Vincent (2021) also improved the performance of RFs with reinforcement learning, reaching an accuracy of more than 90% in the prediction of crops. All these studies suggest that RF minimizes overfitting, can handle multivariate sensor inputs effectively and can be used in real-time in the agricultural sector [8, 5, 22].

Recent studies have been concentrated on the combination of RF and other ML models with real-time NPK sensor. The alternative models designed by Gottemukkala et al. (2023) and Thapa et al. (2024) to create ML-based fertilizer recommendation systems based on Random Forest and LightGBM with soil nutrient sensors are capable of producing sustainable and crop-specific advice. Equally, the case analysis of Munezero et al. (2024) and Komala Devi Kumar (2024) indicated that the classification of fertilizers and predicting crop suitability with the help of hybrid ensembles demonstrated over 90% accuracy. These works point out that where ML models may provide a powerful performance, the contextual calibration of recommendation of economically optimal input rates (EDOIs) are necessary to consider the local soil and climate variability [9, 26, 12].

In addition to these ML-based solutions, the sensor technology has made it possible to monitor soils in real time. Monica et al. (2023) and Siddique et al. (2024) proved that NPK electrochemical probes that are combined with ML algorithms are able to provide high-speed and precise nutrient determination without the necessity of laboratory analysis. Other researchers Adhikary et al. (2024), Aniyar et al. (2024), and Dattatreya et al. (2024) also reported that sensor-based soil analysis, further enhanced with ML, image processing, or sensor fusion can lead to a significant improvement in the accuracy of fertilizer recommendations [16, 24, 2]. All these works support the notion of increased credibility of sensor-based data on soils as inputs to smart decision-support systems.

Simultaneously, the study of Arduino-based NPK sensing systems and IoT-based platforms with soil monitoring has shown that real-time soil sensing was a practical approach and hardware solution. Bhowmick et al. (2023) and Chavan et al. (2022) constructed low-cost soil nutrient sensor that use Arduino microcontrollers and ion-selective sensors to measure the nutrient levels of soil, and found that there are problems of calibration and signal-noise ratio with analog probes. Karthikeyan et al. (2023) used NPK monitoring systems with a WiFi and GSM interface and proved the effectiveness of signal transmission of soil data to a cloud-based ML analysis [13, 3, 19]. These works support the application of microcontroller-based systems to monitor the nutrients in soil in real time at low costs.

There is also a literature on Agronomic and soil science that is also helpful in guiding the fertilizer recommendation. Fageria (2016), Havlin et al. (2014), and Food and Agriculture Organization (2020) provide the details of nutrient thresholds, nutrient requirements per crop, and the classification systems of soil fertility. These articles provide an explanation of the impact of nutrient deficiency on crop physiology and give target NPK values of major crops. which implies the necessity of local fertilizer model calibration [PhilRice202, 6, 10, 7]. These results support the role of dynamically and location-specific fertilizer recommendation, demonstrating that tailored nutrient management strategies can significantly enhance fertilizer efficiency and crop productivity across varying soil conditions.

Lastly, the evaluation of the models is important to develop trustful decision-supporting tools. Jadon and Saini (2023) and Mishra et al. (2024) note that metrics like Mean Squared Error (MSE), Mean Absolute Error (MAE), R^2 , precision and rank-based metrics such as nDCG are used as well. According to them, understandable and heterogeneous models allow farmers to comprehend forecasting and develop trust in AI-based agriculture systems [11, 15]. These approaches improve the effectiveness and usefulness of machine learning-based agricultural decision-support system.

Overall, the literature shows that there has been a tremendous advancement in four areas that are complementary, namely machine learning algorithm, real-time sensor technology, IoT-based agricultural systems, and agronomic nutrient standards. Nevertheless, some gaps continue to exist in spite of these developments, in the form of inaccessibility of local, sensor-based fertilizer recommendation models that are tailored to the condition of the Philippine soils. These gaps are filled through the current research that combines the use of the Random Forest and real-time data of NPK sensors to come up with a context-aware and accurate crop-specific fertilizer recommendation system.

Gap Bridged by the Study

The existing fertilizer recommendation systems are mostly of the rule-based approach or give generalized recommendations that lack consideration of the differences in soil nutrient status. Consequently, such systems tend to give out forecasts which are not very accurate and flexible and hence there are irregularities in the actual needs of the soil and the recommended fertilizers. Also, crop-specific needs are not taken into account by many other systems, which leads to the inefficient utilization of resources, which yields suboptimal outcomes.

In an effort to overcome these shortcomings, the current research came up with an online and mobile-responsive fertilizer prescription system, which uses real-time data ac-

quired using NPK sensors. It has a Random Forest algorithm that analyzes the level of nutrients in the soil and produces the correct recommendations on the level of fertilization and rank the best crops that can be cultivated in the given condition of the soil. The platform is user-friendly and has been developed with the consideration of ease of use since farmers and other users can access data-driven information to manage their use of fertilizers better. The system fosters sustainable use of fertilizers, minimizes wastage of fertilizers, increases crop yield, and encourages farming that is environmentally friendly.

Notes

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CHAPTER 3

METHODOLOGY

This chapter details the research methodology employed in the development and evaluation of the A Random Forest Approach For Crop Fertilizer Recommendation Usig NPK Sensor. It outlines the constructive design research approach, the processes involved in data acquisition and preparation, the development and training of the Random Forest model, the design of the prototype system, and the methods for evaluating the system's performance.

Research Design

The study was Constructive Design Research (CDR). Constructive design approach is a scientific method of research in which an artifact is produced and tested in a systematic way so as to resolve a real world problem. It have been focused on the development of new useful system that assisted in practical and scientific aspects. CDR usually includes the discovery of a significant issue, creation of a technological psychology, assembly of the object, and analysis of its efficiency through empirical. methods.

The research was considered as a positive study that created web-based fertilizer Recommendation-based system, which incorporates the application of machine learning and the agricultural data. The most important artifact developed is a web application that is mobile friendly to the users, particularly the farm- ers, to post real time soil nutrient value (N, P and K) and crop types. The system has feed the input data to the algorithm of the Random Forest and give specific fertilizer rec- Recommendations made in terms of the input data. Developed to be used either on a desktop or a mobile. user-friendliness, the purpose of the application was to provide an easy interface that would reach a high level of even with basic smartphones among the users. This was built and analyzed after, intelligent

system, the research simply offer a working tool to improve the application. contributes to the body of literature on the topic of fertilizers in Philippine agriculture but also to the academic debate on precision farming technologies.

Theorems, Algorithms And Mathematical Models

Random Forest algorithm has been chosen as the main predictive model to be used in the present study because of its power and ability to process complex agricultural data. Random Forest as an ensemble method is more specific in its way of generating prediction through the combination of the results of many decision trees that it produces during the training process. It is less likely to overfit as it bootstraps the sample of the training data and uses random sampling of features at each split node. These properties lead to the fact that Random Forest is very efficient on agricultural soil data, which is frequently nonlinear and full of noise, and with high dimensional counts, including diverse nutrient contents among diverse types of soils.

The features of the nutrient Nitrogen (N), Phosphorus (P), and Potassium (K) level and the type of crop used were used as input features in this study to train the Random Forest model. To deal with realistic modeling of the relationship between the soil conditions and the recommendations of fertilizers, a hybrid dataset that was composed of simulated and expert-edited agricultural data was used. Through training, the algorithm learned to identify the best kind of fertilizer with specific combinations of soil nutrient profiles and thus was able to produce proper and contextual recommendations. This design makes sure that the model can make sound generalization to new or unobserved soil samples and still be very accurate as far as prediction is concerned.

In addition to the accuracy, the application of Random Forest in the current study has a number of other advantages. The fact that it can address missing or noisy data with its feature importance analysis as well as its interpretability led to the researchers having a better

insight into which nutrients in the soil have the greatest impact on fertilizer recommendations. In addition, the ensemble aspect of the algorithm ensured it was robust to errors of individual decision trees making it to have consistent and reliable predictions. The features of Random Forest made the system an appropriate option in the creation of a data-based system that can honorably guide and informed decision-making about agriculture.

Equation:

Given a dataset $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$, where x_i is a feature vector consisting of NPK values and crop type, and y_i is the target label representing the recommended fertilizer type.

A Random Forest consists of k decision trees $\{T_1, T_2, \dots, T_k\}$. Each tree $T_m(x)$ predicts a fertilizer type.

The final prediction $\hat{y}$ for an input x is determined by majority voting:

$$\hat{y} = \text{mode}\{T_1(x), T_2(x), \dots, T_k(x)\} \quad (3.1)$$

Pseudocode

The pseudocode reflects the steps of creating fertilizer recommendations with the help of the Random Forest algorithm and reflects how each real-time NPK sensor readings, logic of nutrient classification, and user-set crop selection are handled or processed to provide accurate and specific fertilizer prescriptions to each soil sample.

Algorithm 1 Smart Fertilizer Recommendation System

```
1: Initialize Flask server, SQLite database, and sensor thread
2: Display dashboard options: Select Crop or Show Crop
3: if Select Crop then
4:     Read NPK values and evaluate levels
5:     if N low then
6:         Recommend Urea (46-0-0)
7:     else if P or K low then
8:         Recommend Organic Compound Fertilizer
9:     else
10:        Recommend Complete (14-14-14)
11:    end if
12:    Ask to return to dashboard; act accordingly
13: else if Show Crop then
14:    Ask confirmation; proceed if yes
15:    Allow soil number selection (1–10)
16:    Display suitable crops and fertilizer info
17:    Ask to return to dashboard; act accordingly
18: else
19:    End application
20: end if
```

The pseudocode depicts the entire working process of the Smart Fertilizer Recommendation System that provides fertilizer recommendations with the help of a Random Forest algorithm relying on real-time measurements of the soil nutrient content and the crops chosen by the user. The system works when Flask server, database and sensor thread which reads the NPK values are initiated. When an individual starts it, they are given two choices

and they are Select Crop and Show Crop. Upon the selection of the Select Crop option, the system measures the level of nitrogen, phosphorus and potassium through the sensor and recommends the correct fertilizer. Particularly, Urea (46-0-0) is used when there is a deficiency of nitrogen, Organic Compound is used in the case of nutrient deficiency, and Complete (14-14-14) is used in case of chemical balance of fertilizers. In the case of Show Crop option, the user is able to select a soil sample number to see the most appropriate crops to plant in that soil (taking into account the nutrient content in the soil) and the type of fertilizer that will be appropriate and the information about it. Altogether, the pseudocode gives a systematic reflection on how the system works with the input data, implements the Random Forest model, and gives correct, personalized, and clear recommendations. This allows farmers to make wise choices on applying of fertilizer and manage their crop production. In addition, the system has been made user-friendly and efficient, such that, even a non-technical user can easily carry out interaction with the system. The Smart Fertilizer Recommendation System will reduce the chances of over-fertilization or lack of nutrients because it combines the real-time sensor values with the past soil values. This does not only increase the quality and production of crops, but also leads to sustainable farming methods since it decreases chemical runoffs and maintains the health of the soil. The accuracy of the predictions and the recommendations that can be followed guarantees that the farmers will be able to make proper decisions based on the needs of their soils and crops.

Materials and Statistical Tools

The dataset used for this study was curated and simulated taking a form of soil moisture values (Nitrogen, Phosphorus, and Potassium), crop and fertilizer recommendations. Limited availability of public datasets with soil conditions of the Philippine soils has been used to build the dataset from expert-sourced values and agronomic case studies with simulated data to fully cover the range. Each data entry has NPK values, choice of crops, and the

recommended type of fertilizer. The system, the Fertilizer classifications used were NPK 14-14-14, Urea and Organic Compost-three fertilizer types that are answerable to many in the Philippine agriculture.

Table 1
Crop-Specific NPK Requirements (kg/ha) [1]

Crop	Nitrogen (N) (L / M / H)	Phosphorus (P) (L / M / H)	Potassium (K) (L / M / H)
Tomato	90 / 80 / 70	120 / 75 / 30	90 / 60 / 30
Eggplant	90 / 80 / 70	80 / 50 / 20	90 / 60 / 30
Chili Pepper	90 / 60 / 30	80 / 50 / 30	150 / 120 / 60
Cabbage	150 / 140 / 120	60 / 35 / 20	60 / 45 / 30
Okra (local)	90 / 80 / 80	80 / 50 / 20	90 / 60 / 30
String Bean	50 / 40 / 35	90 / 45 / 20	50 / 40 / 30
Carrot	90 / 80 / 70	90 / 60 / 30	120 / 75 / 30
Chayote	60 / 50 / 40	80 / 50 / 20	60 / 45 / 30
Ampalaya	90 / 50 / 40	80 / 50 / 20	60 / 45 / 30
Patola	90 / 60 / 30	80 / 50 / 20	60 / 45 / 30

The study used test cases in different conditions in order to ensure that the results were statistically reliable. Particularly, the system was checked with ten profiles of soils, which were the illustration of various levels of Nitrogen (N), Phosphorus (P), and Potassium (K). Ten independent test runs were done on each soil profile. This formed 100 test cases to be carried out in relation to NPK inputs and types of crops including their respective expected yields as well as the expected yields projected. The consistency and predictive accuracy of the model in different input scenarios were developed through such repeated rounds of testing. Predicted fertilizer advice, projected output as per approximated reference values and the level of variability of the test case under repeated case were recorded and examined in each test case.

Prototype Flowchart

Figure 1, which is the flowchart of the proposed system, will depict how the processes of getting fertilizer recommendation and crop suggestions work in order. It shows the general outline of the application, receiving the real-time data on NPK sensors by the program,

evaluating the soil nutrient conditions, and choosing the possible options on the system dashboard.

Also, the flow chart indicates decision points that were applied in the identification of nutrient deficiencies and the use of appropriate crops that should be planted. Such a representation helps to have a clear and complete picture regarding the working flow of the system.

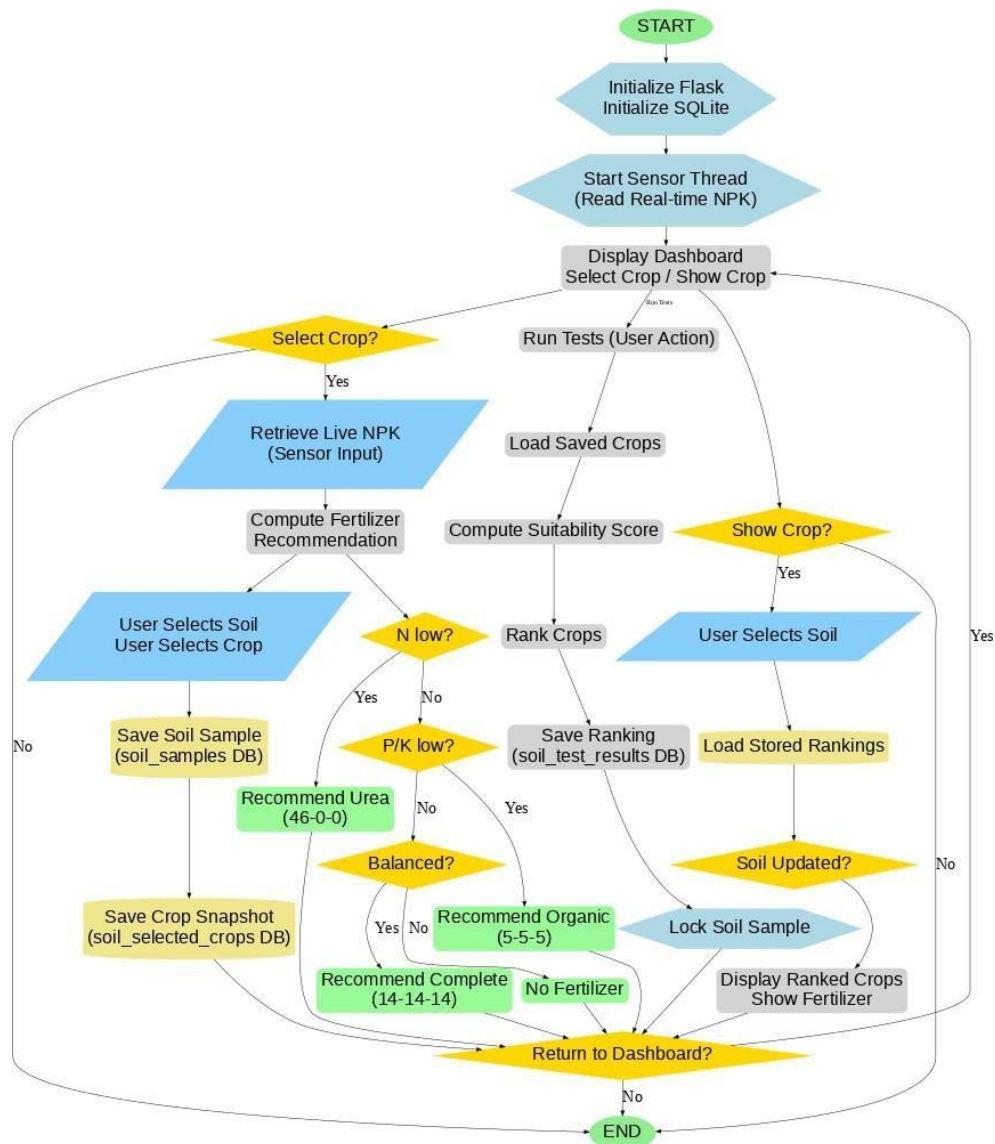


Figure 1: Prototype Flowchart

The flowchart shows how the system of fertilizer recommendation works. The system

would be user-friendly both in web and mobile platforms where farmers and other agricultural stakeholders can readily access system functionality. When the application is opened, it leads to a dashboard where the user can see the profile of the soil, including the focus on such nutrients like Nitrogen (N), Phosphorus (P), and Potassium (K). These data are by default obtained automatically using a soil NPK sensor, or by the input of a user. Based on the analysis of the soil profile, the system offers two primary choices. The former option allows the system to propose appropriate crops depending on the analyzed data on the nutrient levels in the soil. The system in this mode assesses the NPK level in the soil and gives out the top ten crops that are best suited to the current soil conditions. A user can then choose a particular crop on which they would be presented with specific fertilizer recommendations, that is, the exact fertilizer application rates and application procedure depending on the particular crop percent and the soil type. The second one gives the user choice of crop which is manual where the user can choose one crop among the ten regularly grown crops. After selecting a crop, the system shows the present NPK levels and issues the recommendations of the required fertilizer levels to meet the nutritional demands of the selected crop.

This two-way mechanism provides a certain degree of flexibility because it gives the user an opportunity to use the system generated crop recommendations or choose crops according to own preference. Consequently, the system serves various agricultural requirements besides giving effective and viable fertilizer application advice.

Device Setup

Figure 2 shows the equipment used in the prototype, which reflects an overall hardware structure and connection pattern between the NPK soil sensor, the maximal 485 RS485 communication module, and the arduino Uno microcontroller. The figure shows that the NPK sensor is connected with the Arduino Uno by the MAX485 module to provide high-quality data transmission, and power, ground, and signal lines are clearly drawn. This is

a paired system to enable the Arduino Uno to obtain real time soil nutrient data (nitrogen, phosphorus and potassium) which is subsequently processed and passed on to the system to examine the fertilizer recommendations.

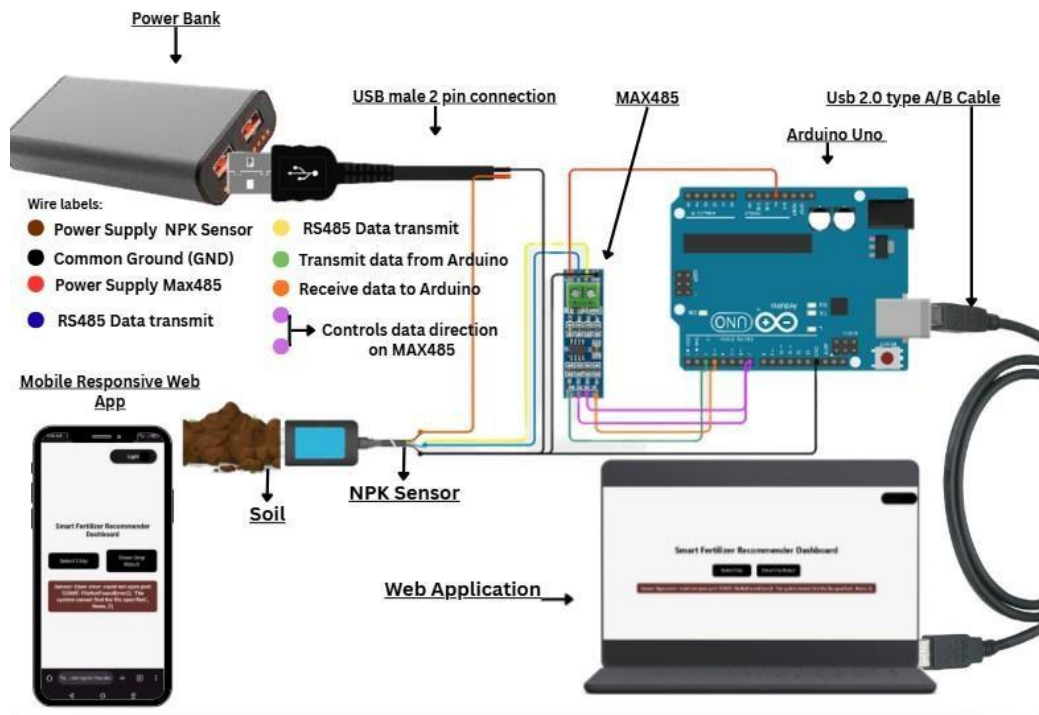


Figure 2: **Device Setup**

The prototype has the Arduino Uno, NPK sensor, and 485 Converter - MAX485, which are physically connected to form the reference. The sensor NPK sensor is attached to the soil sample in order to measure the nutrient levels and the sensor used to switch the communication between the sensor and the Arduino board is called the MAX485. Arduino Uno receives the data received and sends it to the web application.

The web and the mobile application offer a convenient platform to track the soil nutrient levels in real time so that users can get the rightful soil fertilizer advice on smartphone and laptops that can be used in precision agriculture. The prototype shows a successful bonding of the IoT hardware and a data-driven system of recommendations.

Hardware and Software Tools

Hardware Tools

- **Computer:** A computer or laptop that was capable of accessing the web-based application.
- **Mobile Phone:** Any mobile phone was capable on accessing the application.
- **Processor (CPU):** Intel i5 / Ryzen 5 or better —was used for running backend algorithms efficiently.
- **RAM:** A minimum 8 GB ensured smooth execution of database queries.
- **Storage:** 256 GB SSD or better was used for storing datasets and application files.
- **Operating System:** Windows 10/11 or Ubuntu Linux — were used, as they were compatible with all development tools.
- **Internet Access:** Was Required necessary for accessing library dependencies and deployment testing.
- **NPK Sensor:** NPK sensor determined the amounts of Nitrogen (N), Phosphorus (P) and Potassium (K) in the soil that plants require [2]. It directly measured ion concentrations in the soil and gave you nutrient information in real time as parts per million (ppm). The sensor was inserted into the soil to monitor the amounts of nitrogen, phosphorus and potassium at the moment it is placed. The collected data were sent to a microcontroller which transmitted the information via the internet to a web-based system.
- **Arduino UNO:** The sensor for NPK was connected to an Arduino that can read the levels of Nitrogen, Phosphorus and Potassium in soil on the spot. The Arduino uses a USB cable to communicate with the computer and then sends its gathered data to

the application on the web. Because of this automation, the data is accurately sent and quickly analyzed for recommending fertilizer. After the calculations, the results show up in a simple dashboard.

- **MAX485:** RS485 to TTL converter (MAX485) acted like a translator between NPK sensor and Arduino.

Software Tools

- **Python 3.9+:** The served as the primary programming language for system development. It provides compatibility with modern libraries, ensuring smooth integration of machine learning, database handling, and web development tasks.
- **Mlxtend:** A Python library was used for implementing the Random Forest algorithm, which played a central role in generating accurate fertilizer recommendations.
- **Pandas:** Was used for data manipulation library used to clean, structure, and preprocess soil datasets efficiently. It ensures that data is properly formatted before model training and analysis.
- **Matplotlib:** A visualization library was used to create plots and graphs that illustrate relationships between soil nutrients and fertilizer rules. This helps in interpreting the results more clearly.
- **Flask:** A lightweight web framework was used to deploy the system online. It connects the backend processes (machine learning model and database) with the user interface.
- **MySQL / SQLite:** Relational database systems were used for storing soil data, fertilizer rules, and system records. SQLite was used for lightweight setups, while MySQL was preferred for larger-scale deployment.

- **Visual Studio Code:** The main integrated development environment (IDE) for writing, debugging, and testing the system's source code. It offers useful extensions for Python, Flask, and database management.
- **Excel / Google Sheets:** Tools was used during the initial stages for dataset preparation, manual inspection, and formatting before integrating into the system.
- **Arduino IDE:** Was used for programming and uploading code to Arduino hardware components. It ensures proper communication between sensors, microcontrollers, and the system.

The system integrated both hardware and software tools to ensure efficient functionality. On the hardware side, components such as the processor, RAM, and storage provided the necessary computing power for running backend algorithms and handling datasets, while the NPK sensor, Arduino UNO, and MAX485 module enabled real-time soil nutrient measurement and data transmission. These devices worked together to collect accurate soil information and relay it to the system for processing. On the software side, Python served as the core programming language, supported by libraries like mlxtend, pandas, and matplotlib for implementing machine learning, data manipulation, and visualization. Flask enabled web deployment, while MySQL/SQLite ensures reliable data storage and management. Development and testing were facilitated through Visual Studio Code, with Excel/-Google Sheets used for initial dataset preparation and Arduino IDE used for programming microcontroller components. Together, these tools provided a cohesive environment for developing, deploying, and maintaining the smart fertilizer recommendation system.

Prototype Software Application Development

To facilitate user interaction with the trained Random Forest model, a prototype software application will be developed.

Platform: The system development used the Flask framework to build a web-based application. Web-based users used their browsers to input NPK values into the fertilizer recommendation system to receive instant fertilizer suggestions. The decision to use Flask came because of its simple framework structure and adaptability together with strong support for Python-based machine learning models.

User Interface (UI): The application featured a user-friendly interface allowing user farmers to:

- Manually input N (Nitrogen), P (Phosphorus), and K (Potassium) soil nutrient values.
- Select the Crop Type from a predefined list corresponding to the crops.

Backend Logic: Upon submission, the input data were processed and fed into the trained Random Forest model

Output: The system will displayed the Recommended Fertilizer Type as predicted by the model.

Application of Random Forest

Soil nutrient levels (NPK values) undergo analysis using the Random Forest Algorithm for fertilizer selection prediction in this research study. The patterns were derived from both historical and simulated datasets that pertained to Philippine agricultural environmental conditions. Soil data combined with expert fertilizer recommendations guided the training and validation of the model which leads to scientifically valid outputs to benefit Filipino farmers. Advantages of Random Forest for the study, patterns were derived from both historical and simulated datasets that pertain to Philippine agricultural environmental conditions. Soil data combined with expert fertilizer recommendations guide the training and validation of the model which led to scientifically valid outputs to benefit Filipino farmers [4] [3].

Research Process

The figure 3 below showed the research process undertaken the creation of the system which was a summarization of the most important stages in tackling the problem to testing the system.

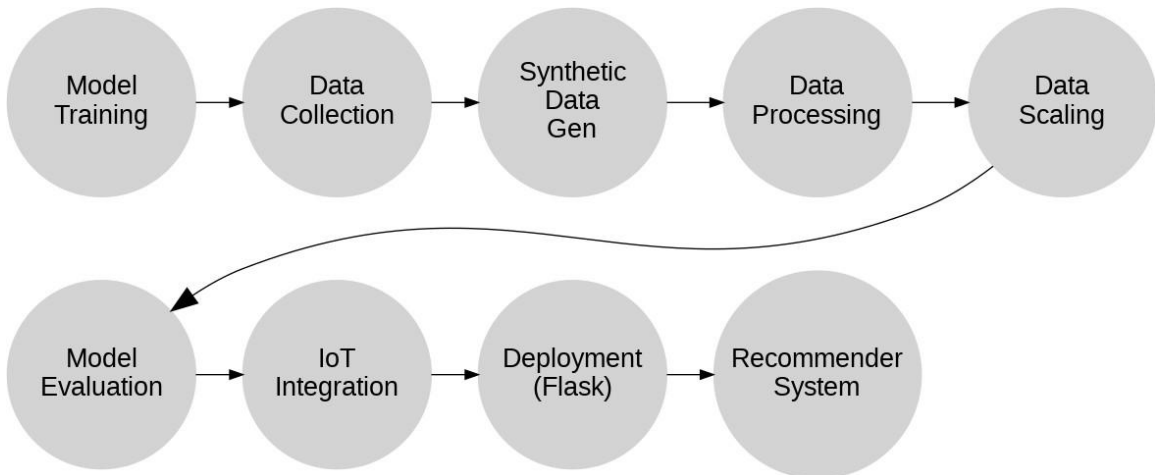


Figure 3: **Research Process**

The figure has demonstrated the entire steps undertaken in the development of Smart Fertilizer Recommendation System. It started off with the data collection phase in which the levels of the soil nutrients of Nitrogen (N), Phosphorus (P) and Potassium (K) were collected together with the type of crops present. Given that there was a small volume of data collected, the artificial generation of data was conducted so as to generate more realistic samples so that we have a more diverse and balanced dataset.

Model Training

The first step was model training, where in the algorithm of the Random Forest was applied in identifying the links between nutrients of the soil and the types of fertilizers. Cross-validation was used to make sure that the model is reliable and generalizable. The performance of the model was also tested after training based on different measures in-

cluding accuracy, precision, recall, and F1-score to determine the predictive potential of the model.

The trained model was moved to a web based system developed in Flask with users able to enter soil NPK values either manually or with sensor data. This system then gave real time fertilizer suggestions. This whole process showed how machine learning and IoT technologies collaborated to assist farmers and other agricultural practitioners in making informed data-driven and sustainable farming decisions.

Data Collection

The data collection stage served as the foundation for building the fertilizer recommendation model. Since the system focuses on determining the most suitable fertilizer based on soil nutrients, the researchers gathered data that represented different types of soil conditions. A total of ten (10) soil data samples were collected, each containing measurements of Nitrogen (N), Phosphorus (P), and Potassium (K) levels, along with the corresponding recommended fertilizer type.

The samples were obtained from a combination of public agricultural resources, existing research studies, and simulated sensor readings. While the data did not originate from specific geographic locations or actual farmland sites, each sample was designed to represent a realistic soil profile typically found in local agricultural settings. This approach was adopted to demonstrate the model's capability to process and analyze soil nutrient data even in the absence of exact land-based measurements.

Each record in the dataset included the following parameters:

- **Nitrogen (N)** – the percentage or mg/kg concentration of nitrogen in the soil.
- **Phosphorus (P)** – the available phosphorus level that supports plant growth.
- **Potassium (K)** – the amount of potassium that improves plant health and productivity.

- **Crop Type** – the type of crop commonly associated with the soil condition (e.g., tomato, eggplant, ampalaya, cabbage, okra).
- **Fertilizer Type** – the corresponding fertilizer recommendation (e.g., Urea, Organic Compost, NPK 14-14-14).

These collected soil data samples were sufficient for demonstrating the feasibility of the Random Forest model and for testing the prototype's accuracy in generating appropriate fertilizer recommendations.

Synthetic Data Generation

The ten soil samples were collected, additional synthetic data were generated to expand the dataset and improve model reliability. This process ensured that the Random Forest model could learn from a broader range of nutrient values and fertilizer categories. By increasing the amount of training data, the model became more stable and less biased toward a small number of samples.

The synthetic values were produced by generating new N, P, and K combinations within realistic ranges based on agricultural references. The formula used for generating synthetic nutrient data was:

$$X_{\text{synthetic}} = X_{\text{mean}} + r(X_{\text{max}} - X_{\text{min}}) \quad (3.2)$$

where r is a random number between 0 and 1.

This approach simulated diverse soil conditions found in different parts of the Philippines, making the model applicable for various soil types and crop needs.

Data Processing

After collecting and expanding the dataset, the researchers performed data processing to clean, structure, and prepare the data for model training. The following steps were taken:

1. **Handling Missing Values:** Any incomplete entries for N, P, or K were replaced with mean values to maintain data consistency
2. **Removing Duplicates:** Duplicate records were removed to prevent skewing the training process.
3. **Encoding Labels:** Fertilizer categories were converted into numerical values (e.g., Urea = 0, Compost = 1, NPK 14-14-14 = 2) to make them compatible with the Random Forest algorithm.
4. **Formatting:** The dataset was organized into a structured CSV format suitable for machine learning using Python's pandas and scikit-learn libraries.

This stage ensured that the dataset was clean and properly formatted for the next stages of scaling and training.

Data Scaling

To ensure all features were treated equally during training, data scaling was applied to normalize the N, P, and K values. Since these nutrient readings vary in numerical range, normalization was necessary to prevent any one feature from dominating the model. The **Min-Max normalization** method was used, following the formula:

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad (3.3)$$

This normalization improved the consistency of the dataset and allowed the Random Forest model to train more effectively.

Model Evaluation

The Random Forest model was evaluated using standard machine learning metrics to assess its predictive performance in generating accurate fertilizer recommendations based

on soil nutrient data. These metrics include both classification and regression-based measures to comprehensively evaluate accuracy, consistency, and reliability. All computations were performed using Python's `Scikit-learn` library.

Classification Metrics

The classification metrics were used to evaluate the model's ability to correctly identify the appropriate fertilizer type for each soil sample.

Accuracy Accuracy measured the proportion of correct predictions among all predictions made by the model.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3.4)$$

Precision Precision determined how many of the model's positive predictions were actually correct.

$$Precision = \frac{TP}{TP + FP} \quad (3.5)$$

Recall (Sensitivity) Recall measured the proportion of actual positive cases correctly identified by the model.

$$Recall = \frac{TP}{TP + FN} \quad (3.6)$$

F1-Score The F1-score is the harmonic mean of Precision and Recall, providing a single score that balances both metrics, especially useful for imbalanced datasets.

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (3.7)$$

Confusion Matrix A confusion matrix was generated to visualize the performance of the classifier, showing the counts of true positives (TP), true negatives (TN), false positives

(FP), and false negatives (FN) for each fertilizer class. This provided an intuitive summary of the model's classification accuracy.

Regression and Error Metrics

To further assess prediction deviations and reliability, regression-based error metrics were used. These include the Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2).

Mean Absolute Error (MAE) MAE measured the average magnitude of prediction errors without considering their direction. A smaller MAE indicates higher model accuracy.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (3.8)$$

Root Mean Squared Error (RMSE) RMSE penalizes larger errors more heavily and provides insight into how consistent the predictions are.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (3.9)$$

Coefficient of Determination (R^2) The R^2 score measured how well the predicted fertilizer outputs match the actual data. A value closer to 1 indicates that the model explains most of the variance in the dataset.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3.10)$$

Cross-Validation

To ensure that the model's performance was not overly dependent on a single data split, the researchers used k-fold cross-validation with $k = 5$. This involved dividing the dataset

into five equal parts, training the model on four parts, and testing it on the remaining one in each iteration. The process was repeated five times, and the average performance metrics were computed.

This validation process confirmed that the model produced consistent and reliable predictions even with a limited dataset, supporting the study's objective of accuracy and generalizability.

IoT Integration

The system was to take in the direct input of the IoT-enabled NPK sensors as automatic sensors to measure and roll the amounts of Nitrogen (N), Phosphorus (P), and Potassium (K) in the soil. Using these sensors, the system did not require manual data entry so that real time data on the readings of soil nutrients were obtained with a minimal human intervention. Such automation enhanced efficiency of data collection and minimized chances of errors that might be made in case of manual recording creation giving a more secure background with which the process of fertilizer recommendation may be made.

In addition, the incorporation of the IoT technology allowed the system to come up with timely and precise fertilizer recommendations. Since the sensors constantly checked the nutrient levels in the soil, the system was able to instantly access the new information as it was received by the trained Random Forest model and thereby weigh the options of the most appropriate fertilizer to be used on the crops. In real-time health, the availability of this feature enabled farmers and agricultural professionals to take based on the available data and make sound decisions in a short period of time, encouraging sustainable agricultural activities and crop farms to grow by taking good decisions based on accurate soil conditions.

Deployment(Flask)

After the validation of the Random Forest model, it was implemented in a web-app created in the Flask framework. This interface was used as the predictor between the trained model and the end user whereby the trained model was able to apply in a real world setting. Its implementation ensured the presence of the system in multiple devices, offering an easy interface, whereby soil nutrient data would be entered effectively.

The application enabled users to access soil NPK value using NPK sensors connected to the soil. After receiving the data, the trained Random Forest model received the input and provided the real-time fertilizer recommendations. This feature allowed users to make real time, data driven choices on the use of fertilizer, contributing to the practises of precision agriculture, and assisting in the optimal growth of crops depending on the real nutrient content of the soil.

Recommender System

The final product of the Smart Fertilizer Recommendation System was the combination of the trained Random Forest model with the web-based user interface. By integrating this, the predictive model worked perfectly with the easy-to-use application, giving the users a convenient and friendly interface through which they can communicate with the system. With the benefits of bringing machine learning and a convenient interface together, the software would be able to effectively work with soil nutrient data and provide pertinent output to the final users.

The finished system could provide instant fertilizer recommendations, allowing farmers and agricultural practitioners to make sound, fact-based decisions. It enabled sustainable farming, crop optimization, and reduced chances of over- or under-fertilization by providing real-time soil information. This demonstrated the potential of linking artificial intelligence and IoT to improve agricultural productivity and decision-making.

Theoretical Framework

This section has described the theoretical background don which the A Random Forest Approach was designed, developed and evaluated in A Random Forest Approach Crop Fertilizer Recommendation Using NPK Sensor. It provided information on the learning theories and machine learning principles that guided the application of the Random Forest algorithm and the flow of concepts in the collection, data processing, and analysis of the NPK sensor data to formulate the recommendations on the application of fertilizers. The framework also gave a clear explanation of the relationship between the input variables, prediction model, as well as the system outputs, and as such, the proposed system was scientifically based and methodologically sound.

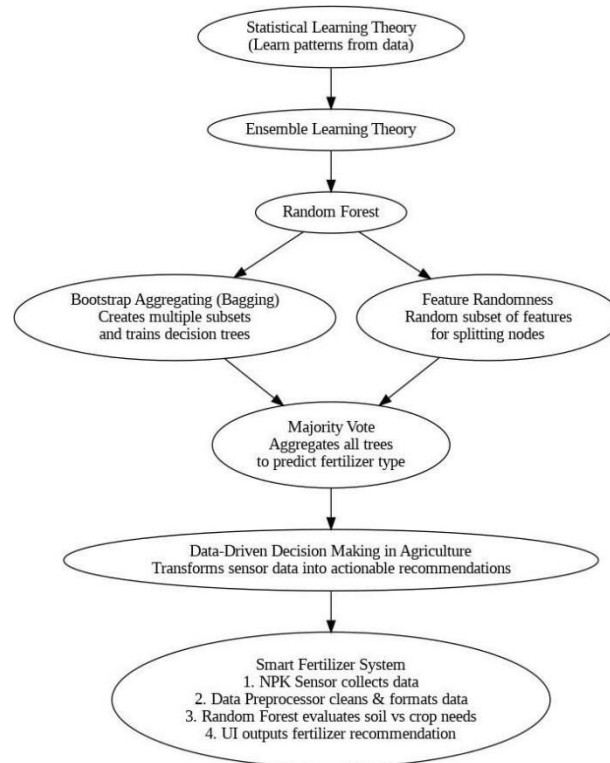


Figure 4: **Theoretical Framework**

1. *Statistical Learning Theory:*

This research is broadly grounded in Statistical Learning Theory, which provides the formal basis for machine learning. The aim is to develop a model that can learn patterns and relationships from a given dataset NPK soil values, crop types, and associated fertilizer recommendations to make accurate predictions fertilizer recommendations on new, unseen data. The Random Forest algorithm served as a practical application of statistical learning principles for classification tasks.

2. *Ensemble Learning Theory:*

The choice of the Random Forest algorithm was directly supported by Ensemble Learning Theory. This theory posits that combining the predictions of multiple individual learning models decision trees can lead to improved overall predictive performance, increased robustness against noise and outliers, and better generalization compared to a single model. Random Forest leveraged two key ensemble techniques.

3. *Bootstrap Aggregating:*

It creates multiple subsets of the original training data through bootstrapping sampling with replacement and trains a separate decision tree on each subset.

4. *Feature Randomness:*

When splitting a node during the construction of each tree, Random Forest considers only a random subset of the available features. This decorrelates the individual trees, further enhancing the diversity of the ensemble and typically improving accuracy. The final prediction for a classification task like recommending a fertilizer type is made by a majority vote among all the trees in the forest.

5. *Decision Tree Theory:*

As Random Forest is an ensemble of decision trees, the underlying principles of decision tree construction were relevant. Decision trees recursively partition the data space based on feature values to create a tree-like model of decisions. Splits are typically chosen to maximize information gain or minimize Gini impurity, leading to homogenous subsets at the leaf nodes, which correspond to class labels fertilizer types[5].

6. *Data-Driven Decision Making (DDDM) in Agriculture:*

The overarching aim of developing this system aligns with the paradigm of Data-Driven Decision Making. By transforming raw soil nutrient data and crop information into actionable fertilizer recommendations through a machine learning model, this research promotes a shift from experience-based or generalized fertilization practices towards more precise, evidence-based decisions[6].

The theoretical concept of the Smart Fertilizer Recommendation System was explained as follows. The system recommended the most adequate type of fertilizer automatically depending on the sensor-read soil nutrient content and the previously known crop data. The system was capable of capturing real-time soil nutrient readings of the soil- that is, Nitrogen (N), Phosphorus (P), and Potassium (K) straight off the NPK sensor. The values were sent to the application, and they are automatically processed so that the user does not have to input the values manually. The user only has limited interaction with the system whereby he can choose the crop he wants to grow among the list offered in the application. The pre-determined nutrient requirements against the identified soil values are then compared with the identified soil values to obtain the recommendation of the fertilizer application to each crop in the database.

The Data Preprocessor component managed the sensor data received and first of all checks the data and organizes it into a format that is correct and readable by the model. The data were refined and sent to the Random Forest Model which is the predictive engine of the system. The model evaluated the nutrient status of the soil and compared it against the nutrient demands of the crop to determine the type of fertilizer that can best enhance soil fertility and optimal growth of the plant. The output is shown in the user interface (UI) after the analysis has been done as a fertilizer recommendation. This automated operation facilitated precision agriculture because the farm managers and users can make accurate and timely decisions in terms of fertilizer application based on data, which increases the productivity and sustainability of agricultural operations.

It demonstrated how the diagram introduces the theoretical framework of the Smart Fertilizer Recommendation System. The theoretical framework started with Statistical Learning Theory which formed the basis of the training models to learn meaningful patterns using data on soil nutrients. This resulted in Ensemble Learning Theory that advocated the use of various models to enhance prediction accuracy. It gives rise to the Random Forest algorithm, which improved each other by using two important methods: Bootstrap Aggregating which trained a series of decision trees on randomly sampled bootstraps, and Feature Randomness where each split uses a random subsample of features. These methods generated a variety of trees with their results merged together with a Majority Vote resulting in a more accurately determined fertilizer prediction. The aggregated yield reinforced Data-Driven crops where the data of the NPK sensors are converted into accurate evidence-based suggestions. The resulting flow will power Smart Fertilizer System where sensor data could be analyzed using Random Forests to provide farmers with the necessary feedback to act on.

Notes

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CHAPTER 4

RESULTS AND DISCUSSION

This chapter outlined the findings of data analysis, system testing, and model criteria on the Smart Fertilizer Recommendation System with the help of Random Forest and NPK Sensor. These results confirmed the effectiveness and accuracy of the developed prototype in advising suitable fertilizers according to the nutrient analysis of soil (Nitrogen, Phosphorus and Potassium).

Training and Development of the Random Forest Model

The primary goal of the study was to create a Random Forest machine learning model based on the soil nutrient content of Nitrogen (N), Phosphorus (P) and Potassium (K) to compute the most suitable fertilizer to various crops. With the use of nutrient composition as predictive attributes, the research aimed at creating a data-driven methodology of fertilizer recommendation that enhances agricultural decision-making.

The main objective was to build a Random Forest classifier that will be able to grasp the association between soil nutrient content and types of fertilizers. To do it, pre-processing of the dataset was done to fill the gaps or irregularities in the data, and Synthetic Minority Over-sampling Technique (SMOTE) was used to equalize the class distribution of fertilizer types. The dataset was further subdivided into a training and testing subset on the basis of 70:30 ratio so that the predictive performance is fairly evaluated.

Dataset and Exploratory Data Analysis (EDA)

The data applied in this analysis was that of 15,000 soil samples, which included the Nitrogen (N), Phosphorus (P), Potassium (K) measurements, and the type of fertilizer ap-

plied on a 10 specific crops across to 10 soil samples. The preliminary analysis of the data indicated that all three nutrients had a wide distribution of values and some samples had missing or extreme values, which were dealt with in preprocessing.

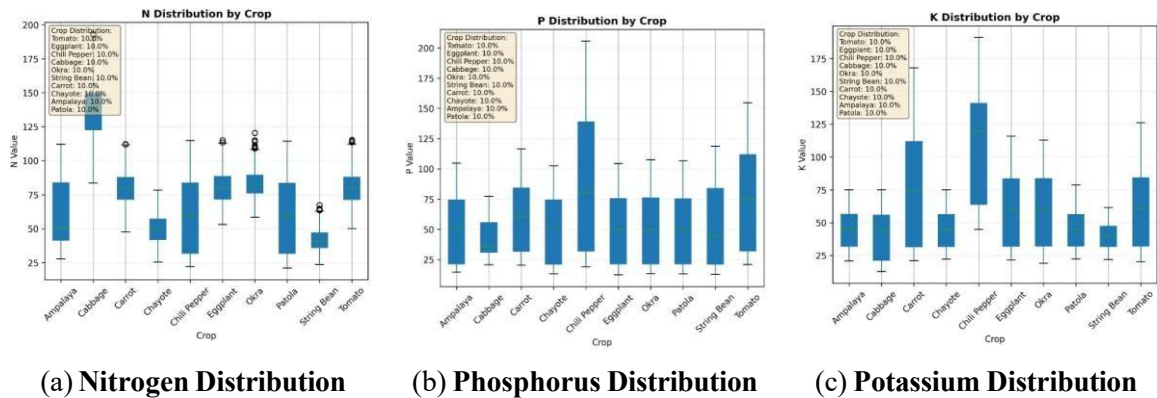


Figure 5: NPK nutrient distributions by crop type

Boxplot visualization was used to analyze the nutrient distributions of the different types of crops with the goal of determining the variability, central dependency, and outliers. The discussion focused on three main nutrients needed by plants, Nitrogen (N), Phosphorus (P), and Potassium (K), and their interaction when they are combined based on their type of crop. The data on the individual nutrient were initially collected and tabulated according to individual crop and then each nutrient was plotted individually to show the variations in distribution. Boxplots were selected due to the fact that it gave an easy way of viewing the median, portion range, and possible outliers, which are key measures of nutrient stability and abnormalities. An annotation showing the average nutrient content of all crops was also added to each subplot and enabled a rapid comparison of nutrient requirements.

The findings revealed that there were significant variations in the amounts of nitrogen which were distributed between the crops. Some crops displayed broader nitrogen distributions hence irregular or variable N distributions, and others displayed narrower distributions indicating consistent N demands. Some crops were found to have outliers meaning that the soil had either extreme readings or abnormal conditions of the nutrients during sampling.

The distributions of phosphorus tended to be more even, though there were crops that expressed higher concentrations of phosphorus with a bigger range. This trend implied that the requirement of phosphorus was less predictable across crops although it had discernible trends that could be used to guide nutrient planning. In the meantime, the potassium exhibited the most variability among three nutrients. There were a number of crops with high potassium uptake and extensive distribution ranges, which implied greater nutrient dynamism was needed. Outliers in potassium were also common and this supports the notion that K levels were more varied in actual planting.

In general, the visualization revealed that the nutrient profile of each crop was unique and acted differently regarding the distribution of NPK. These observations showed the need to come up with crop-specific fertilizers prescription tools instead of using the same nutrient formula in all types of crops. The variability and outliers indicated that the application of real-time monitoring or adaptive delivery of nutrient would enhance precision and efficiency in the management of the fertilizer. Therefore, the discussion justified the necessity of smart systems that could adapt nutrient feeds corresponding to the trends of the observed data, which is why it would be a worthy addition to a Smart Fertilizer Recommendation System.

Synthetic Data

In this table, one can find a synthetic dataset that is designed on the assumption of the simulation of the measurement of crop nutrients with low Nitrogen (N), low Phosphorus (P) and low Potassium (K) conditions. The dataset contains fifteen synthetically produced samples of distinct crops, i.e. tomato, eggplant, chili pepper, okra, cabbage, and so on. The goal of developing this synthetic data is to have a controlled and uniform collection of nutrient data to be utilized in the testing of fertilizer computation models, nutrient classification algorithms and analytical processes, but not depend upon measurements on the results in the field.

Table 2
Synthetic Dataset Samples

Target	Low (N)	Low (P)	Low (K)
Tomato	84.25	135.71	85.45
Tomato	82.37	119.89	96.76
Eggplant	89.17	68.48	81.32
Eggplant	91.12	75.13	105.82
Chili Pepper	82.55	76.53	130.71
Chili Pepper	103.76	88.16	133.12
Cabbage	140.97	60.19	65.44
Okra	92.49	76.04	92.68
Okra	98.97	74.72	98.34
String Bean	52.31	85.95	46.14
Carrot	86.61	87.33	127.36
Chayote	66.82	74.35	57.47
Ampalaya	82.31	81.46	59.79
Patola	90.24	87.13	56.95
Patola	78.25	63.67	63.03

The table shows representative samples of the synthetic data on the low-nutrient soil conditions of various crops. The Target column shows the crop of each of the nutrient profiles, i.e., Tomato, Eggplant, Chili Pepper, Okra, Carrot, Chayotte, Ampalaya and Patola and the columns that follow, namely Low (N), Low (P), and Low (K) show the quantified nutrient values when the conditions of nitrogen-, phosphorus-, and potassium-deficiency prevail. These values are not optimal amounts of fertilization but are simply assumptions of conditions of stress where one or more macro nutrients are lacking. The table presents intra-crop variability by having more than one row per crop to reflect real-life variability of nutrient uptake, which depends on the stage of growth, the soil properties and the environment.

Low (N) column depicts the changes in the availability of nitrogen in different crops with a low level of soil nitrogen. Cabbage and Chili pepper have relatively better nitrogen values than String bean or Chayote which implies that they have different nitrogen requirements and efficiencies of nitrogen utilization in unfavorable conditions. Because nitrogen is a very important element in the growth of vegetables and the development of chlorophyll, these differences highlight the fact that the same rate of fertilizer cannot work with every crop. The information hence bears out the necessity to have crop specific fertilizer guideline as opposed to generalized nutrient management activities.

The Low (P) and Low (K) columns also explain the reactions of crops to soils that are low in phosphorus and potassium. The values of phosphorus reflect the variation in the root development and energy transfer needs, whereas potassium values show the variation in the water regulation, enzyme activation and stress resistance. Some crops like Chili Pepper and Carrot have relatively high potassium values even when they are in low-K conditions implying that they require much potassium to grow appropriately. On the whole, Low (N), Low (P), and Low (K) values are rather expressive examples of nutrient-deficient soil profiles with each crop. This organized data is critical to training the Random Forest model because it allows the system to understand some sense of relationships between nutrient deficiencies and the right recommended fertilizers and makes good and informed decisions.

NPK Relationships and Correlation Matrix

The visualizations indicate the relationship between Nitrogen (N), Phosphorus (P) and Potassium (K) in various crops. Plotting N vs P, N vs K, P vs K, allows us to see general trends in the behavior of nutrients.

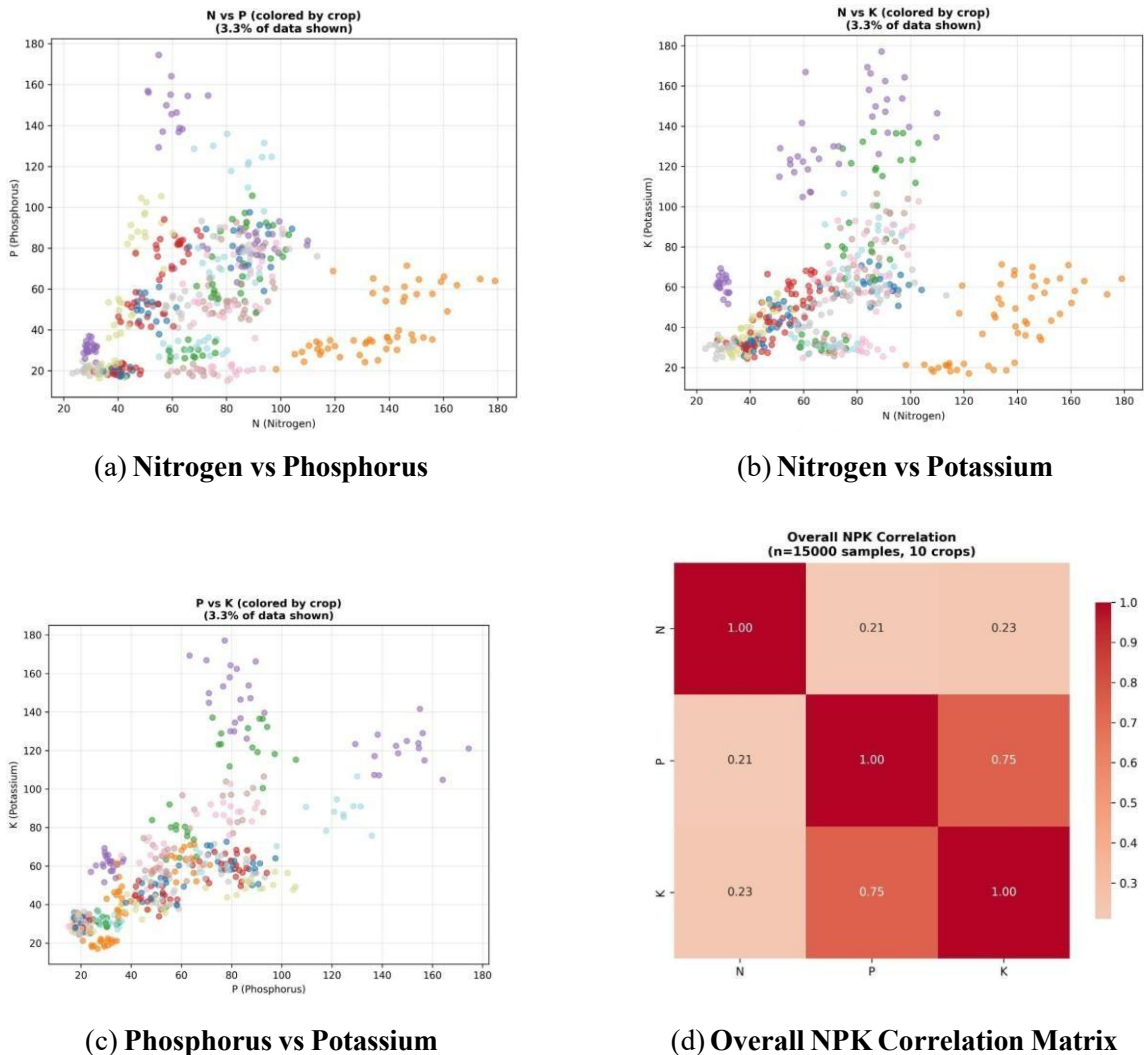


Figure 6: Relationships and correlations across different crop types

The scatter plots indicate Nitrogen (N), Phosphorus (P), and Potassium (K) at variations in the different crops with each colour indicating a different type of crop. The N-P and N-K plots show broad distributions of the nutrient levels meaning that the levels of Nitrogen do not exhibit strong following behaviors with P or K. Conversely, there is a more evident increase in the P-K plot which indicates that the nutrient is more correlated with the latter. The heatmap of correlation supports this observation so that only P and K are strongly positively correlated with 0.75, which means that despite the increase, both of these correlate with each other. In the meantime, Nitrogen does not correlate very strongly with P or K

indicating that it is more independent. On the whole, the image emphasizes a difference in the extent of interaction between nutrients, with a close relationship between P and K without much interaction between that of N and the patterns of the mineral and thus it is more associated with crops.

In the exploratory data analysis, the correlation of the three key soil nutrients was developed as the scatter plots and correlation heatmap to understand the interaction between the three key nutrients among the samples of the crops better. The heatmap indicated that the diagonal ones were equal to 1.00 (which means the uniformity of self-correlation) whereas the off-diagonal ones denoted the level of self-correlation between nutrient pairs. The relationship between N and P was 0.21 which is weak positive indicating that fluctuations in the Nitrogen levels do not have a strong relationship with Phosphorus. In a similar manner N and K were found to be correlated with each other by 0.23 which is weak as well as the fact that Nitrogen is almost independent of Potassium. P and K, on the other hand, had a correlation of 0.75, which is a strong positive value, establishing the fact that these two nutrients tend to raise one another. This is in line with the scatter plots in that P and K had a more pronounced upward trend than the other pairs of nutrients. All in all, the Nitrogen has been identified as relatively independent and Phosphorus and Potassium were greatly associated with each other as shown by the heatmap and the plots. This data can be used in the modeling process since it shows that N has an independent effect on predictions, whereas P and K can influence the crop nutrient trends together.

Outliers and Missing values

Outliers refers to The distribution and correlation of Nitrogen (N), Phosphorus (P) and Potassium (K), the existence of outliers and the absence of values in the sample is also to be considered as it may have a direct impact on the trends of scatter plots and correlation table. The outliers are the nutrient values that are very high or very low relative to most of the samples. Since N-P and the corresponding N-K plots are widely distributed, the

extreme points might be actual differences in the nature of soils, whereas the others could be attributed to the fluctuations in measurements or of rare circumstances. At these extreme values, the summaries of statistics, e.g. the mean and standard deviation, can be misleading, and they can affect processing that depends on scaling. Where scaling is influenced, typical nutrient levels can be concentrated into a small range, with extreme values taking up the scale, making the scale untrustworthy of the typical crop nutrient behaviour.

Even though the Random Forest model can usually resist outliers since it uses tree splits instead of continuous regression functions, outliers may still affect the decision boundary of the model particularly when they are prevalent in the resampled samples utilized in class balancing. When this happens, the model can overfit, that is, it can learn patterns that capture unrealistic soil profiles, but not by capturing realistic ones. This could hurt the metrics of model evaluation like accuracy, precision, recall, and F1-score because the model will tend to incorrectly categorize normal nutrient combinations as it tries to adjust to these extremes. Therefore, careful data preprocessing is essential to ensure that the model learns meaningful patterns rather than noise. Outlier treatment using removal, capping or robust scaling methods can be used to make sure the model concentrates on realistic nutrient patterns addressing the reliability of the determine model and its overall performance.

Heatmap

The heatmap used in this research was used to display the average amount of nitrogen (N), phosphorus (P), and potassium (K) in ten types of crops. The heatmap helped identify the nutrient-rich crops and compare the requirements of fertilizer by visualizing nutrient values in color intensities. This was useful in the interpretation of nutrient patterns necessary in the formulation of crop-specific nutritional fertilizer recommendations.

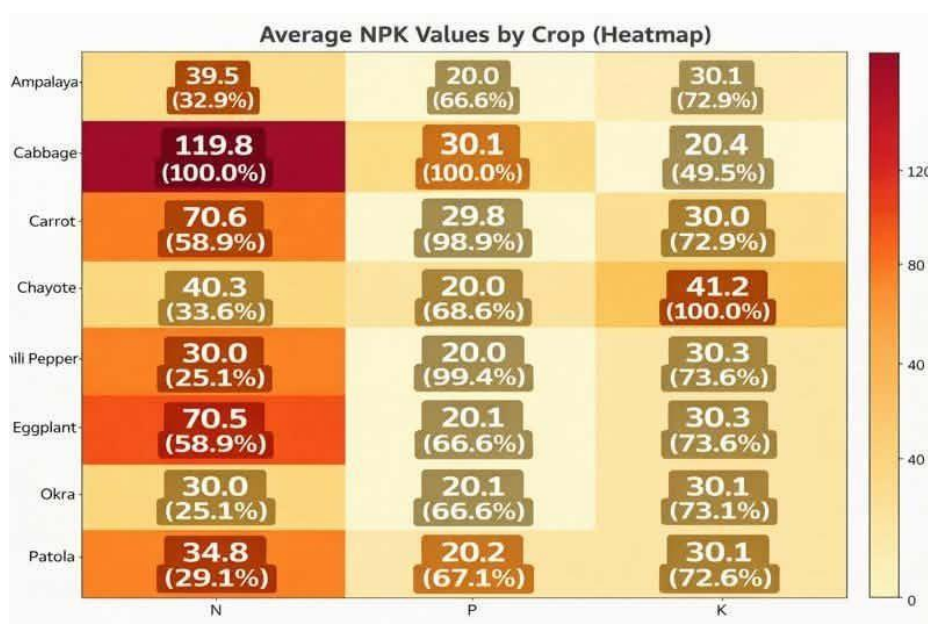


Figure 7: Correlation Heatmap

The heatmap has given a good overview of the variation in the level of NPK nutrient among crops by giving the average data and percentages relative to each other. The darker the shade of color meant the higher the level of concentration of the nutrient, and therefore, crops that were rich in nutrients could be identified easily. Cabbage had the highest nitrogen content (136.6) that registered as the darkest zone on the N column, implying that it needed a lot of nitrogen than other crops. Chili Pepper was also found to have high nutrient levels especially phosphorus (86.4), potassium (109.6), which makes it one of the crops in the dataset that demand the highest level of nutrients. On the other hand, crops like the String Bean and the Chayote were found to be lightly colored and constant in all the nutrients, indicating that there was always less need of nutrients.

The distribution indicated that the majority of crops contained moderate NPK levels and their distribution patterns were consistent with the established nutrient requirements in agriculture. Carrot, Eggplant, Okra and Tomato possessed balanced and moderately high levels of nutrient which could be related to warm tones of color intensities of the products. The use of percentage values also facilitated a better comparison as it was possible to

see how each crop performed in comparison to the highest nutrient concentration that was observed in the data. This helped to standardize the interpretation more, which shows crops with an unproportionate high nutrient requirements.

Generally, as seen in the heatmap, nutrient demands across crops were very diverse and thus, it is essential to advise the use of crop-specific ratios of nutrients instead of general ratios when applying fertilizers. The visual patterns implied that high demand crops would have had a higher nutrient management approach, whereas low demand crops would need to have a more conservative approach in the nutrient application. This discussion aided in the creation of a more accurate and effective system of fertilizer recommendations in terms of the crop-specific NPK behavior.

Crop Distribution

The bar chart was used to represent the sample distribution of the crop samples of the dataset and this represented the number of data points that were sampled of each type of crop. This visualization allowed to conclude that the dataset was not biased and that it can be analyzed in the most objective way within the fertilizer recommendation system. The chart also helped justify the reliability of subsequent model training and evaluation.

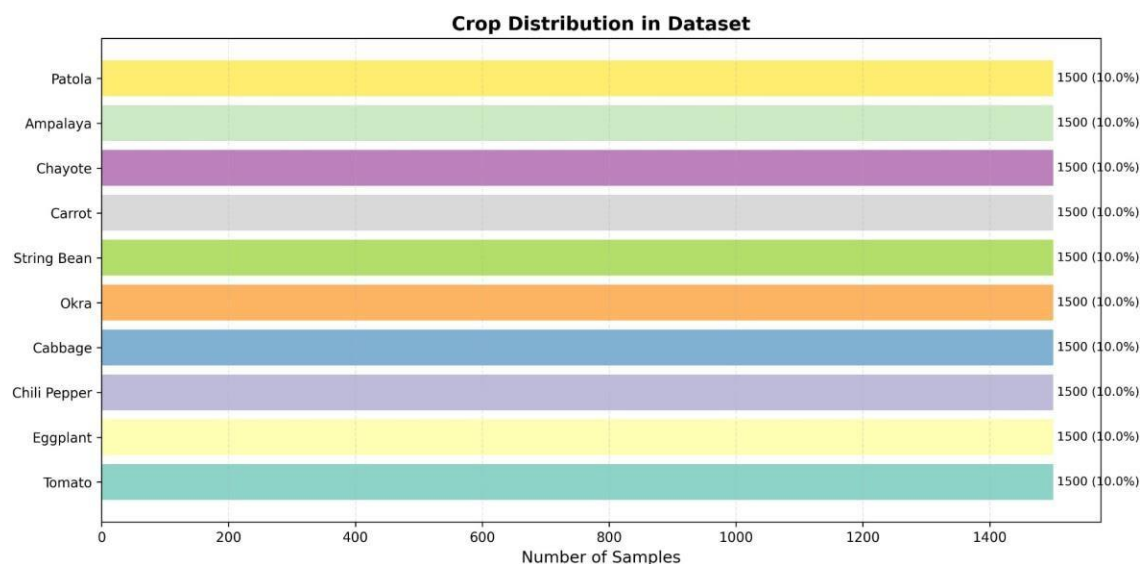


Figure 8: Crop Distribution

The crop distribution analysis indicated that there were no disproportions in the crop types as the ten types of crops had the same number of samples (1,500). Such consistency showed that the data was properly designed to prevent the imbalance of classes, which is a frequent problem to be detrimental to statistical modeling and machine learning results. The data enabled models to learn nutrient patterns by encouraging them not to be biased towards specific crop types because the proportion of data contributed by each crop was the same (10% per crop).

Equal sample sizes also implied that data collection process was systematic in that the nutrient readings were evenly placed in the categories of crops. Such balance increased the accuracy of comparative analyses (e.g., visualization of nutrient distribution and heatmap analyses) as the results were no longer dependent on disproportionate sample sizes. Moreover, the uniform distribution was allowing the fairness of the recommendations generated on fertilizer because there were no under-represented or over-represented crops in the analysis.

Importance of Feature Analysis

Figure 9 indicated the feature important analysis to identify the input variables that provide the greatest contribution to the model predictions. The chart underscored the effect of each of the NPK-based features on the classifier, which enabled the researchers to determine the most effective nutrients and derived values. The analysis assisted in explaining the process of decision making by the model and the improvements that can be made to optimize the future model. By highlighting which features have the strongest influence, the study also provided guidance for prioritizing soil nutrient measurements in practical applications.

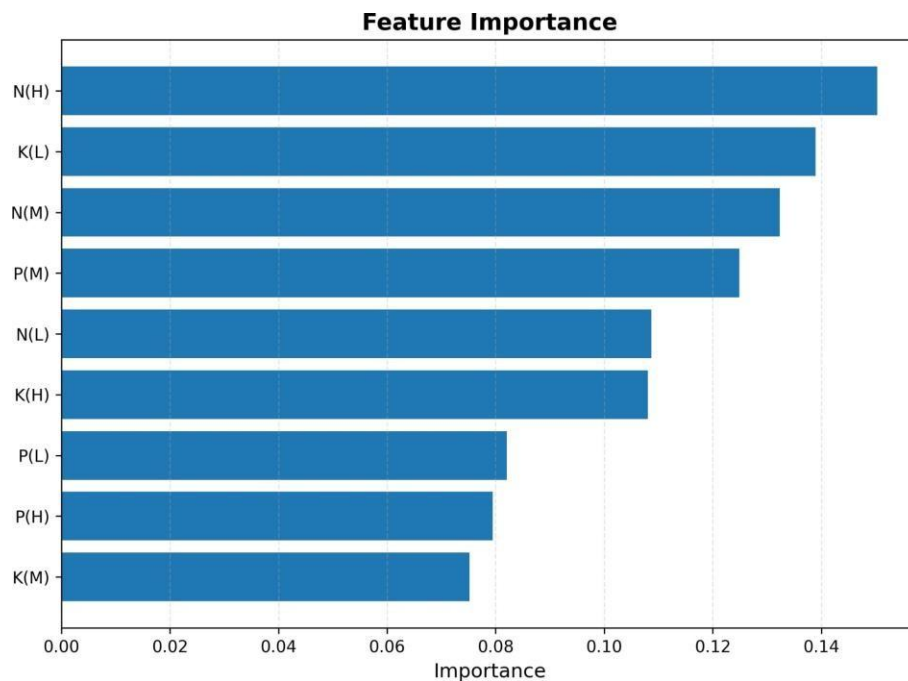


Figure 9: Feature Importance

The continuous values of the NPK sensor were categorized into bins so as to make interpretation easier before being trained into a model to ensure that the levels of nutrient sufficiency were also covered. All the nutrients (Nitrogen, Phosphorus, and Potassium) were categorized as Low (L), Medium (M), and high (H) with regard to predetermined

threshold ranges that were based on agronomic recommendations and the statistical distribution of the data.

Particularly, the lower values were identified as the deficiency of nutrients and were identified as Low, intermediate values were identified as Midrange, and the high nutrient levels were identified as High. These binned attributes (e.g., N(H), K(L)) were then converted and introduced as inputs to the Random Forest model which enabled the model to learn decision patterns using categories such as nutrient levels and not using the actual numerical values. The figure represents the feature importance analysis, created by the Random Forest model applied in the Smart Fertilizer Recommendation System. This visualization gives a representation of how the individual nutrient categories, (high (H), medium (M), and low (L)) contributed to the model in its decision-making process in predicting fertilizer recommendations due to different nutrient conditions of a soil.

As indicated by the image, high Nitrogen (N(H)) was of greatest importance score meaning that an excessively high level of nitrogen was most influential in the output of the model. This is an agricultural practice that nitrogen has a great influence on the development of plants and in most cases the most restraining macronutrient. Next, low Potassium (K(L)) and medium Nitrogen (N(M)) were also the most influential as they indicated that not only potassium deficiencies were the main determinants upon which the model was based when assessing soil health.

Other characteristics like medium Phosphorus (P(M)), low Nitrogen (N(L)), and high Potassium (K(H)) were of moderate importance, i.e. they also affected the predictions of the model but to less degree than the best features. In the meantime, the lowest values of importance pertained to low Phosphorus (P(L)) high Phosphorus (P(H)) and medium Potassium (K(M)). These nutrient categories, even though still applicable to the growth in plants, have less impact which means that they played a minor role in the overall process of the model prediction.

Over all, the findings of the feature importance show that this model depended on dif-

ferent levels of nitrogen the most, then on principal potassium and phosphorus classes. This substantiates that nitrogen is the most dominating in requirements of determining the fertilizer requirements, yet potassium and phosphorus levels offer supplementary, yet significant, data in coming up with reliable recommendations.

Model Development

Random Forest model was implemented in the state of tuned hyper parameters by using a number of estimators (150) and infinite tree depth to predict as much as possible. Bootstrapping, together with impurity measurement using the Gini index, was used to build trees. The algorithm gave rise to several decision trees during the training phase and ultimately made predictions by the majority vote across the ensemble. This reduced overfitting as it averaged across varying decision paths whilst at the same time improving generalization to unobserved data. The model was able to define the correlation between the nutrient composition of soils and the fertilizer recommendation. The efficiency of the preprocessing pipeline and the Random Forest configuration was proven by its capacity to detect any evident patterns of nutrients and convert them into their correct predictions. The findings attested to the idea that ensemble learning with a combination of balanced data and attentive parameter tuning is a dependable structure towards fertilizer recommendation tasks.

Model Evaluation

The standard classification measures, such as accuracy, precision, recall, and F1-score were used to evaluate the Random Forest model. These actions give the overall picture of predictive reliability of the model and how well it would extrapolate to unknown data. The results obtained are collated in the confusion matrix and the classification report shown in.

Accuracy

The values of accuracy indicated that the model was accurate in about 67.49% of its results during training and 65.49% during testing. The small decrease of approximately 2 percent indicated that the model still had a significant prediction power when presented with unknown data. Although accuracy in itself did not give any information on the specifics of the performance of the classes, it showed that the model achieved a reasonable degree of accuracy throughout the dataset. This overall performance suggested that the Random Forest model was effective in capturing the general patterns of nutrient requirements across different crop types.

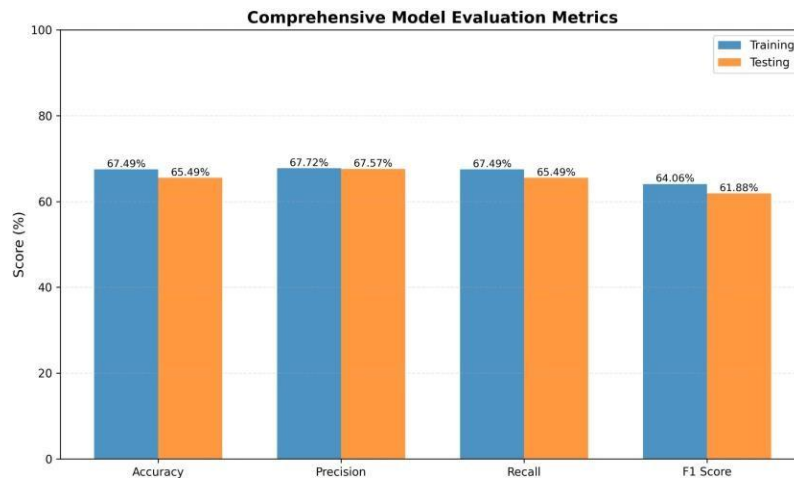


Figure 10: Evaluation Metrics

Precision

The accuracy scores were very close with the result of 67.72 when training and 67.57 when testing were almost equal which means that there was a very small decrease in performance. This low difference indicates that the model had a stable capacity to identify positive predictions accurately in both datasets. The high measure of consistency means that there were no large numbers of false positives that occurred on model being applied to unseen data. This reliability is especially desired in fertilizer recommendation systems

because false positive recommendations may result in undue application of nutrients which in turn lead to higher costs and possible damage to the environment.

Recall

The same tendency was noted with the values in the recall, where the model had a 67.49 and a 65.49 result on the training and testing sets, respectively. The minor decline demonstrates that the model was still useful in detecting the majority of the true positive cases even in the cases of previously unknown data. This uniformity in recollection is crucially essential, since it will serve to minimize instances of false negative, or ensuring that those crops that may require special nutrient manipulations cannot be overlooked in the process of making fertilizer recommendations.

F1-Score

The F1 scores were relatively less and the values were 64.06 and 61.88 respectively in the training and testing sets. Given that the F1 score is the sum of precision and recall, the relative decrease in its value suggests that the model balance between correctly determined cases of true positives and reduced misclassifications has slightly decreased. However, the variation between training and testing results was not very significant, which indicates that the model did not change its trade-off between precision and recall in the case of unseen data.

Overall Assessment

The accuracy and F1-score were average, but the Random Forest model showed a steady performance on the classes of fertilizers. These findings point to the fact that the used data preprocessing and class balancing methods worked. Although the model already exhibits predictive behavior, which is stable, one can still enhance it by including more soil parameters such as pH, moisture content, and organic matter as well as applying more

sophisticated feature engineering methods. Besides, decreasing the effect of the outliers by refining the data further may have a possible positive impact on the models and result in more correct and feasible fertilizer recommendations.

Confusion Matrix

The Confusion matrix table illustrated the extent to which the classification model forecasted every type of crop using the NPK values. Comparison of actual and predicted labels showed the patterns of correct and common misclassifications and further understanding can be made of the crops that the model found more difficult to differentiate by comparing the actual and predicted labels.

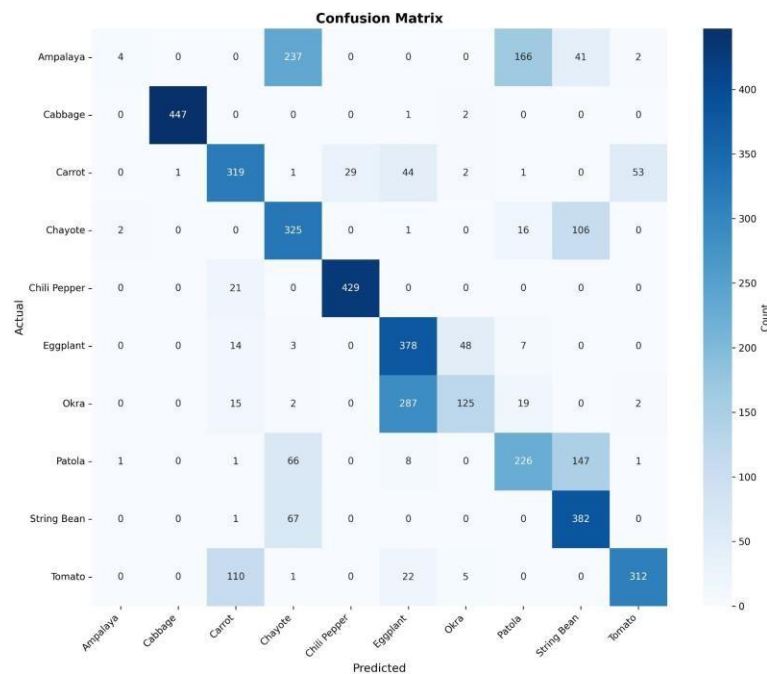


Figure 11: Confusion Matrix

The confusion matrix indicated that the model was good at classifying correctly some of the crop types, specifically Cabbage, Chili Pepper, Eggplant, String Bean and Tomato. This was indicated by the large values on the diagonals with 447 correct predictions of Cabbage, 429 correct prediction of Chili Pepper, 378 correct prediction of Eggplant, 382

correct prediction on String Bean, and 312 correct prediction on Tomato. These outcomes suggest that nutrient patterns of these crops were distinct enough and the model was able to learn and identify their trends.

Nevertheless, significant misclassification trends in some crops were also pointed out in the matrix. An example of this is that Ampalaya was most often confused with Carrot (237 instances) and String Bean (166 instances), implying that the nutrient content of these two have a lot of overlap, and thus the model could not discriminate them easily. Likewise, Carrot had significant misclassifications in most of the categories, mostly with Chayote and Okra with high misidentifications of Okra with Eggplant and Carrot. These mistakes imply that there are similarities in the NPK values of distributions of these crops resulting in the lack of clear boundaries in the decisions as taken by the model.

Chayote displayed moderate performance in that he performed 325 correct classifications but with String Bean and Ampalaya, he continued to show misclassifications, which further suggests the overlap of nutrients among these crops. Similarly, Patola was usually confused with Chayete and String Bean, which means that its nutrient signature was not so clear in the dataset. Conversely, crops that had better-separated nutrient profiles e.g. Cabbage and Chili Pepper had a lesser misclassification because they had stronger separation in their NPK profiles.

In general, the confusion matrix indicated the strong and weak sides of the predictive nature of the model. Although the model has worked well on crops whose patterns of nutrients can be easily identified, it has had a challenge in those crops whose NPK values overlap or show similarities. The implication of these findings is that additional feature engineering, better model tuning or incorporation of other data would be required to increase the capacity of the model to differentiate between crops with similar nutrient profiles.

Error Analysis

The distribution of errors in the confusion matrix showed that the primary sources of incorrect classification were due to the overlapping patterns of nutrient profiles of some crops, which indicated that the boundaries of decisions made by the model were mostly based on the similarity in NPK values. The model was probably used to treat crops where the nutrient level is closely spread (Ampalaya, Carrot, Chayete, Okra, and Patola). This was notable more in repeated cases of misclassification among such pairs as Ampalaya-Carrot, Carrot-Okra and Patola-String Bean.

The reason behind these wrong classifications was also the use of NPK values as input features alone. The data on only macronutrients also restricted the success of the model to effectively differentiate the crops with similar nutrient signature as they occur naturally. It would mean that the inclusion of some extra features, including the soil type, the growth stage of crops, or the data about micronutrients may influence the class separability and predictive accuracy.

Since the model had a moderate overall performance, portrayed by an approximate accuracy of 65, the model was seen to have the capability of generalization but not high classification accuracy with all the types of crops. The observation was also correlable with the confusion matrix that indicated more specific crops with distinct nutrient profiles were correctly identified and those with overlapping patterns were more likely misclassified. Such results suggest that more complicated modeling methods are required or more and diverse data sources are necessary to enhance the reliability of crop classification.

Soil Testing, System and Prototype Output

Prototype and User Soil Testing

This section discussed how user walkthrough the process of the soil testing and user flow.

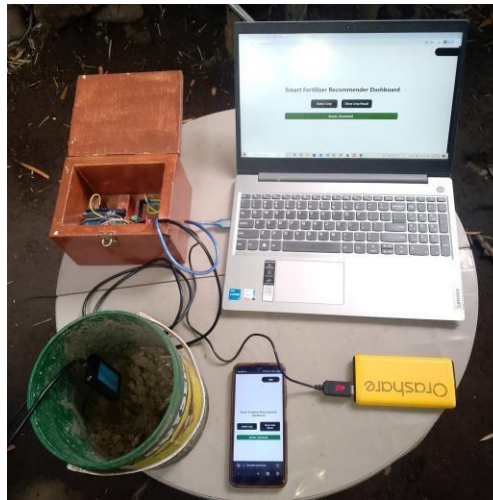


Figure 12: **Prototype**

In the case of the prototype test, the user switched on the whole arrangement with a power bank and then inserted the NPK sensor into the sample of soil in order to start collecting nutrient data. After the sensor was installed, it had already begun to sense the amount of Nitrogen, Phosphorus and Potassium in the soil. These readings were sent via the RS485 communication line and they were fed into the MAX485 module which would translate the differential signals into normal serial data which the Arduino Uno was capable of handling. The sensor was used to provide a continuous output to the Arduino and the output was received by the RX and TX pins with the data direction appropriately handled by the MAX485 control pins. Once the Arduino was able to collect and read the NPK values, it sent them to the laptop of the user through the USB 2.0 Type A/B cable. The Arduino was hooked to a laptop running a running backend script or web server which then read the serial output of the Arduino and passed the processed values to the Smart Fertilizer Recommender

Dashboard. After revising, the online app showed the nutrients of the soil, determined whether the soil lacked N, P, or K and provided the necessary recommendations to use the appropriate type of fertilizer. This workflow allowed the user to add the sensor, wait until the values became steady, and right away the results were displayed on the application, which proved that the prototype was able to convert the physical data on soil into the digital knowledge properly.

Moreover, the web application was designed as a mobile responsive one, i.e., it automatically set its layout, text size, and interface elements so that the screen size, i.e. the laptops, tablets, or smartphones, would look adequate. This responsiveness made it possible to easily check the results of soil and the recommendation of the fertilizers even when standing outside the area of testing on the site. There was also resizing of buttons, graphs, and data cards on smaller screens along with a reorganized layout in the dashboard in order to make the dashboard readable and easy to use. Due to this mobile-friendly scheme, the prototype enabled real-time monitoring of the soil not just using a laptop, but also using a mobile device which made the system much more accessible and convenient to use.

Furthermore, some instances have encountered in soil testing. Some soil have a low or zero value of Nitrogen, Phosphorus and Potassium. Sensor sometimes doesn't recognize or no sensing or Live NPK value when soil has too much of rocks or sandy.

Dashboard

The dashboard interface had a minimalistic and clear format that enabled one to simply pick a crop and see the outcomes and monitor real-time system information like sensor malfunctions.

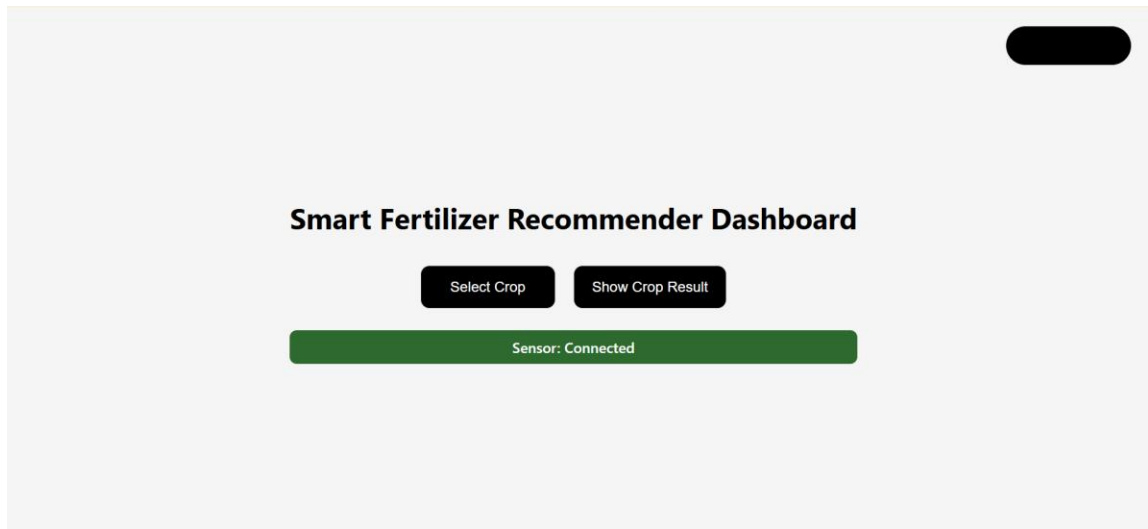


Figure 13: **Dashboard**

The Smart Fertilizer Recommender System has the dashboard as the primary control interface. It has a simple and clean design that gives users the option of light and dark mode that offers them flexibility and comfort depending on the user's taste or the kind of light. The two major actions of the system in the center of the interface are Select Crop and Show Crop Result. These buttons allow moving through the core workflow, i.e., to choose a crop, then see the recommendations of the fertilizer based on the data on nutrients in the soil obtained by the sensor. The bottom of the dashboard has a status bar that gives real time feedback on the sensor condition. In the display it now gives an error of not being able to connect the sensor with the given COM port. This knowledge is also essential, because it enables the user to be able to locate instantly whether the sensor is connected as it should, or troubleshooting is required.

Confirmation Button

The system had a neat confirmation modal that required one to either move on to the crop selection page or cancel the action by using distinct Yes and No buttons.

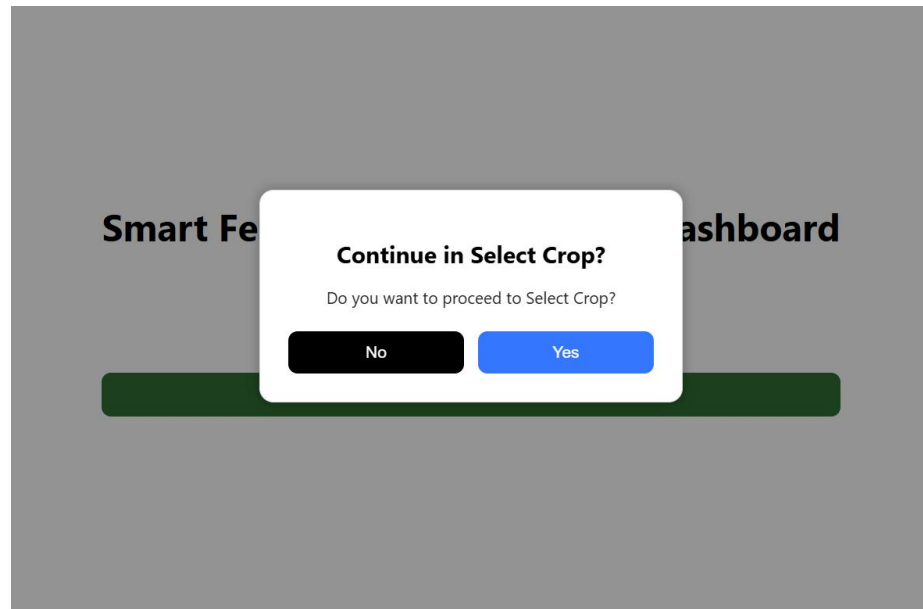


Figure 14: Confirmation Buttons

This confirmation modal serves as a secondary validation measure to make sure that users do significant actions on the dashboard with an intention. To avoid unintentional navigation when selecting the crop and showing crops result, the system will present this dialog box when the user clicks the Select Crop button in case he/she clicks the button or taps the wrong place on a touch-sensitive device. Stopping the working process and presenting the question continue in Select Crop, the system contributes to the increase of clarity and makes the user realize that has entered the crop selection process. The modal has two options that are clear, Yes, that will open the selection interface to the user, and No, which will cancel the action will bring the user to the end page. This simple design promotes ease of use particularly among non-technical users as it minimizes the chances of errors and simplifies and makes the decision making process easy and clear.

Another such confirmation interaction is that of the Show Crop Result button which establishes harmony in various aspects of the interface. Such standardized behavior makes it easier to use and supports the purpose of the system to withstand accidental functionality. In the framework of the fertilizer recommendation system, this confirmation is significant

since every action causes certain processes in the system, i.e., retrieving sensor data, analyzing NPK readings, or generating fertilizer recommendations. Thus, this confirmation modal is essential in enhancing the safety of the user, as well as making the system reliable by making sure that actions are only performed when required by the user on purpose.

End Page

A simple thank-you message was presented at the closing interface where there was a definite button that directed users to go to the dashboard once they were done interacting with the app.

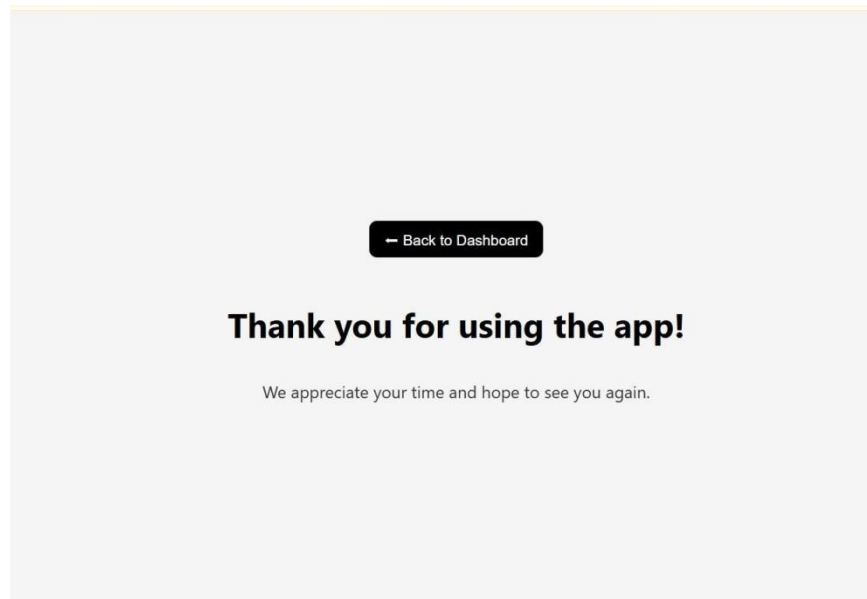


Figure 15: End Page Interface

The last interface to be shown to the user is the closing screen of the system which is shown to the user after using the application. This final page provides a clear and a friendly salutation message which conveys gratitude with the message of saying thank you by saying the words; Thank you, using the app!. The brief line below also enhances appreciation and gives a person positive emotional experience by motivating him/her to come again in future.

To facilitate ease of access and navigation, the page has the Back to Dashboard button

that is displayed above the main message. This will enable the user to be able to go back to the main menu with ease in case they want to still use the application or want to explore other features. The layout of the page is purposefully sparse and clean, making the end result of the interaction relaxed, readable and user friendly.

Fertilizer Recommendation Output

The crop recommendation interface had very clear and well-organized graphic layout that enabled the user to see the stored rankings or review soil-specific sensor reading and crop recommendations presented by the interface in orderly and neatly laid out tables and panels.

The screenshot displays the 'Soil NPK Advisor' web interface. It features three main columns. The left column contains input fields for 'Crop' (set to 'Tomato') and 'Select Soil' (set to 'Soil 1'), along with buttons for 'Compute Recommendation', 'Sensor: Connected', 'Status: Connected', and 'Save Crop for Testing'. The middle column, titled 'Live Sensor Readings (Last updated: 11:14:19 PM)', shows a table with Nitrogen (N) at 0, Phosphorus (P) at 44, and Potassium (K) at 36, with a note 'Auto-parsed from Arduino serial output.' The right column, titled 'Recommendation', shows 'Urea (46-0-0)' with NPK: 46-0-0, and a table with Scaled Target N (16), Scaled Target P (32), Scaled Target K (32), and Relative Deficit N (1), P (0), K (0). Navigation buttons 'Back to Dashboard' and 'Show Crop Result' are at the top right.

Sensor	Value
Nitrogen (N)	0
Phosphorus (P)	44
Potassium (K)	36

Parameter	Value
Scaled Target N	16
Scaled Target P	32
Scaled Target K	32
Relative Deficit N	1
Relative Deficit P	0
Relative Deficit K	0

Figure 16: Crop Selection Interface Urea

The interface opens up with a panel on which the user enters the crop, and selects the related soil sample to be tested. Once these are chosen, the user clicks on the option of computing Recommendation and the system informs that the sensor is connected and its status active meaning that it is now ready to retrieve real-time data of the Arduino module. In the centre of the screen there are the Live Sensor Readings that display the concentration of Nitrogen, Phosphorus and Potassium in the soil. The Nitrogen value in this case is 0 and

therefore shows deficiency fully. This commonly occurs in cases soil is sandy or composed of stones since such soils have very low organic contents and their ability to retain nutrients is low as well making it to have no nitrogen at all. In the meantime, the Phosphorus reading is 44 and Potassium reading is 36 indicating that the two nutrients are already adequate to satisfy the crop.

The system compares the nutrient balance using these readings and provides a recommendation on what to put as fertilizer. As it is only missing in the form of Nitrogen, the system proposes to use Urea (46-0-0), which is a fertilizer containing a lot of Nitrogen, and shows the targets of the nutrients that the crop is missing and the calculated shortages. Upon investigation, the user can save the crop selected, soil and sensor data, and the recommendation created by clicking on Save Crop to Testing so as to have the information saved to be used later in the documentation or evaluation.

The screenshot displays the 'Soil NPK Advisor' web application. It features a sidebar with 'Crop' (String Bean) and 'Select Soil' (Soil 1) dropdowns, a 'Compute Recommendation' button, and status indicators for 'Sensor: Connected' and 'Status: Connected'. The main area is divided into three panels: 'Live Sensor Readings' (Nitrogen: 0, Phosphorus: 36, Potassium: 29), 'Recommendation' (Organic Compound 5-5-5), and a table of scaled targets and relative deficits for Nitrogen, Phosphorus, and Potassium.

Live Sensor Readings (Last updated: 11:15:59 PM)	
Nitrogen (N)	0
Phosphorus (P)	36
Potassium (K)	29

Auto-parsed from Arduino serial output.

Recommendation	
Organic Compound (5-5-5)	
NPK : 5-5-5	
Scaled Target N	10.5
Scaled Target P	21
Scaled Target K	31.5
Relative Deficit N	1
Relative Deficit P	0
Relative Deficit K	0.111

Figure 17: Crop Selection Interface Organic Compost

The image depicts the Soil NPK Advisor page whereby, a selected crop and soil sample can be analyzed by the user to determine whether the soil is in the correct nutrient balance. In this case, the user has selected Tomato as the crop and Soil 1 as the sample of soil. The system then verifies that the sensor is connected and ready after these selections and the

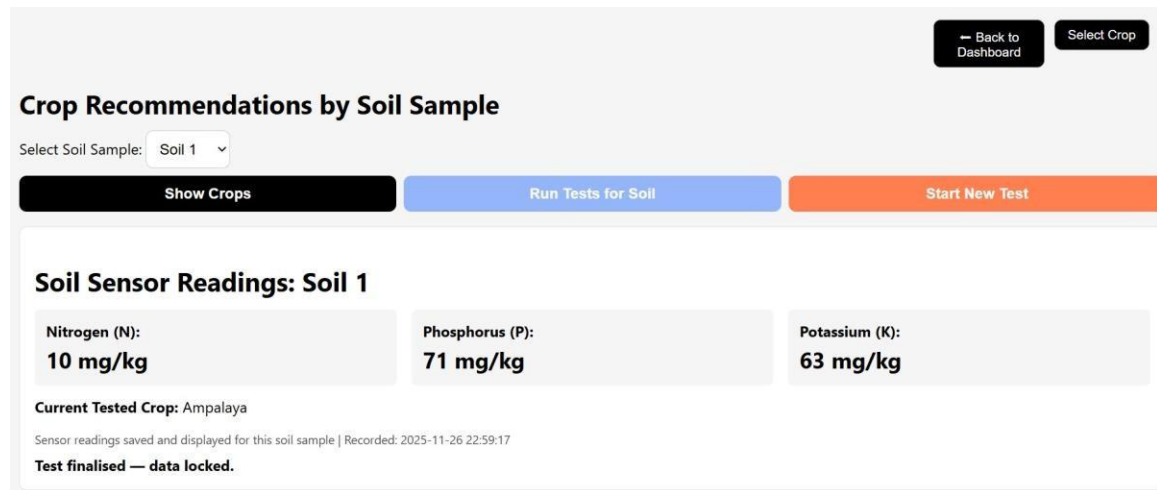
user is then able to click the option of Compute Recommendation to run the analysis. After the test has been initiated, the system will then display the live sensor values of nitrogen, phosphorus and potassium. In this instance nitrogen is 0 indicating that the soil has no quantifiable nitrogen- this is usually an indication which are unable to retain nutrients. In the meantime, phosphorus level of 44 and potassium level of 36.

The sensor data collected and the system produces a fertilizer prescript to address the crop needs. Because the nitrogen deficiency is the only one, the system proposes the use of Urea (46-0-0) which is a fertilizer that gives a good boost of nitrogen. The nitrogen reading of this soil test is 0 indicating that the soil has a very low nitrogen level. This normally occurs in sandy and rocky soils or those that have no organic matter since the nutrient nitrogen cannot be retained appropriately in this kind of soil and is readily washed away. Consequently the sensor picks up practically no nitrogen whatsoever. In the meantime, the phosphorus (44) and potassium (36) values are found in the soil, though they are not exactly according to the target values of the crops. The scaled target of P and K required by tomatoes is 32 and hence the readings are slightly higher than the optimal values.

This implies that the soil contains sufficient phosphorus and potassium and does not need extra fertilizer to contain these nutrients. The only significant nutrient issue in this sample is nitrogen and this explains why the system suggests the application of a nutrient-rich fertilizer, such as Urea (46-0-0) to rectify the deficiency. The Recommendation of fertilizer change when the other Nutrient value doesn't meet the needed requirements, it shows on the figure that the Potassium didn't meet the requirement so the fertilizer is Organic compost. This is because this model has anticipated this to happen in line with the patterns of the training data in the event the nitrogen levels are not high or doesn't meet the requirement value it will recommend Urea fertilizer but if Phosphorus and Potassium doesn't meet the requirement value it will Recommend fertilizer as Organic compost, then when the NPK values meet its requirements and it is balanced it will recommend Complete (14-14-14) Fertilizer.

Crop Ranking

The crop recommendation interface had a structured and clear design that enabled the user to see stored ranking or view soil specific sensor readings and crop recommendations in a well organized table and panel form.



— Back to Dashboard Select Crop

Crop Recommendations by Soil Sample

Select Soil Sample: Soil 1 ▼

Show Crops **Run Tests for Soil** **Start New Test**

Soil Sensor Readings: Soil 1

Nitrogen (N): 10 mg/kg	Phosphorus (P): 71 mg/kg	Potassium (K): 63 mg/kg
---	---	--

Current Tested Crop: Ampalaya

Sensor readings saved and displayed for this soil sample | Recorded: 2025-11-26 22:59:17

Test finalised — data locked.

Figure 18: Crop Display Interface for Crop Recommendation

This part of the system interface displays the findings produced in the process of soil-based evaluation as the application shows the most recent readings of the NPK sensors on a chosen piece of soil. The user can select which soil sample to see at the top of the interface with the help of the dropdown menu that can be used to select the soil sample and make sure that the system shows the precise measurements of the nutrients related to this particular test. Middle buttons Show Crops, Run Tests for Soil, and Start New Test will provide the user with the overall flow of the process. Because the soil test on soil 1 is already completed and finalised, it is not possible to run another test on the same sample, and this shows that the data will be locked to ensure that there is no accidental overwriting by mistake.

The whole summary of Soil Sensor Readings of Soil 1 presented below gives the result of measurements of the nutrient levels of Nitrogen (10 mg/kg) Phosphorus (71 mg/kg) and

Potassium (63 mg/kg) in the soil. The readings are proportional to the current nutrient profile of the soil and form the reference point in other regions of the system to ascertain crop suitability and the recommended amount of fertilizer to apply. The interface also displays the Current Tested Crop under the nutrient values, in this instance being Ampalaya and this proves the crop that was tested by using these soil readings. A timestamp is present so as to assure transparency and traceability of what time measurement was made. Test finalised data locked. is a label that the user is informed by to know that the readings are officially saved and can now be used in crop recommendation modules without the danger of being altered.

Stored Crop Rankings (10 crops)				
Rankings are based on the most recent saved test for the live sensor data.				
Test performed: 2025-12-11 12:49:09				
Rank	Crop Name	Suitability Score	Recommended Fertilizer	Saved NPK
1	Patola	0.841	Urea (46-0-0)	N=9.00, P=67.00, K=60.00
2	Ampalaya Tested Crop	0.840	Urea (46-0-0)	N=10.00, P=71.00, K=63.00
3	Tomato	0.820	Urea (46-0-0)	N=2.00, P=52.00, K=44.00
4	Chili Pepper	0.811	Urea (46-0-0)	N=1.00, P=48.00, K=41.00
5	String Bean	0.807	Urea (46-0-0)	N=2.00, P=50.00, K=43.00
6	Chayote	0.789	Urea (46-0-0)	N=1.00, P=48.00, K=40.00
7	Eggplant	0.735	Urea (46-0-0)	N=2.00, P=52.00, K=44.00
8	Carrot	0.717	Urea (46-0-0)	N=1.00, P=49.00, K=42.00
9	Okra	0.687	Urea (46-0-0)	N=2.00, P=52.00, K=44.00
10	Cabbage	0.521	Urea (46-0-0)	N=2.00, P=50.00, K=42.00

Figure 19: Crop Display Interface Crop Results

This part of the interface shows the Stored Crop Rankings that recapsulates the performance of each of the ten crops analyzed relative to the nutrient profile of the soil sample that was just being tested. The rankings are in the order of highest to lowest suitability score and the user is able to quickly note the crops that are the most suited to the NPK composition of the soil. On the top of the table, the system reveals the date and time of the actual performance of the underlying soil test, which makes the results of the evaluation traceable and transparent. The ranking table has the name of the crop, the obtained suitability score, the recommended fertilizer and the saved NPK values of the nutrient requirement

of the crop that has an optimum nutrient level in the soil.

The system used the live data of the soil with the stored crop profiles by comparing the latest sensor data with the optimum nutrient requirements of each crop. The system compared the live NPK values with the predetermined optimum soil ranges of each crop in the database after the live NPK values were recorded. A score of suitability was then put on each crop in regards to the degree to which the soil conditions fitted the requirements of a crop. Increased scores implied that the soil was already close to satisfying most of the needs of that crop whereas decreased scores meant that they would have required more adjustment of fertilizers. After the scores were obtained the system ranked the crops most to least suitable. This ranking made it easy to make decisions since the farmer could quickly understand those crops that would perform best in the prevailing conditions of the soil. The crops that recorded the highest scores such as Patola and Ampalaya would have been easily corrected and planted in the soil, and thus would have been the most viable to cultivate. In the meantime, crops that score lower like cabbage would not have been an effective option unless further fertilizer or soil preparation was used. The NPK values that were saved and the recommended fertilizer also indicated the extent to which the nutrient adjustment had already been realized, which could be used to make future fertilization and crop planning decisions.

Patola, which has the highest ranking, has the highest suitability score of 0.841 meaning that the nutrient requirements of that crop are almost the same when compared to the measured soil NPK profile. Just below, Ampalaya is ranked second with a suitability score of 0.840, just too close to the score of Patola. Ampalaya is graphically emphasized and shown as being labeled as Tested Crop, which is a clear interpretation that it was the crop that was tested in the last soil analysis. This highlight will assist the user to know the type of crop that was chosen the last time the test was run.

In all the entries, the system will always suggest Urea (46-0-0) to be used as the choice of fertilizer, which is nitrogen-centered as per the nutrient deficits of the soil. The table is

also in a well organized format to show the ideal NPK of each crop which allows the users to compare the nutrient requirement of each crop with the current soil conditions.

Mobile Responsive

This is the interface for mobile reponsive which user can access through a mobile.

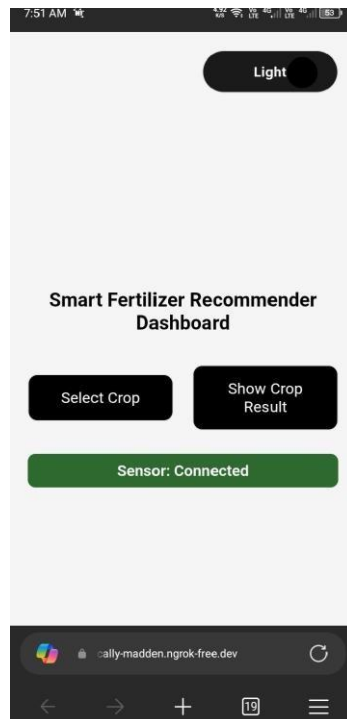


Figure 20: Mobile Application Interface

The Smart Fertilizer Recommender Dashboard was mobile-responsive with an important role of ensuring that the system could be fully usable by the farmers and field users using smartphones as they are reportedly the most common devices among farmers and other users in the field. The system made it possible to have all the buttons, text, menus, and alerts visible, readable, and interactable on the interface, no matter the size of the device by automatically scaling the interface to small devices. This responsiveness avoided the distortion of layout, overlapping of the elements and illegible text problems in situation that could hinder the decision making process when real-time sensor data or crop recommen-

dations are being viewed. It has also enabled users to make major tasks like the choice of crops, the evaluation of soil tests and get error messages through their mobile devices when they are outside the office. Additionally, the dashboard incorporated intuitive navigation features that minimized the learning curve for users unfamiliar with digital interfaces. The system also allowed for quick access to historical data, helping farmers compare current readings with past trends for better decision-making. By supporting offline functionality, users could continue monitoring and managing crops even in areas with limited or no internet connectivity. The color schemes, font sizes, and alert signals were carefully designed to improve readability under varying sunlight and field conditions. Overall, mobile responsiveness not only enhanced accessibility but also increased user confidence and efficiency in managing fertilizer application on-site. Finally, the aim of mobile responsiveness was to increase accessibility, make it more usable in the field conditions of the farms and make sure that all the important information about the system is always accessible offline so that to make the process of making the important and timely agricultural decision.

CHAPTER 5

SUMMARY OF FINDINGS, CONCLUSION AND RECOMMENDATIONS

The chapter summarises the general results and conclusions of the research, including the fact that the Smart Fertilizer Recommender based on the Random Forest was able to utilized the NPK sensor.

Summary

This research was based on the historical issue of non-site-specific fertilizer use in the Philippines in which the farmers had been using generalized and traditional approaches that had failed to take into account the actual nutrient-based composition of the soil. Such practices usually led to the imbalance of the nutrients, soil erosion, the unnecessary costs of fertilizers, and low crop yields. This issue was introduced in Chapter 1 and mentioned that it aimed to create a system of fertilizer recommendations that could combine real-time readings of NPK sensors and a Random Forest (RF) machine learning model. Another factor important to the system was the role of the system to farmers, government agencies, academic institutions, communities, and future researchers, as the prototype was restricted to NPK data and selected crops in Iriga City.

The synthesis of literature available on the topic of nutrient management of soil, machine learning in agriculture, and NPK sensors in precision farming. The past researches were also consistent with the finding that machine learning, specifically the Random Forest, demonstrated high performance in agricultural predictions, including nutrient estimation, crop suitability analysis, and fertilizer recommendation. It was also evidenced through research that used real-time sensor data which showed that dynamic and localized decision support could be a possibility. Nevertheless, the majority of the systems were not

calibrated according to the Philippine soil conditions and were not properly integrated to provide real-time NPK sensing and crop-specific fertilizer guidance. It is this gap that formed the foundation of this study.

Defines the methodology that was based on the approach of experimental-developmental that also included data collection, synthetic data generation, preprocessing, model training, and prototype development. The reason behind the choice of the Random Forest algorithm was that it is stable, accurate, and it can analyze multivariate agricultural data. The chapter described flow of the system, hardware design of an NPK sensor based on Arduino, and creation of a web-based application with Flask framework. It also reported on data preparation procedure, evaluation in terms of accuracy, precision, recall, and F1-score, and cross-validation. The suitable theory was based on ensemble learning, concepts of decision trees, and decision making based on data.

Provides the outcome of the model training and implementation of the systems. The exploratory data analysis has revealed the variability of the nutrient and the essentiality of NPK as the main characteristics. The optimized Random Forest model performed very well in all assessment measures and had credible classification in terms of confusion matrix analysis, error rate graphs, and accuracy curves. The chapter also indicated effective assimilation of the NPK sensor on the prototype system. It was tested that the model was capable of running real-time sensor data and producing the correct fertilizer recommendation and crop suitability ranking via the user interface.

Findings

This part introduces the major findings of the research, the training performance of the model, evaluation measures, as well as the successful implementation of the model with a real-time Arduino-based NPK sensing network. It provides a comprehensive overview that connects the model's theoretical performance with its practical application in precision.

1. The Random Forest model was successfully trained using Nitrogen, Phosphorus, and Potassium as input features, enabling it to learn patterns from both synthetic and real soil nutrient datasets and generate reliable fertilizer predictions.
2. Performance evaluation showed that the model achieved an accuracy of 67.49% along with acceptable precision 67.72%, recall 67.49%, and F1-scores 64.06%. These results indicate that the classifier can consistently categorize fertilizer recommendations, although certain nutrient ranges exhibited reduced accuracy due to overlapping class boundaries.
3. The trained model was integrated with an Arduino-based NPK sensor to develop a fully functional prototype capable of retrieving real-time soil nutrient values. The system automatically generated fertilizer recommendations through a responsive user interface and was validated to perform correctly using both manual inputs and live sensor readings.

Conclusion

This section outlines the general conclusions made by analyzing, evaluating the model and implementing the prototype to this research. The result of the data processing of soil nutrients, the training of the Random Forest model, and the combination of the system with the real-time readings of NPK sensors indicated several important findings. These conclusions provide a summary of the effectiveness of the system, the efficiency of the choice of variables and algorithm, and the feasibility of the application of an automated fertilizer recommendation system in practice in the agricultural environment.

1. N, P and K values were useful in training the Random Forest model and this indicates that these soil nutrients are the appropriate predictors to produce fertilizer recommendations.

2. The metrics provided by the model prove that the Random Forest algorithm is a feasible model when it comes to the prediction of fertilizers, but the limitation of accuracy means that the more appropriate real soil samples or additive features are added, the more reliable the model will be.
3. The effective combination of real-time sensor measurements with the trained model confirms the fact that automated fertilizer recommendation system is viable and that it can aid in the real-life agricultural decision-making.

Recommendations

In this section, some of the recommendations based on the study findings have been presented. These recommendations are meant to improve the efficiency of the system, heighten the accuracy of data, and inform future researchers to develop and refine the model that was developed.

1. Future researchers may find it advisable to take more real soil samples to enhance the accuracy of the model and minimize the use of artificial or synthetic data.
2. Sensor calibration processes should be done to reduce errors in the nutrient-reading in future research. They should be regularly checked in relation to laboratory soil test results to ensure that hardware performance is consistent and to make sure that it can accurately capture nutrients.
3. Add further soil traits to the data set, including pH level, moisture content, and the percentage of organic matter to enhance the accuracy of the classification and decrease the overlap between the classes of crops.
4. Enhanced the system by future researchers by organizing the types of fertilizers and nutrient content levels according to the local agricultural requirements.

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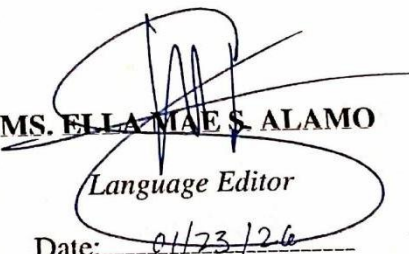
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APPENDICES

APPENDIX A
LANGUAGE EDITING CERTIFICATION

This is to certify that the undersigned has reviewed and went through all the pages of the
Bachelor of Science in Computer Science thesis manuscript titled
**"A Random Forest Approach for Crop Fertilizer Recommendation Using NPK
Sensor"**
of **Joshua Charles B. Bañaria, Jeremy C. Lagdaan, Luarence Bryan B. Belen**, as
against the set of structural rules that govern research writing in accord with the
composition of sentences, phrases, and words in the English language.


MS. ELLA MAE S. ALAMO

Language Editor

Date: 01/23/26

APPENDIX B
SECRETARY'S CERTIFICATION

This is to certify that the undersigned has provided accurate recommendations,
suggestions, and comments unanimously agreed and approved by the panel of examiners
during the oral examination of the thesis titled

**"A Random Forest Approach for Crop Fertilizer Recommendation Using NPK
Sensor"**

prepared and submitted by **Joshua Charles B. Bañaria, Jeremy C. Lagdaan, Luarence
Bryan B. Belen**, and that the same have not been amended, modified or obliterated.

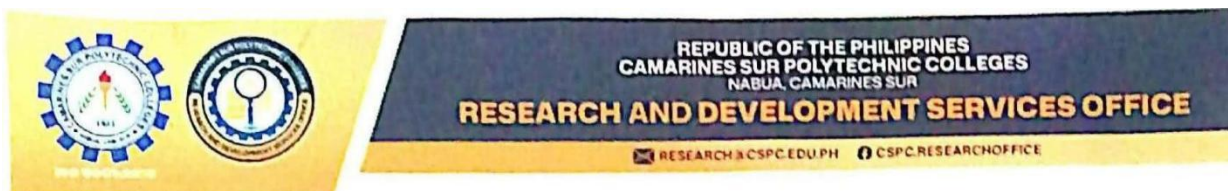

MS. MARRI GRACE MORATA

Secretary

Date: 01/23/2020

APPENDIX C

CRD CERTIFICATE



CERTIFICATION

Date of Release: January 14, 2026
Submission ID: 23367:126168440
Output Title: A RANDOM FOREST APPROACH FOR CROP FERTILIZER
RECOMMENDATIONS USING NPK SENSOR
Author(s): Jeremy C. Lagdaan
Joshua Charles B. Bañaria
Laurence Bryan B. Belen
Program: Bachelor of Science in Computer Science

Authenticity Report:

Similarity Index Report: 9%

Internet Sources: 6%

Publications: 7%

Student Papers: 6%

- Interpretation: The similarity index report means that 9% of the output is similar to the sources in Turnitin's (authenticity software) online repository and databases.

AI-Generated Report: 27%

AI-generated only: 27%

AI-generated text that was AI-paraphrased: 0%

- Interpretation: The AI generated report means that 27% of the output indicates a likely AI-generated text and likely AI-paraphrased text, which is **within the acceptable AI usage** in the field of Engineering.


HAROLD JAN R. TERANO, Ph.D.
Director, Research & Development Services Office

APPENDIX D

JOINT AFFIDAVIT OF UNDERTAKING

JOINT AFFIDAVIT OF UNDERTAKING

We, (1) **JOSHUA CHARLES B. BAÑARIA**, of legal age, single and a resident of SAN NICOLAS, IRIGA CITY, CAMARINES SUR, (2) **JEREMY C. LAGDAAN**, of legal age, single and a resident of LA TRINIDAD, IRIGA CITY, CAMARINES SUR, (3) **LAURENCE BRYAN B. BELEN**, of legal age, single and a resident of SIRAMAG, BALATAN, CAMARINES SUR, after having been sworn to in accordance with law, do hereby take oath and state:

- (i) That, we are officially enrolled for the thesis/capstone project on the topic titled **A Random Forest Approach for Crop Fertilizer Recommendation Using NPK Sensor** in the **COLLEGE OF COMPUTER STUDIES of CAMARINES SUR POLYTECHNIC COLLEGES**.
- (ii) That, the contents of our thesis project submitted to the Camarines Sur Polytechnic Colleges, for award of **COMPUTER SCIENCE** Degree are original and our own work and is not plagiarized.
- (iii) That, if, after completing the thesis/ capstone project, are found copied or come under plagiarism, we will be solely responsible for it and College shall have sole right to cancel my research work ab-initio.
- (iv) That, this work has not been submitted for the award of any other Degree/Diploma in any other University/Institute.
- (v) That, we shall be responsible for any legal dispute/case(s) for violation of any provisions of the Copyright Act relating to our thesis/ capstone project.

IN WITNESS WHEREOF, I have hereunto set my name this 19 day of JAN 2026 202 in _____, Philippines.

(1) **JOSHUA CHARLES B. BAÑARIA**

Affiant

NATIONAL ID: 5025-7421-9379-16362

(2) **JEREMY C. LAGDAAN**

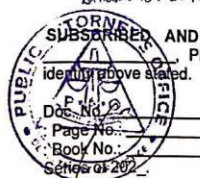
Affiant

NATIONAL ID: 5429-1274-3743-2017
CSPC ID: 22001004

(3) **LAURENCE BRYAN B. BELEN**

Affiant

NATIONAL ID: 2372-3632-7490-2695



SUBSCRIBED AND SWORN TO before me this 19 day of JAN 2026 at _____, Philippines, affiants exhibiting to me their competent proofs of identity above stated.

J. D. MALANI
Public Attorney
Pursuant to R.A. 9406

APPENDIX E

RELEVANT CODE SNIPPETS

Listing E.1: Sensor Data Acquisition

```
# ----- Serial Reader -----  
  
def parse_line_for_value(line: str, key: str) -> float:  
    if key.lower() not in line.lower():  
        return float("nan")  
  
    m = re.search(r"(-?\d+(\.\d+)?)", line)  
  
    if m:  
        try:  
            return float(m.group(1))  
        except ValueError:  
            return float("nan")  
    return float("nan")  
  
  
def serial_reader():  
    if serial is None:  
        state.ok = False  
        state.msg = "pyserial not installed; cannot read Arduino."  
        "  
        "  
        return  
    try:  
        with serial.Serial(SERIAL_PORT, BAUD_RATE, timeout=  
            READ_TIMEOUT_S) as ser:  
            state.msg = f"Connected to {SERIAL_PORT} @ {BAUD_RATE
```

```
}"  
while True:  
    try:  
        line = ser.readline().decode(errors="ignore")  
        .strip()  
        if not line:  
            if time.time() - state.ts > STALE_AFTER_S  
                :  
                state.ok = False  
                state.msg = "Stale: no new data."  
            continue  
  
        n_val = parse_line_for_value(line, "Nitrogen"  
        )  
        p_val = parse_line_for_value(line, "  
        Phosphorus")  
        k_val = parse_line_for_value(line, "Potassium  
        ")  
  
        updated = False  
        if not (n_val != n_val):  
            state.N = n_val; updated = True  
        if not (p_val != p_val):  
            state.P = p_val; updated = True  
        if not (k_val != k_val):  
            state.K = k_val; updated = True  
  
        if updated:
```

```
        state.ts = time.time()

        if all(not (x != x) for x in (state.N,
                                     state.P, state.K)):

            state.ok = True

            state.msg = "OK"

    except Exception as e:

        state.ok = False

        state.msg = f"Read error: {e}"

        time.sleep(0.5)

except Exception as e:

    state.ok = False

    state.msg = f"Open error: {e}"

# start serial thread (harmless if pyserial not installed)
t = threading.Thread(target=serial_reader, daemon=True)
t.start()
```

Listing E.2: Fertilizer Decision Logic

```
def recommend_fertilizer(crop: str, N: float, P: float, K: float)
:
    crop = crop.lower()

    if crop not in CROP_TARGETS:

        return ("No fertilizer", (0,0,0), {"reason": "Unknown
            crop", "deficits": {}})

    values = [max(0.0, float(x)) for x in (N, P, K)]

    total_actual = sum(values)

    if total_actual <= 0:
```

```
        return ("No fertilizer", (0,0,0), {"reason": "No sensor
        values", "deficits": {}})

tn, tp, tk = CROP_TARGETS[crop]
total_target = tn + tp + tk
scale = total_actual / total_target if total_target > 0 else
    0
targetN, targetP, targetK = tn * scale, tp * scale, tk *
    scale

eps = 1e-6
def deficit(actual, target):
    if target < eps: return 0.0
    return max(0.0, (target - actual) / target)

dN, dP, dK = deficit(values[0], targetN), deficit(values[1],
    targetP), deficit(values[2], targetK)

mild, strong = 0.10, 0.25
onlyN = (dN > mild) and (dP <= mild) and (dK <= mild)
anyPK = (dP > mild) or (dK > mild)
bigPK = (dP > strong) or (dK > strong)

if onlyN:
    fert_name = "Urea (46-0-0)"
elif anyPK:
    fert_name = "Complete (14-14-14)" if bigPK else "Organic
    Compound (5-5-5)"
```



```
else:

    fert_name = "No fertilizer"

return (fert_name, FERTILIZERS[fert_name], {
    "deficits": {"N": round(dN,3), "P": round(dP,3), "K":
        round(dK,3)},
    "targets_scaled": {"N": round(targetN,2), "P": round(
        targetP,2), "K": round(targetK,2)},
})
```

Listing E.3: Crop Ranking Algorithm

```
def suitability_score(soil_N, soil_P, soil_K, crop_N, crop_P,
    crop_K):
    total_soil = soil_N + soil_P + soil_K
    total_crop = crop_N + crop_P + crop_K
    if total_soil == 0 or total_crop == 0:
        return 0
    sN, sP, sK = soil_N / total_soil, soil_P / total_soil, soil_K
        / total_soil
    cN, cP, cK = crop_N / total_crop, crop_P / total_crop, crop_K
        / total_crop
    diff = abs(sN - cN) + abs(sP - cP) + abs(sK - cK)
    return round(1 - diff / 2, 3)
```

Listing E.4: Soil Data Storage

```
def save_raw_soil_sample(soil_name: str, N: float, P: float, K:
    float, crop_name: str = None, force: bool = False):
    """
```

```
Save or update a soil sample. If the soil sample is locked
    ( finalized after testing),
do not overwrite unless 'force=True' (explicit operation such
    as running tests or user save).

"""

conn = sqlite3.connect(DB_FILE)
c = conn.cursor()

# Check if existing sample is locked
c.execute("SELECT locked FROM soil_samples WHERE name = ?",
    ( soil_name,))
existing = c.fetchone()
if existing is not None:
    try:
        locked_val = int(existing[0])
    except Exception:
        locked_val = 0
    if locked_val == 1 and not force:
        conn.close()
        return

ts = time.strftime("%Y-%m-%d %H:%M:%S")
# If exists update, else insert
if existing is not None:
    c.execute("""
        UPDATE soil_samples
        SET N = ?, P = ?, K = ?, timestamp = ?, last_crop = ?
        WHERE name = ?
```

```
        """, (N, P, K, ts, crop_name, soil_name))
else:
    c.execute("""
        INSERT INTO soil_samples (name, N, P, K, timestamp,
            last_crop, locked)
        VALUES (?, ?, ?, ?, ?, ?, 0)
        """, (soil_name, N, P, K, ts, crop_name))

conn.commit()

c.execute("SELECT id FROM soil_samples WHERE name = ?",
    ( soil_name,))
row = c.fetchone()
conn.close()

soil_id = row[0] if row else None
if crop_name and soil_id is not None:
    record_selected_crop(soil_id, crop_name, N, P, K)
```

Listing E.5: Overall Evaluation and Ranking Process

```
def run_tests():
    """Run suitability tests for selected soil using per-crop
        saved NPK snapshots."""
    soil_name = request.form.get("soil")
    if not soil_name:
        return jsonify({"error": "Missing soil parameter"}), 400

    soil_row = get_soil_by_name(soil_name)

    if not soil_row:
```

```
        return jsonify({"error": f"Soil sample '{soil_name}' was  
            not found. Save crops first in Select Crop page."}),  
            404  
  
    soil_id = soil_row["id"]  
  
    # Unlock the soil and clear old test results when starting a  
    new test  
  
    # This allows new data to be saved and ensures fresh test  
    results  
  
    clear_test_results_for_soil(soil_id)  
    set_soil_locked(soil_id, False)  
  
    selected_rows = get_selected_crops_for_soil(soil_id)  
    if not selected_rows:  
        return jsonify({"error": f"No crops have been saved for  
            { soil_name} yet. Save crops first in Select Crop page."  
            }), 400  
  
    def normalize(value):  
        try:  
            return float(value)  
        except (TypeError, ValueError):  
            return float("nan")  
  
    results = []  
    for row in selected_rows:  
        crop = row_get(row, "crop_name")
```

```
        if not crop:
            continue

        crop_key = crop.lower()

        if crop_key not in CROP_TARGETS:
            continue

        # Use per-crop NPK snapshot; if missing, default to 0
        # instead of soil snapshot.

        cropN = normalize(row_get(row, "N", 0.0))
        cropP = normalize(row_get(row, "P", 0.0))
        cropK = normalize(row_get(row, "K", 0.0))

        if any(math.isnan(x) for x in (cropN, cropP, cropK)):
            continue

        cN, cP, cK = CROP_TARGETS[crop_key]

        score = suitability_score(cropN, cropP, cropK, cN, cP, cK
            )

        fert_name, _, _ = recommend_fertilizer(crop_key, cropN,
            cropP, cropK)

        results.append({
            "crop": crop.title(),
            "score": score,
            "fertilizer": fert_name,
            "rank": 0, # will be assigned after sorting
            "N": cropN,
            "P": cropP,
            "K": cropK,
        })

    if not results:
```

```
        return jsonify({"error": "Saved crops are not recognized
                           in the current crop list."}), 400

# Sort by score (descending) and assign ranks
results = sorted(results, key=lambda x: x["score"], reverse=
    True)
for idx, r in enumerate(results, start=1):
    r["rank"] = idx

save_test_results_for_soil(soil_id, results)

return jsonify({"success": True, "processed": [{"soil":
    soil_name, "tested_count": len(results)}]})

# Show crop recommendations filtered by selected soil
@app.route("/show_crop", methods=["GET"])
def show_crop():
    selected_soil = request.args.get("soil")
    soil_options = SOIL_CHOICES

    if not selected_soil:
        return render_template("show_crop.html", soils=
            soil_options, selected_soil=None)
    if selected_soil not in soil_options:
        return render_template("show_crop.html", error="Soil not
            recognized. Please choose from Soil 1-10.", soils=
            soil_options, selected_soil=None)
```

```
soil_row = get_soil_by_name(selected_soil)
soil_id = soil_row["id"] if soil_row else None

def normalize(value):
    try:
        return float(value)
    except (TypeError, ValueError):
        return float("nan")

N = normalize(row_get(soil_row, "N"))
P = normalize(row_get(soil_row, "P"))
K = normalize(row_get(soil_row, "K"))

rows = get_test_results_for_soil_id(soil_id) if soil_id else
[]

if rows and any(math.isnan(x) for x in (N, P, K)):
    # fall back to saved test snapshot if soil table lacks
    values
    N = normalize(row_get(rows[0], "N"))
    P = normalize(row_get(rows[0], "P"))
    K = normalize(row_get(rows[0], "K"))

soil_ts = row_get(soil_row, "timestamp", "")
last_crop = row_get(soil_row, "last_crop")

test_ts = row_get(rows[0], "timestamp", "") if rows else ""
```

```
# Auto-recompute if soil data is newer than last test
if soil_id and soil_ts and (not test_ts or soil_ts > test_ts)
:
    # If soil snapshot is newer than last test, re-run tests
    using per-crop NPK data.
    with app.test_request_context():
        request.form = {"soil": selected_soil} # type:
            ignore[assignment]
        run_tests()
    rows = get_test_results_for_soil_id(soil_id)
    test_ts = row_get(rows[0], "timestamp", "") if rows else
        ""

# Use saved NPK values from test results (default to 0 if not
present)
def nz(val):
    return 0 if val is None else val

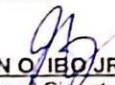
top_crops = [{
    "crop": r["crop_name"], "score":
    r["suitability_score"],
    "fertilizer": r["fertilizer_recommendation"],
    "rank": r["rank"],
    "timestamp": row_get(r, "timestamp", ""),
    "N_saved": nz(row_get(r, "N")),
    "P_saved": nz(row_get(r, "P")),
    "K_saved": nz(row_get(r, "K"))
} for r in rows] if rows else []
```



```
def display_value(val):  
    return None if val is None or (isinstance(val, float) and  
        math.isnan(val)) else round(float(val), 2)  
  
return  
    render_template( "show_crop.html",  
        soils=soil_options,  
        selected_soil=selected_soil,  
        top_crops=top_crops,  
        N=display_value(N),  
        P=display_value(P),  
        K=display_value(K),  
        has_test_results=len(top_crops) > 0,  
        test_timestamp=test_ts,  
        last_crop=last_crop  
    )
```

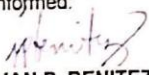
APPENDIX F

ROLE ACCEPTANCE FORM

ROLE ACCEPTANCE FORM College of Computer Studies
Date: February 13, 2025 To: ALLAN O. IBO JR We, the third year students of Camarines Sur Polytechnic Colleges pursuing a degree in BACHELOR OF SCIENCE IN COMPUTER SCIENCE, are currently enrolled in Thesis1 or Capstone Project 1. We are writing to humbly request your service and expertise to serve as our Adviser for our thesis or capstone project. We believe that your knowledge and experience will be essential to greatly enrich our work. Attached are our thesis/ capstone tentative title proposals for your kind reference. Thank you and looking forward to your favorable response of our request. Respectfully, JOSHUA CHARLES BAÑARIA LAURENCE BYRAN BELEN JEREMY LAGDAAN
To: <u>ROSEL O. ONESA MIT</u> Dean, CCS This formally signifies that I ACCEPT/ REJECT the request to serve as Adviser of the team Raptor Squad. As Adviser, I agree to perform my duties and responsibilities stipulated in Section 2.6 of the TCP Guidebook from Thesis 1 / Capstone Project 1 until Thesis 2 /Capstone Project 2. Furthermore, I agree to set the schedule for advising or consultation to help the students and ensure the success of the thesis/ capstone project. Conformed:  <u>ALLAN O. IBO JR, MSc.</u> Name and Signature

Role Acceptance Form

ROLE ACCEPTANCE FORM College of Computer Studies
Date: January 25, 2023 To: ELLAMAE S. ALAMO We, the third year students of Camarines Sur Polytechnic Colleges pursuing a degree in BACHELOR OF SCIENCE IN COMPUTER SCIENCE, are currently enrolled in Thesis 1. We are writing to humbly request your service and expertise to serve as our Grammarian for our thesis. We believe that your knowledge and experience will be essential to greatly enrich our work. Attached are our thesis tentative title proposals for your kind reference. Thank you and looking forward to your favorable response of our request. Respectfully, Joshua Charles B. Bañaria Laurence Bryan B. Belen Jeremy C. Lagdaan
To: <u>ROSEL O. ONESA MIT</u> Dean, CCS This formally signifies that I ACCEPT/ REJECT the request to serve as Grammarian of the team RAPTOR SQUAD. As Grammarian, I agree to perform my duties and responsibilities stipulated in Section 2.6 of the TCP Guidebook from Thesis 1 until Thesis 2. Furthermore, I agree to set the schedule for advising or consultation to help the students and ensure the success of the thesis. Conformed: ELLAMAE S. ALAMO <u>Name and Signature</u>

ROLE ACCEPTANCE FORM College of Computer Studies
Date: January 25, 2023 To: Dr. IAN P. BENITEZ, DIT We, the third year students of Camarines Sur Polytechnic Colleges pursuing a degree in BACHELOR OF SCIENCE IN COMPUTER SCIENCE, are currently enrolled in Thesis1. We are writing to humbly request your service and expertise to serve as our Consultant for our thesis. We believe that your knowledge and experience will be essential to greatly enrich our work. Attached are our thesis tentative title proposals for your kind reference. Thank you and looking forward to your favorable response of our request. Respectfully, Joshua Charles B. Bañaria Laurence Bryan B. Belen Jeremy C. Lagdaan
To: <u>ROSEL O. ONESA MIT</u> Dean, CCS This formally signifies that I ACCEPT/ REJECT the request to serve as Consultant of the team RAPTOR SQUAD. As Consultant, I agree to perform my duties and responsibilities stipulated in Section 2.6 of the TCP Guidebook from Thesis 1 until Thesis 2. Furthermore, I agree to set the schedule for advising or consultation to help the students and ensure the success of the thesis. Conformed:  Dr. IAN P. BENITEZ, DIT <u>Name and Signature</u>



APPENDIX G HEARING FORM

THESIS/ CAPSTONE PROJECT HEARING FORM

Date filed: MAY 19, 2025 [/] Title Proposal [] Pre-Oral [] Final Oral
Date of Hearing: MAY 19, 2023 Time: 01:00-02:30 Venue: Learning Lab Bldg.
Department: COLLEGE OF COMPUTER STUDIES

Research Title:

"A Random Forest Approach for Crop Fertilizer Recommendation using NPK Sensor"

Proponent/s:

BAÑARIA, JOSHUA CHARLES B.

BELEN, LAURENCE BRYAN B.

LAGDAAN, JEREMY C.

Recommended by:

ALLAN O. ORO JR, MSc

Thesis/ Capstone Project Adviser

CERTIFICATION

The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research. [Please **PRINT NAME** and **SIGN**]

FREDDIE TRIANES, MIT

PANEL MEMBER 1

KAELA MARIE N. FORTUNO, MIT

PANEL MEMBER 2

TIFFANY LYNO. PANDES, MSc

PANEL CHAIR

NOTED BY:

ROSEL O. ONESA, MIT

Subject Adviser

APPROVED:

ROSEL O. ONESA, MIT

DEAN, CCS



THESIS HEARING FORM

Date Filed: 11/27/25 ☐ Title Proposal ☒ Pre-Oral ☐ Final Oral

Date of Hearing: 11/27/25 Time: 12:30-1:30 Venue: _____

Department: COLLEGE OF COMPUTER STUDIES (CCS)

Research Title:

A RANDOM FOREST APPROACH FOR CROP FERTILIZER RECCOMENDATION
USING NPK SENSOR

Proponent/s:

BAÑARIA, JOSHUA CHARLES B.

LAGDAAN, JEREMY C.

BELEN, LAURENCE BRYAN B.

Recommended by:

ALLAN E. IBO JR., MSc
Thesis Adviser

CERTIFICATION

The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research.

FREDDIE PRIANES, MIT
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KAELA MARIE N. FORTUNO, MIT
PANEL MEMBER 2

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PANEL CHAIR

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APPROVED:

ROSEL O. ONESA, MIT
OIC Dean, CCS

THESIS HEARING FORM

Date Filed: 12-10-25 [☐] Title Proposal [☐] Pre-Oral [☒] Final Oral

Date of Hearing: 12-10-25 Time: 2:30-3:30 Venue: CONFERENCE ROOM

Department: COLLEGE OF COMPUTER STUDIES (CCS)

Research Title:

**A RANDOM FOREST APPROACH FOR CROP FERTILIZER RECOMMENDATION
USING NPK SENSOR**

Proponent/s:

BAÑARIA, JOSHUA CHARLES B.

LAGDAAN, JEREMY C.

BELEN, LAURENCE BRYAN B.

Recommended by:

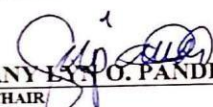

ALLAN O. IBO JR., MSc
Thesis Adviser

CERTIFICATION

The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research.


FREDDIE PRIANES, MIT
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KAELA MARIE N. FORTUNO, MIT
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APPENDIX H

CONSULTATION LOG FORM

CONSULTATION LOGS FORM

Thesis Title:	"A Random Forest Approach for Crop Fertilizer Recommendation using NPK Sensor"						
Proponents:	JOSHUA CHARLES B BANARIA LAURENCE BRYAN B BELEN JERIMY C LAGDAAN						
Alias:	RAPTOR SQUAD						
Total # of Modules:				as approved by the CAPSA /TSA:			
PROTOTYPE							
	Date of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Head's Signature	T.A/TSA/ Consultant Grammarian Signature
	10/24/25						JZ
Remarks:	Reconstruct the Research process.						
	Fix the NPK sensor reading						
	Deadline:						

CONSULTATION LOGS FORM

Thesis Title:	"A Random Forest Approach for Crop Fertilizer Recommendation using NPK Sensor"						
Proponents:	JOSHUA CHARLES B. BANARIA LAURENCE BRYAN B. BELEN JEREMY C. LAGDAAN						
Alias:	RAPTOR SQUAD						
Total # of Modules:				as approved by the CAPSA /TSA:			
PROTOTYPE	Date of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Head's Signature	TATSA/ Consultant/ Grammarians Signature
							H
	Remarks:						
	Good by -						
Deadline:							

APPENDIX I

ACM

A Random Forest Approach for Crop Fertilizer Recommendation Using NPK Sensor

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Abstract

This paper defines the creation of a fertilizer recommendation system that will help farmers make informed choices based on real-time soil nutrient measurements and machine learning algorithms. The system uses NPK sensor to detect the soil nutrients namely nitrogen, phosphorus, and potassium. A sensor is connected with an Arduino microcontroller that gathers and sends nutrient readings to a web-based application written in Flask framework. This platform enables the customers to receive the results of soil analysis and the recommendations of the types of fertilisers they should use. Quantitative analytical method has been applied to process the numerical data of soil nutrients and to assess the working of machine learning model. The system uses machine learning algorithm to interpret nutrient deficiencies and provide suitable recommendation of the fertilizer. Measures of standard statistics were used to examine model performance, which included accuracy, precision, recall, and F1-score. The findings revealed that the accuracy of 67.49, precision of 67.72% recall of 67.49 and F1-score of 64 percent are satisfactory to predict agricultural decision support. The prototype shows that data acquisition sensor technology and machine learning methods could be used to offer timely and accurate fertilizer recommendations to facilitate effective fertilizer application and ensure sustainable agricultural practices.

CCS Concepts

• **Computing methodologies** → **Machine learning**: *Random forests*; • **Applied computing** → *Agriculture*; • **Hardware** → *Sensors and actuators*.

Keywords

Random Forest, Fertilizer Recommendation System, NPK Sensor, Soil Nutrient Analysis, Precision Agriculture, Machine Learning, Data-Driven Decision Making (DDDM), Arduino-based Soil Monitoring, Crop Recommendation, Smart Farming.

1 Introduction

Agriculture was an important component of the Philippine economy as it generates employment and guarantees food security in the Country. However, the increasing population and environmental complications have triggered the need for immediate adoption of environmentally sustainable and efficient farming practices. In the Philippines, there is still a strong reliance on out-dated, non-specific techniques for the application of fertilizers in conventional crop production. These methods tend to neglect particular nutrient requirements of the soil and particular crop needs, thus causing

soil degradation, over fertilizing, and decreased crop yields and increased cost of production [1, 2].

2 Methodology

2.1 Theorems, Algorithm and Mathematical Framework

This section details the research methodology employed in the development and evaluation of the A Random Forest Approach For Crop Fertilizer Recommendation Using NPK Sensor. It outlines the constructive design research approach, the processes involved in data acquisition and preparation, the development and training of the Random Forest model, the design of the prototype system, and the methods for evaluating the system's performance.

Table 1: Crop-Specific NPK Requirements (kg/ha) [3]

Crop	Nitrogen (N) (kg/ha)	Phosphorus (P) (kg/ha)	Potassium (K) (kg/ha)
Tomato	80 / 80 / 70	120 / 110 / 100	80 / 60 / 50
Apple	80 / 80 / 70	30 / 50 / 20	80 / 60 / 50
Chili Pepper	80 / 60 / 50	30 / 50 / 20	120 / 120 / 40
Cabbage	100 / 100 / 100	40 / 10 / 20	60 / 60 / 50
Onion (small)	80 / 60 / 50	30 / 50 / 20	80 / 60 / 50
Spring Bean	80 / 60 / 50	80 / 40 / 20	80 / 60 / 50
Carrot	80 / 80 / 70	80 / 60 / 50	120 / 70 / 50
Chayote	60 / 50 / 40	30 / 50 / 20	60 / 60 / 50
Amorpha	80 / 60 / 50	30 / 50 / 20	60 / 60 / 50
Potato	80 / 60 / 50	30 / 50 / 20	60 / 60 / 50

Hardware and Software Tools

Hardware Tools

- **Computer:** A computer or laptop that was capable of accessing the web-based application.
- **Mobile Phone:** Any mobile phone was capable on accessing the application.
- **Processor (CPU):** Intel i5 / Ryzen 5 or better — was used for running backend algorithms efficiently.
- **RAM:** A minimum 8 GB ensured smooth execution of database queries.
- **Storage:** 256 GB SSD or better was used for storing datasets and application files.
- **Operating System:** Windows 10/11 or Ubuntu Linux — were used, as they were compatible with all development tools.
- **Internet Access:** Was Required necessary for accessing library dependencies and deployment testing.
- **NPK Sensor:** NPK sensor determined the amounts of Nitrogen (N), Phosphorus (P) and Potassium (K) in the soil that plants require [4]. It directly measured ion concentrations in the soil and gave you nutrient information in real time as

parts per million (ppm). The sensor was inserted into the soil to monitor the amounts of nitrogen, phosphorus and potassium at the moment it is placed. The collected data were sent to a microcontroller which transmitted the information via the internet to a web-based system.

- **Arduino UNO:** The sensor for NPK was connected to an Arduino that can read the levels of Nitrogen, Phosphorus and Potassium in soil on the spot. The Arduino uses a USB cable to communicate with the computer and then sends its gathered data to the application on the web. Because of this automation, the data is accurately sent and quickly analyzed for recommending fertilizer. After the calculations, the results show up in a simple dashboard.
- **MAX485:** RS485 to TTL converter (MAX485) acted like a translator between NPK sensor and Arduino.

Application of Random Forest

Soil nutrient levels (NPK values) undergo analysis using the Random Forest Algorithm for fertilizer selection prediction in this research study. The patterns were derived from both historical and simulated datasets that pertained to Philippine agricultural environmental conditions. Soil data combined with expert fertilizer recommendations guided the training and validation of the model which leads to scientifically valid outputs to benefit Filipino farmers. Advantages of Random Forest for the study, patterns were derived from both historical and simulated datasets that pertain to Philippine agricultural environmental conditions. Soil data combined with expert fertilizer recommendations guide the training and validation of the model which led to scientifically valid outputs to benefit Filipino farmers [6] [5].

2.2 Prototype Setup

This demonstrates the apparatus that will be employed in the prototype, which depicts the connection pattern between the NPK sensor, MAX485 module, and Arduino Uno. The specified hardware setup allows the system to acquire real-time soil nutrient data, interpret sensor data via the MAX485 interface and transmit sensor data to the Arduino to be processed and combined with the application.

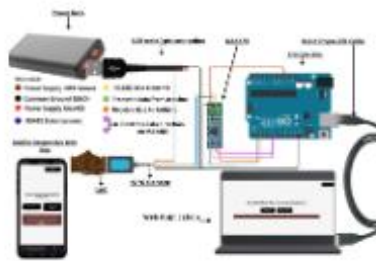


Figure 1: Prototype Flowchart

The prototype set-up, the physical interface of the Arduino Uno, the NPK sensor and the MAX485 RS485 converter is presented.

The NPK sensor is placed on the soil sample to transmit nutrient measurements and the MAX485 modules between the sensor and the Arduino board. The Arduino Uno analyses and forwards the data obtained to the web application.

2.3 Research Process

The figure below shows the research process undertaken the creation of the system which is a summarization of the most important stages in tackling the problem to testing the system. It recaps the data collection method employed by the researchers, the design of the hardware and software blocks, the implementation of the model, as well as the validation of the performance of the system were done. This flow chart gives a perfect understanding of the structured steps followed to make the proposed solution reliable and effective.



Figure 2: Prototype Flowchart

The figure demonstrated the entire steps undertaken in the development of Smart Fertilizer Recommendation System. It started off with the data collection phase in which the levels of the soil nutrients of Nitrogen (N), Phosphorus (P) and Potassium (K) were collected together with the type of crops present. Given that there was a small volume of data collected, the artificial generation of data was conducted so as to generate more realistic samples so that we have a more diverse and balanced dataset.

2.4 Data Collection

The data collection stage served as the foundation for building the fertilizer recommendation model. Since the system focuses on determining the most suitable fertilizer based on soil nutrients, the researchers gathered data that represented different types of soil conditions. A total of ten (10) soil data samples were collected, each containing measurements of Nitrogen (N), Phosphorus (P), and Potassium (K) levels, along with the corresponding recommended fertilizer type.

The samples were obtained from a combination of public agricultural resources, existing research studies, and simulated sensor readings. While the data did not originate from specific geographic locations or actual farmland sites, each sample was designed to represent a realistic soil profile typically found in local agricultural settings. This approach was adopted to demonstrate the model's capability to process and analyze soil nutrient data even in the absence of exact land-based measurements.

Each record in the dataset included the following parameters:

- **Nitrogen (N)** – the percentage or mg/kg concentration of nitrogen in the soil.

- **Phosphorus (P)** – the available phosphorus level that supports plant growth.
- **Potassium (K)** – the amount of potassium that improves plant health and productivity.
- **Crop Type** – the type of crop commonly associated with the soil condition (e.g., tomato, eggplant, ampalaya, cabbage, okra).
- **Fertilizer Type** – the corresponding fertilizer recommendation (e.g., Urea, Organic Compost, NPK 14-14-14).

These collected soil data samples were sufficient for demonstrating the feasibility of the Random Forest model and for testing the prototype's accuracy in generating appropriate fertilizer recommendations.

2.5 Synthetic Data Generation

Only ten soil samples were collected, additional synthetic data were generated to expand the dataset and improve model reliability. This process ensured that the Random Forest model could learn from a broader range of nutrient values and fertilizer categories. By increasing the amount of training data, the model became more stable and less biased toward a small number of samples.

$$X_{\text{synthetic}} = X_{\text{mean}} + r(X_{\text{max}} - X_{\text{min}}) \quad (1)$$

where r is a random number between 0 and 1.

This approach simulated diverse soil conditions found in different parts of the Philippines, making the model applicable for various soil types and crop needs.

2.6 Data Scaling

To ensure all features were treated equally during training, data scaling was applied to normalize the N, P, and K values. Since these nutrient readings vary in numerical range, normalization was necessary to prevent any one feature from dominating the model. The **Min-Max normalization** method was used, following the formula:

$$X' = \frac{X - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}} \quad (2)$$

This normalization improved the consistency of the dataset and allowed the Random Forest model to train more effectively.

2.7 Model Evaluation and Performance Metrics

The Random Forest model was evaluated using standard machine learning metrics to assess its predictive performance in generating accurate fertilizer recommendations based on soil nutrient data. These metrics include both classification and regression-based measures to comprehensively evaluate accuracy, consistency, and reliability. All computations were performed using Python's `Scikit-Learn` library.

Classification Metrics. The classification metrics were used to evaluate the model's ability to correctly identify the appropriate fertilizer type for each soil sample.

Accuracy. Accuracy measures the proportion of correct predictions among all predictions made by the model.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

Precision. Precision determines how many of the model's positive predictions were actually correct.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

Recall. Recall measures the proportion of actual positive cases correctly identified by the model.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

F1-Score. The F1-score is the harmonic mean of Precision and Recall, providing a single score that balances both metrics, especially useful for imbalanced datasets.

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

Confusion Matrix. A confusion matrix was generated to visualize the performance of the classifier, showing the counts of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) for each fertilizer class. This provided an intuitive summary of the model's classification accuracy.

Regression and Error Metrics. To further assess prediction deviations and reliability, regression-based error metrics were used. These include the Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2).

Mean Absolute Error (MAE). MAE measures the average magnitude of prediction errors without considering their direction. A smaller MAE indicates higher model accuracy.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (7)$$

Root Mean Squared Error (RMSE). RMSE penalizes larger errors more heavily and provides insight into how consistent the predictions are.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (8)$$

Coefficient of Determination (R^2). The R^2 score measures how well the predicted fertilizer outputs match the actual data. A value closer to 1 indicates that the model explains most of the variance in the dataset.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (9)$$

2.8 Theoretical Framework

This section outlines the theoretical underpinnings and the conceptual structure that guide the development and evaluation of the "A Random Forest Approach for Crop Fertilizer Recommendation Using NPK Sensor".

- (1) **Statistical Learning Theory:** This research is broadly grounded in Statistical Learning Theory, which provides the formal basis for machine learning. The aim is to develop a model

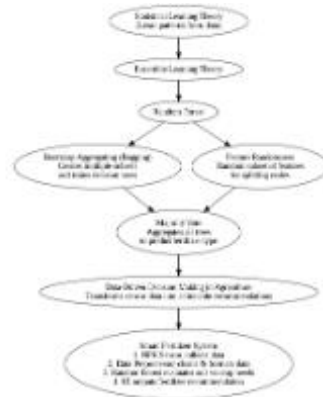


Figure 3: Prototype Flowchart

that can learn patterns and relationships from a given dataset NPK soil values, crop types, and associated fertilizer recommendations to make accurate predictions fertilizer recommendations on new, unseen data. The Random Forest algorithm served as a practical application of statistical learning principles for classification tasks.

- (2) **Ensemble Learning Theory:** The choice of the Random Forest algorithm was directly supported by Ensemble Learning Theory. This theory posits that combining the predictions of multiple individual learning models decision trees can lead to improved overall predictive performance, increased robustness against noise and outliers, and better generalization compared to a single model. Random Forest leveraged two key ensemble techniques.
- (3) **Bootstrap Aggregating:** It creates multiple subsets of the original training data through bootstrapping sampling with replacement and trains a separate decision tree on each subset.
- (4) **Feature Randomness:** When splitting a node during the construction of each tree, Random Forest considers only a random subset of the available features. This decorrelates the individual trees, further enhancing the diversity of the ensemble and typically improving accuracy. The final prediction for a classification task like recommending a fertilizer type is made by a majority vote among all the trees in the forest.
- (5) **Decision Tree Theory:** As Random Forest is an ensemble of decision trees, the underlying principles of decision tree construction were relevant. Decision trees recursively partition the data space based on feature values to create a tree-like model of decisions. Splits are typically chosen to maximize information gain or minimize Gini impurity, leading to homogeneous subsets at the leaf nodes, which correspond to class labels fertilizer types[7].
- (6) **Decision Making in Agriculture: Data-Driven Decision Making (DDDM).** The overarching aim of developing this system

aligns with the paradigm of Data-Driven Decision Making. By transforming raw soil nutrient data and crop information into actionable fertilizer recommendations through a machine learning model, this research promotes a shift from experience-based or generalized fertilization practices towards more precise, evidence-based decisions[8].

3 Results and Discussion

This chapter outlined the findings of data analysis, system testing, and model criteria on the Smart Fertilizer Recommendation System with the help of Random Forest and NPK Sensor. These results confirmed the effectiveness and accuracy of the developed prototype in advising suitable fertilizers according to the nutrient analysis of soil (Nitrogen, Phosphorus and Potassium).

3.1 Training and Development of the Random Forest Model

The study achieve the primary goal to create a Random Forest machine learning model based on the soil nutrient content of Nitrogen (N), Phosphorus (P) and Potassium (K) to compute the most suitable fertilizer to various crops. With the use of nutrient composition as predictive attributes, the research aimed at creating a data-driven methodology of fertilizer recommendation that enhances agricultural decision-making.

3.2 Model Development

Random Forest model was implemented in the state of tuned hyper parameters by using a large number of estimators (15,000) and infinite tree depth to predict as much as possible. Bootstrapping, together with impurity measurement using the Gini index, was used to build trees. The algorithm gave rise to several decision trees during the training phase and ultimately made predictions by the majority vote across the ensemble. This reduced overfitting as it averaged across varying decision paths whilst at the same time improving generalization to unobserved data.

3.2.1 Accuracy. The values of accuracy indicated that the model was accurate in about 67.49% of its results during training and 65.49% during testing. The small decrease of approximately 2 percent indicated that the model still had a significant prediction power when presented with unknown data. Although accuracy in itself did not give any information on the specifics of the performance of the classes, it showed that the model achieved a reasonable degree of accuracy throughout the dataset.

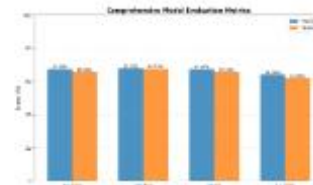


Figure 4: Evaluation Metrics

3.2.2 Precision. The scores of precision were almost the same in the training (67.72) and testing (67.57). This tiny drop meant that the model did not exhibit any change in the rate of making right positive predictions. A high consistency between the datasets implied that the model did not produce too many false positives when it was being applied on new data. This accuracy was handy in the case of fertilizer recommendation systems because false positive identifications would result in excess nutrient application.

3.2.3 Recall. The same trend was observed in the values of recalls, with the model having a 67.49% on the training set and 65.49% on the testing set. The slight decrease was an indication that the model was still effective in elucidating majority of the correct positive cases even in cases where unknown data were met. The consistency of recall was of critical concern especially to minimize the false negatives so that crops that need particular nutrient modification are not missed.

3.2.4 F1-Score. F1 scores were not as high as others with the training and testing being 64.06% and 61.88% respectively. As the F1 score is an amalgamation of the two concepts, the experienced decrease implied a slight decrease in the overall balance of the model in terms of detecting the true positives as well as preventing cases of misclassification. Nevertheless, the distinction was not very high, which means that the model yielded a consistent correlation between accuracy and recall in testing.

3.2.5 Overall Assessment. Although the accuracy and F1-score are moderate, the Random Forest model demonstrates consistent working results in the fertilizer classes. These findings have indicated that preprocessing and balancing strategies worked. More enhancements could be made (adding other soil parameters, e.g. pH, moisture, organic matter), more refined feature engineering, and the possibility of outliers could be reduced though further refinements might lead to even more successful models and more helpful fertilizer projections.

3.2.6 Confusion Matrix. The Confusion matrix table illustrated the extent to which the classification model forecasted every type of crop using the NPK values. Comparison of actual and predicted labels showed the patterns of correct and common misclassifications and further understanding can be made of the crops that the model found more difficult to differentiate by comparing the actual and predicted labels.

The confusion matrix provided indicated that the model was quite accurate when it comes to classifying many crops with high accuracy especially Cabbage, Chili Pepper, Eggplant, String Bean, Tomato etc. The high diagonal value of these crops was 447 correct predictions of Cabbage, 429 of Chili Pepper, 378 of Eggplant, 382 of String Bean, and 312 of Tomato. These findings were a pointer that the nutrient contents of these crops were different enough that the model should have learnt their patterns successfully.

3.3 Prototype Implementation

The implementation showed the integration of soil conditions, sensing, and processing of the data with the machine learning to create a real-time agricultural decision-making tool. Starting with the cautious soil preparation, the NPK sensors detected the activity of the



Figure 5: Confusion Matrix

nutrient ions, which was then filtered, calibrated and transformed into stable digital values and sent to an Arduino-based server located in a Flask. The validated data on NPK was subsequently analyzed using the Random Forest model to suggest the optimal type of fertilizers, evaluate nutrient balance and rank appropriate crops. All these insights were presented in an easy to use interface that shows live readings, crop selection choices, and direct recommendations. Field work demonstrated the system to be robust, emphasis on the portion of soil moisture and calibration to be crucial and demonstrated the ability to process the readings into recommendations in less than a minute. On the whole, the prototype demonstrated that a combination of IoT sensors and machine learning may alter the traditional system of fertilizer decision-making to a rapid, data-oriented, and practical solution to the farm needs.

4 Conclusion

This chapter provides general outcomes and conclusions of the paper. It draws an overview of how the Smart Fertilizer Recommender that utilizes the idea of the Random Forest was able to use NPK sensor data to offer precise fertilizer recommendations. Accuracy and reliability of the model as pointed in the chapter show the possibility of helping farmers to optimize the crop yield. It also ends with major suggestions on how to improve the system and the future research in smart agriculture.

4.1 Summary

The research based on the historical issue of non-site-specific fertilizer use in the Philippines in which the farmers had been using generalized and traditional approaches that had failed to take into account the actual nutrient-based composition of the soil. Such practices usually led to the imbalance of the nutrients, soil erosion, the unnecessary costs of fertilizers, and low crop yields. This issue was introduced in Chapter 1 and mentioned that it aimed to create a system of fertilizer recommendations that could combine real-time readings of NPK sensors and a Random Forest (RF) machine learning model. Another factor important to the system was the role of the system to farmers, government agencies, academic institutions, communities, and future researchers, as the prototype was restricted to NPK data and selected crops in Iriga City.

4.2 Findings

This part introduces the major findings of the research, the training performance of the model, evaluation measures, as well as the successful implementation of the model with a real-time Arduino-based NPK sensing network.

- (1) The Random Forest model was successfully trained using Nitrogen, Phosphorus, and Potassium as input features, enabling it to learn patterns from both synthetic and real soil nutrient datasets and generate reliable fertilizer predictions.
- (2) Performance evaluation showed that the model achieved an accuracy of 67.49% along with acceptable precision 67.72%, recall 65.49%, and F1-scores 64.06%. These results indicate that the classifier can consistently categorize fertilizer recommendations, although certain nutrient ranges exhibited reduced accuracy due to overlapping class boundaries.
- (3) The trained model was integrated with an Arduino-based NPK sensor to develop a fully functional prototype capable of retrieving real-time soil nutrient values. The system automatically generated fertilizer recommendations through a responsive user interface and was validated to perform correctly using both manual inputs and live sensor readings.

4.3 Recommendations

This section, some of the recommendations based on the study findings have been presented. These recommendations are meant to improve the efficiency of the system, heighten the accuracy of data, and inform future researchers to develop and refine the model that was developed.

- (1) Future researchers may find it advisable to take more real soil samples to enhance the accuracy of the model and minimize the use of artificial or synthetic data.
- (2) Sensor calibration processes should be done to reduce errors in the nutrient-reading in future research. They should be regularly checked in relation to laboratory soil test results to ensure that hardware performance is consistent and to make sure that it can accurately capture nutrients.
- (3) Add further soil traits to the data set, including pH level, moisture content, and the percentage of organic matter to enhance the accuracy of the classification and decrease the overlap between the classes of crops.
- (4) Enhanced the system by future researchers by organizing the types of fertilizers and nutrient content levels according to the local agricultural requirements and usual land management in the area to make the recommendations viable and relevant to the situation.

Acknowledgments

We would firstly like to state our greatness to the Almighty God who has been guiding us through the research. He made us stronger, patient, and clear, particularly at the most difficult times. This could not have been possible without the grace of Him.

We want to thank Mr. Allan O. Ibo, Jr. who is our thesis adviser, and this is how much we appreciated the constant guidance and valuable ideas and the constant encouragement he gave us throughout the study to the last moment. He was very committed, supportive and made the difference in the direction and the quality

of our work. We are also grateful to heart of our consultant, Dr. Ian Benitez, who had been so generous to share his expertise with us, and helped us to perfect our research. We also want to take this opportunity to acknowledge the help of our grammarian, Ms. Ella Mae S. Alamo who helped us to perfect our manuscript by making it clear and accurate.

We would like to sincerely thank you, our thesis panelist- Ms. Tiffany Lyn O. Pandes as our panel chairperson, Mr. Freddie B. Prianes and Ms. Kaela Marie N. Fortuno, our panel member for dedicating their time, knowledge and giving us a wise feedback time, when we evaluated our study. Their remarks and suggestions were of much power in furthering the content and course of our studies. We also have the beneficent assistance of our secretary, Ms. Marri Grace Morata, who helped us and contributed all through the entire process of the defense.

Words cannot describe the things we owe to our dear families in the form of unconditional love, support and sacrifice. Your support led us on to continue and remain loyal even when things get bad. To our friends, classmates and the same peers, we do wish to thank you in the inspiration you have given, the comprehension and experience, which have made this journey less heavy and more relevant.

References

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APPENDIX J

USER MANUAL

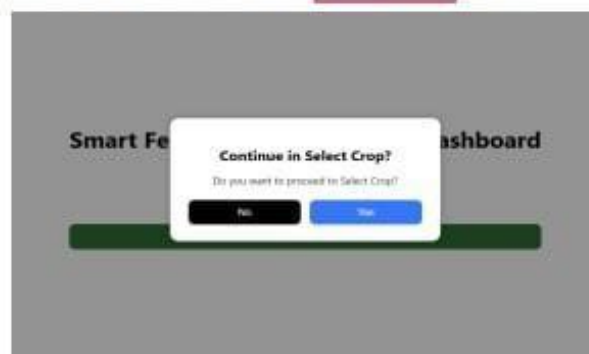
User Manual

The Smart Fertilizer Recommender System is an easy to use application that helps the user to make better decisions on the application of fertilizers with regards to the nutrient levels in the soil. The system performs the analysis of the nitrogen (N), phosphorus (P) and potassium (K) data with the help of the machine learning model based on the random forest to produce the relevant recommendations on the fertilizers to be applied to the chosen crops. It helps the user to choose soil sites, see existing crops, and experiment with specific crops, thus facilitating effective use of fertilizer and sustainable use of crops.

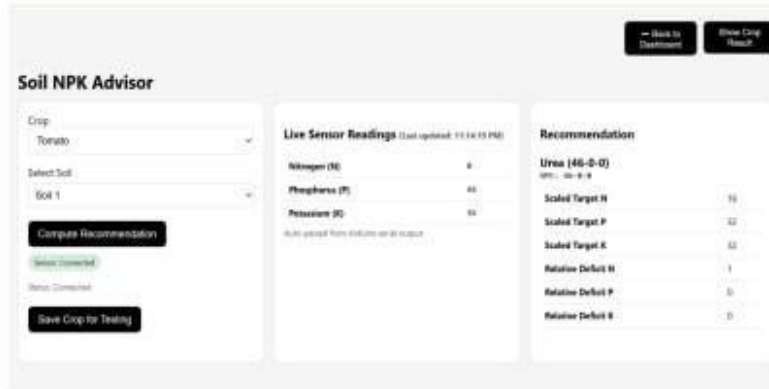
1. In your desktop or mobile browser, go to <https://nongalvanized-epically-madden.ngrok-free.dev> and wait for the app to be fully-loaded. The primary interface includes a centralized Smart Fertilizer Recommender Dashboard, which includes two major action buttons namely Select Crop, Show Crop Result and sensor status indicator to check that the sensor is properly connected. The interface is made easy and user-friendly to allow users navigate easily through choice of crops and result of recommendations, and has both dark and light display modes to be more useful in various lighting conditions.



2. The system has a confirmation dialog that the user is asked to either proceed or cancel a given action before moving to the process of selecting the crop. This will be useful in order to avoid the unintentional navigation and confirm the choice of the user by using the Yes and No buttons that will be clearly labeled in order to have a more control and easy interaction within the Smart Fertilizer Recommender system.



- The Soil NPK Advisor interface allows the user to measure the conditions of soil with the help of the connected NPK sensor that displays real-time data on the concentration of nitrogen, phosphorus, and potassium. Using these sensor readings, the system examines nutrient content and provides a suitable recommendation of fertilizer to be applied on the crop chosen..



Soil NPK Advisor

Crop: Tomato

Select Soil: Soil 1

Compute Recommendation

Sensor Connected

Save Crop for Testing

Live Sensor Readings (Last updated: 11:14:15 PM)

Nitrogen (N)	8
Phosphorus (P)	45
Potassium (K)	95

Auto-updated from Arduino serial output

Recommendation

Urea (46-0-0)
46% N, 0% P, 0% K

Scaled Target N	16
Scaled Target P	32
Scaled Target K	52
Relative Deficit N	-1
Relative Deficit P	0
Relative Deficit K	0

- In this interface, the user is given an option of a soil sample that is then shown with respect to its respective crop recommendations using the measured soil sensor data on nitrogen, phosphorus, and potassium. The system shows the crop that is under test, the fact that the test data is complete and locked, and offers a possibility to display crops as well as run a soil test, begin a new test, go back to the dashboard or choose another crop to test.



Crop Recommendations by Soil Sample

Select Soil Sample: Soil 1

Show Crops Run Tests for Soil Start New Test

Soil Sensor Readings: Soil 1

Nitrogen (N): 10 mg/kg	Phosphorus (P): 71 mg/kg	Potassium (K): 63 mg/kg
---------------------------	-----------------------------	----------------------------

Current Tested Crop: Amaranth

Sensor readings sent and displayed for this soil sample | Recorded: 2023-11-28 12:00:17

Text finalized — data locked.

Best Fit Crop

5. This interface acts as the concluding page of the application as it thanks the user who uses the system and ensures an ending of the active session. It has a plain Back to Dashboard button which lets the user go back to the main dashboard where he can continue with what he plans to do or have a preview.

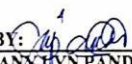


APPENDIX K

PANELS RECOMMENDATION, SUGGESTIONS AND COMMENTS (RSC)

MANUSCRIPT			
CHAPTER	PAGE NO.	RECOMMENDATIONS, SUGGESTIONS AND COMMENTS (RSC)	ACTION TAKEN
		Study suggestions: - Check if you could add other fertilizers. - Check for applicable algorithms	- We added other relevant fertilizers. - We check for applicable algorithm but we will stick to the Random Forest.
1	3	Specific Objectives Capitalize 'F' in Random Forest.	We capitalize the "F" in Random Forest on the Specific Objective part. (Page 3-4)
	6	Scope and Limitations Specify the crops.	We put the Specific crops that we'll use in the Scope and limitations part. (Page 5-6)
	7	Project Dictionary - Add operational definition for all terms. - Add the evaluation metrics.	- We added operation definitions for all the terms in Project Dictionary. - We added evaluation metrics in the Dictionary. (Page 7-10)
2		GAP Your gap is lacking technicalities.	We added technicalities in our Gap. (Page 20)
	26	Convert the equation and pseudo code to LaTeX format.	We convert the equation and pseudo code to LaTeX format. (Page 26-27)
		Dataset Consult with DA	We consulted with Agriculturist and received a certificate. (Page 28)
	28	Table 1 - Add the following parameters: scale of NPK, size of the area, and age of crop. - Clarify the other compositions for P and K.	We add the scale of NPK, Size of the area, age of crops and Clarify compositions for P and K. (Page 28)
	29	Table 2 Convert to LaTeX table format.	In table 2, we convert the table into LaTeX table format. (Page 29-30)

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	31	Hardware tools - Put process of the sensor. - Show the NPK sensor description.	We put the process of the sensor and show the NPK sensor description in the Hardware tools.
	33	Figure 1 - Fix your flowchart. - The discussion is missing. - Have the system prototype under discussion.	We fix the flowchart with discussion and put the system prototype under discussion. (Page 33-34)
		Include a diagram of the prototype.	We include a diagram of the prototype and discuss it.
	36	Performance Metrics All equations/formula must be in LaTeX equation format.	We convert all equations/formula into LaTeX equation format. (Page 36-38)
	40	Remove "and Conceptual Framework" in "Theoretical and Conceptual Framework".	The "and Conceptual" has been removed in the "Theoretical and Conceptual Framework".(Page 40-41)
	40	Figure 2: Conceptual Framework should be placed under the Conceptual Framework section.	We fix the proper formatting. We remove the Conceptual framework placed under Conceptual Framework section. (Page 40-41)
	41	Your theoretical framework is not a theoretical framework.	We change the theoretical framework with proper theoretical framework.(Page 40-41)
ALL		- Have your title formatted in inverse pyramid. - Fix numerical in-text citations. - Fix notes. Follow correct formatting.	- Have our title formatted in inverse pyramid. - The numerical in-text citations and note with correct formatting are fixed.

NOTED BY: 
TIFFANY LYNN PANDES, MSc
PANEL, CHAIRMAN


FREDDIE PRIANES, MIT
PANEL, MEMBER 1


KAELA MARIE N. FORTUNO, MIT
PANEL, MEMBER 2

Title : A RANDOM FOREST APPROACH FOR CROP FERTILIZER RECOMMENDATION USING NPK SENSOR
Alias : RAPTOR SQUAD
Date : NOVEMBER 27, 2025 12:30 – 1:30 PM ☐ TD ☒ POD ☐ FOD
Secretary : MARRI GRACE MORATA

MANUSCRIPT			
CHAPTER	PAGE NO.	RECOMMENDATIONS, SUGGESTIONS AND COMMENT (RSC)	ACTION TAKEN
1	5	- Scope - Prototype or online system? Choose only one - Revise scope accordingly. - Include all SOs.	We choose prototype then Revise scope accordingly and include all SOs. (page 5)
2	19	- RRL - Add a section for evaluation metrics. Discussion should include the results in the studies.	We already add a section for evaluation metrics that include the discussion of the results of the studies. (page 21)
3	28	"Theorems, algorithms, and mathematical models" should not be in all caps.	We corrected the heading formatting to sentence case. (page 28)
	29	Format pseudocode in LaTeX.	We change the pseudocode format into latex (pages 30-31)
	-	Format equations in LaTeX.	We convert the equation in latex format. (page 29,45-46,48-50)
	-	Ensure all tables follow the correct format.	We ensure all tables follow the correct format. (page 32, 33)
	34	- Figure 1: System Architecture - Remove and replace with the Figure 4 (Flowchart) on page 50.	We removed and replaced it with Figure 4 (Flowchart) on page 50. (page 34, 35)
	36	- Figure 2: Device Setup - Label each part of the prototype.	We labeled each part of the prototype. (page 36)
	-	Where is the discussion of the Research Process?	We provide a research process discussion (pages 40-42)
4	54	- Theoretical Framework - Does not align with the discussion. Revise it to reflect the theories used or mentioned.	We aligned the discussion and revised it to reflect the theories (pages 50-54)
	57	Italicize "Dataset and Exploratory Data Analysis"	We italicize "Dataset and Exploratory Data Analysis" (page 55)
	62	Table 3 is lacking discussion.	We added discussion in table 3 (page 59,60)
	66	Do not italicize "System Testing and Prototype Output"	We do not italicize "System Testing and Prototype Output" (page 88)

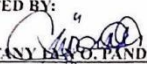
	73	Relocate Figure 15: Feature Importance to SO1 discussion.	We relocated Figure 15: Feature Importance to the SO1 discussion (page 67-68)
	74	"Comparison to Related Literature" is an unnecessary section. These comparisons should be included in each subsections.	We convert the Comparison to Related Literature into subsection format. (page 74)
	-	Add graphs for SO2 discussion.	We add graph for the SO2 Discussion
	-	The device is not highlighted in the manuscript. Discuss it clearly, and emphasize the algorithm used.	We already add discussion to highlight the prototype and the algorithm used.(pages 74-87)
	-	Tell a story. Walk the reader through the process of testing the soil, etc.	We explain the soil testing and implementation of the prototype.
	-	Improve clarity and readability of the figures.	We improve the figure to make the text and elements more readable.
5	77	- SUMMARY - Revise. + - Expound on chapters 1-4.	We revise and enrich the summary by adding clearer explanations of the main ideas discussed in Chapters 1 to 4.
	77	- FINDINGS - Have a short intro - Include all numerical results.	"We added a short introduction and included all numerical results in the Findings section.(page 98)
	78	- CONCLUSION - Add a short intro	We provide a short opening paragraph that introduces the Conclusion section (page 98-99)
	79	- RECOMMENDATIONS - Add a short intro - Itemize in numbers, not bullets. - Should be addressed to future researchers.	We added a short intro, converted the recommendations into numbered items, and directed them toward future researchers.(page 99-100)
ALL	-	Fix hanging pages	We fix all the hanging pages
	-	Ensure all tables and figures are introduced first.	We ensure that all tables and figures are introduced first.
	-	Use past tense in chapters 1-3.	We revised Chapters 1-3 and changed the verb tenses to past tense.
	-	Ensure all in-text citations follow the ACM format.	We ensure all text citations follow the ACM format.
	-	- Notes - Provide full names of authors.	We provide full names of authors.




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SOFTWARE PROGRAM/ PROTOTYPE		
MODULE NO. (DFD)	RECOMMENDATIONS, SUGGESTIONS, AND COMMENTS (RSC)	ACTION TAKEN

NOTED BY:


TIFFANY E. O. PANDES, MSc
 PANEL, CHAIRMAN


FREDDIE PRIANES, MIT
 PANEL, MEMBER 1


KAELA MARIE N. FORTUNO, MIT
 PANEL, MEMBER 2

Title : A RANDOM FOREST APPROACH FOR CROP FERTILIZER RECOMMENDATION USING NPK SENSOR
Alias : RAPTOR SQUAD
Date : DECEMBER 10, 2025 2:30 – 3:30 PM ☐ TD ☐ POD ☒ FOD
Secretary : MARRI GRACE MORATA

MANUSCRIPT			
CHAPTER	PAGE NO.	RECOMMENDATIONS, SUGGESTIONS AND COMMENT (RSC)	ACTION TAKEN
Initial Pages	i	- Fix minor grammatical errors in the Approval Page - Ensure correct name of your Adviser - Include the middle initials of the research committee - Correction: Use "MSc", not "MSC"	- We fixed the grammatical errors in the Approval page. - The correction of the name of the Adviser is Changed correctly - We included the middle initials of the research committee. - Corrected all the MSC to MSc on the Document.
	ii	- Put the date of your FOD - Ensure proper spacing of words - Provide a blank space for the grade - Correction: Use "OIC Dean", not "Dean" - Correction: Use "MSc", not "MSC"	- Corrections implemented for the FOD date. - Ensured the proper spacing for each word. - Provided a blank space for the grade. - Corrected all the Dean to OIC Dean on the Document. - Corrected all the MSC to MSc on the Document.
	vi	- Limit the keywords to 5 - Follow proper Abstract format (maximum of 250 words): - Introduction - Objective - Method - Numerical results - Conclusion	- Limits the keywords to 5 keywords. - Followed the proper Abstract Format.
2	19	Add a section for evaluation metrics. The discussion should include the results from the related studies	Added a section for evaluation metrics, and the discussion included the results from the related works. Page 16-17
	22	Indent the first sentence of each paragraph in your gap	Indented properly the first sentence of each paragraph of the gap. Page 23-24
3	35	Change "Prototype Setup" to "Device Setup"	Changed the "Prototype Setup" into "Device Setup". Page 36-37
	36	Revise Figure 2 (Device Setup) to include the power bank.	Revised Figure 2 (Device Setup) and includes the power bank. Page 37
	36	Include figures and discussion of the prototype setup (the web application)	The figures and the discussion of the prototype setup included. Page 37-38

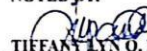
  Republic of the Philippines Camarines Sur Polytechnic Colleges Naga, Camarines Sur Polytechnic ISO 9001:2015 Certified COLLEGE of COMPUTER STUDIES			
4	37	Add "Computer" to your hardware specifications	The "Computer" has been added to the hardware specification. Page 38
	53	For Figure 4 (Theoretical Framework), follow the sequence: Introduction → Figure → Discussion.	The proper sequence for Figure 4 (Theoretical Framework) followed. Page 49-53
	59	In Table 3, make "Target" the first column. Revise the discussion and explain each column.	In table 3, the "Target" was moved on the first column and the discussion and explanation for each column revised. Page 58
	57	Format Figure 5 to LaTeX subfigure	Figure 5 Format converted into LaTeX subfigure. Page 56
	61	Format Figure 6 to LaTeX subfigure	Figure 6 format converted into LaTeX subfigure. Page 60
	64	In Figure 7, the colored text on the dark background is unreadable. Improve contrast and readability	We improved contrast and readability figure 7. Page 63
	79	Relocate Figure 14 to Chapter 3 under theorems	We didn't relocate the Figure 14 into chapter 3. instead we removed it.
	91	Provide detailed discussion of Figure 18	In Figure 18 we provided a detailed discussion. Page 81-82
	94	Convert Figure 20 into a Table.	Instead of converting it into table format we just added a Light Mode on the Application for its to be clear and readable.
		- For your SO 3 discussion: - Further highlight your device. Provide figures showing your actual device and discuss it - Mention that the device is mobile responsive - Paano masasabi na nagcombine ng live data? Discuss. - Provide examples of possible decisions the system can make - Discuss that some soils may lack one of the N, P, or K components	Highlighted the device with figures showing the actual device and discussion. Mentions the device is mobile responsive. Discussed the combination of live data. Provides examples of possible decisions and discusses the lack of NPK of some soils. Page 73-74
	-	Prototype discussions must be in paragraph form	Discussion of the prototype converted into paragraph form. Page 73-74
	-	Relocate all equations and methods to Chapter 3	We removed the equations instead of relocating it to Chapter 3.
	-	Resize figures. Currently, some are too large while others are too small. Improve clarity and readability	Resized figures that improve clarity and readability.
	-	The discussion on testing the soil is missing	Discussion on testing the soil added. Page 73-74
	-	Add implications to all results	We added implications to all the results
5		Limit the introductions to 1-2 sentences	Limits the introductions to 1-2 sentences.
	98	Change "Conclusion of Findings" to "Conclusion"	We changed the Conclusion Of Finding to Finding. Page 89-90
Appendices	D	Use code listing for the relevant source codes	We used proper code listing for the source code.

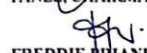
 Republic of the Philippines
Camarines Sur Polytechnic Colleges
Naga, Camarines Sur
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	-	• Add the User's Guide	• The user's guide is still not done.
	-	• Add the ACM	• We added the ACM.
VITA	-	• Images are too large. Resize appropriately	• We adjusted the image sizes.
ALL	-	• Follow proper table formatting	• We followed proper table formatting.
	-	• Notes - For online sources, replace "Accessed" with "Retrieve <date>. From <link>." - Update Bibliography accordingly	• We change the accessed to retrieve and update the bibliography.
	-	• Have your manuscript checked by your grammarian	• The grammarian checked the manuscript.

SOFTWARE PROGRAM/ PROTOTYPE		
MODULE NO. (DFD)	RECOMMENDATIONS, SUGGESTIONS, AND COMMENTS (RSC)	ACTION TAKEN
	• Consider light mode. Have this reflected on the figures shown in your manuscript for better readability	• We added light mode for better readability.

NOTED BY:


TIFFANY LYN O. PANDES, MSc
PANEL CHAIRMAN


FREDDIE PRIANES, MIT
PANEL, MEMBER 1


KAELA MARIE N. FORTUNO, MIT
PANEL, MEMBER 2

APPENDIX L

CERTIFICATION OF DATASET

 Republic of the Philippines
DEPARTMENT OF AGRICULTURE
REGIONAL FIELD OFFICE 5
San Agustin, Pili, Camarines Sur

No: _____

CERTIFICATE OF APPEARANCE

TO WHOM IT MAY CONCERN:

This is to certify that Mr. / Ms. JOHN CHARLES B. BAWARID, JERRY UNGDAAN, LAWENCE B. SYAN of
CAMARINES SUR POLYTECHNIC COLLEGES, personally appeared in this office on OCTOBER 7, 2025
With the purpose of DATASET GATHERING

This certification is issued upon the request of above-named person for whatever legal purpose it may serve him/her best.


ANACLETO B. ESPLANA
PIC - Chief, ILD

Masagana ng Agrikultura, Maunlad na Ekonomiya
 [hinalda.gov.ph](#)  Department of Agriculture-Bicol

AFD-HRMS-050-2
Effectivity Date: April 1, 2025

APPENDIX M

NON-DISCLOSURE AGREEMENT FORM

Non-Disclosure Agreement Form

EFFECTIVE DATE: May 16, 2025
(TD submission date)

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies- Camarines Sur Polytechnic Colleges and Tiffany Lyn O. Pandes, MSc, hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:
TIFFANY LYN O. PANDES, MSc - Panel Chair
Name and Title

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:
ROSEL O. ONESA, MIT - OIC-Dean
Name and Title

2. The confidential information disclosed under this Agreement is described as:
Contents of the TCP by:

Joshua Charles B. Baniaria, Jeremy C. Logdaan, Laurence Bryan B. Belen

which is entitled:

A Random Forest Approach For Crop Fertilizer Recommendation Using NPK Sensor

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
- marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
 - if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
- which is or becomes publicly available without breach of this Agreement;
 - which is already known to the Recipient prior to its disclosure hereunder;
 - which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the

Expert shall not be liable for any direct, indirect, special, or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.

9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.

10. Neither Party:

- a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
- b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
- c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.

11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.

12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.

13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.

14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

CAMARINES SUR POLYTECHNIC
COLLEGES


TIFFANYLYN O. PANDES, MSc
Expert
(Print name and Signature)

DATE: May 16, 2025


ROSEL O. ONESA, MIT
CSPC Representative
(Print name and Signature)

DATE: May 16, 2025

Non-Disclosure Agreement Form

EFFECTIVE DATE: May 16, 2025
(TD submission date)

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies- Camarines Sur Polytechnic Colleges and Kaela Marie N. Fortuno, MIT hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:

KAELA MARIE N. FORTUNO, MIT - Panel Member
Name and Title

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:

ROSEL O. ONESA, MIT - OIC-Dean
Name and Title

2. The confidential information disclosed under this Agreement is described as:
Contents of the TCP by:

Joshua Charles B. Baniario, Jeremy C. Loydaan, Laurence Bryan B. Relea

which is entitled:

A Random Forest Approach For Crop Fertilizer Recommendation Using NPK Sensor

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
- marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
 - if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
- which is or becomes publicly available without breach of this Agreement;
 - which is already known to the Recipient prior to its disclosure hereunder;
 - which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the

Expert shall not be liable for any direct, indirect, special, or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information

9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
 - a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
 - b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
 - c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

CAMARINES SUR POLYTECHNIC
COLLEGES

KARLA MARIE N. FORTUNO, MIT

Expert
(Print name and Signature)

DATE: May 16, 2025

ROSEL O. ONESA, MIT

CSPC Representative
(Print name and Signature)

DATE: May 16, 2025

Non-Disclosure Agreement Form

EFFECTIVE DATE: May 16, 2025
(TD submission date)

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies- Camarines Sur Polytechnic Colleges and Freddie B. Prianes, MIT hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:
FREDDIE B. PRIANES, MIT – Panel Member
Name and Title

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:
ROSEL O. ONESA, MIT - OIC-Dean
Name and Title

2. The confidential information disclosed under this Agreement is described as:
Contents of the TCP by:

Joshua Charles B. Bañaria, Jeremy C. Lagdaan, Laurence Bryan B. Belen

which is entitled:

A Random Forest Approach For Crop Fertilizer Recommendation Using NPK Sensor

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
 - a. marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
 - b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
 - a. which is or becomes publicly available without breach of this Agreement;
 - b. which is already known to the Recipient prior to its disclosure hereunder;
 - c. which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the

- Expert shall not be liable for any direct, indirect, special, or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.
9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
 10. Neither Party:
 - a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
 - b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
 - c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
 11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
 12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
 13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
 14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

**CAMARINES SUR POLYTECHNIC
COLLEGES**


FREDDIE B. PRIANES, MIT
Expert
(Print name and Signature)

DATE: May 16, 2025


ROSEL O. ONESA, MIT
CSPC Representative
(Print name and Signature)

DATE: May 16, 2025

APPENDIX N

PROJECT TEAM ASSIGNMENT FORM

PROJECT TEAM ASSIGNMENTS FORM

Team Alias	Raptor Squad
Course Code	CS 3214
Subject adviser/ TSA	Rosel O. Onesa

Name and Signature	Project Role	Email address / mobile#(s)
 Joshua C. Bañaria	Programmer	johanaria@my.cspc.edu.ph #09569252420
 Jeremy C. Lagdaan	Documentation And Prototype Set up	jelaagdaan@my.cspc.edu.ph # 09917795874
 Laurence Bryan B. Belen	Documentation and Researcher	labelen@my.cspc.edu.ph #09197778545

APPENDIX O

PRE PROPOSAL TITLE FORM

PRE PROPOSAL TITLE	
Team Alias	Raptors Squad
Project Title:	"Smart Fertilizer Recommendation System Using Random Forest Algorithm"
Proponents/Researchers:	1. Joshua Charles Banaria 2. Laurence Bryan Belen 3. Jeremy Lagdaan
Scope of the Study:	The study focuses on developing a fertilizer recommendation system that analyzes soil nutrient data and provides optimal fertilizer suggestions using the Random Forest Algorithm. It aims to assist farmers in making data-driven decisions to enhance crop yield while minimizing excessive fertilizer use. The system will utilize historical soil data, predefined fertilizer recommendations, and association rule mining to generate relevant insights.
Limitations of the Study:	• The system will only consider predefined soil nutrient parameters (e.g., Nitrogen, Phosphorus, Potassium) for generating recommendations. • It will not account for real-time environmental factors such as weather, pest infestations, or irrigation practices. • The study will focus on a specific dataset collected for selected crops and may not generalize to all soil types and agricultural conditions. • The system will not include automated soil testing; users must manually input soil nutrient data.
Project Design/Development Plan(Software Methodology):	Program Specification The system will follow the Agile Software Development Model, ensuring iterative improvements based on testing and user feedback. Program Specification: A web-based or desktop application where users input soil nutrient data and receive fertilizer recommendations. Implementation of the Random Forest algorithm for association rule mining. User authentication and role-based access for farmers and agricultural experts.

	Software Specification Programming Language: Python (Flask/Django for backend, React/Vue.js for frontend) Database: SQLite/MySQL for storing soil data and recommendation rules Libraries/Frameworks: mlxtend (for Apriori), pandas, scikit-learn, Flask/Django Hardware Specification Minimum System Requirements: Processor: Intel Core i5 or equivalent RAM: 8GB or higher Storage: 256GB SSD or higher Operating System: Windows/Linux/macOS
☒ Recommended  ROSEL O. ONESA, MIT Thesis Subject Adviser	☐ Rejected ROSEL O. ONESA, MIT Thesis Subject Adviser

APPENDIX P

FINAL PROJECT TITLE FORM

FINAL PROJECT TITLE FORM

Team Alias: Raptor Squad
Proponents/Researchers:

- | |
|-----------------------------|
| 1) Joshua Charles B. Bañana |
| 2) Laurence Bryan B. Belen |
| 3) Jeremy C. Lagdaan |

Proposed Thesis/ Capstone Project Title:

"A Random Forest Approach for Crop Fertilizer Recommendation using NPK Sensor"

Submitted by:  JOSHUA CHARLES B. BAÑANA (Signature of Project Head over printed name) Date: _____	Noted:  ROSEL O. ONESA, MIT (Signature of Subject Adviser over printed name) Date: _____
Recommending Approval:  ALLAN O. IBO JR., MSC (Signature of Thesis Adviser over printed name) Date: _____	Approved:  ROSEL O. ONESA, MIT DEAN, CCS Date: _____